

For the Greater Good: Network Citizenship Behavior and Innovative Behavior in
American Local Government Leaders

by Jing Burgi-Tian

B.S. in Economics, July 2009, Fudan University, P.R. China
M.S. in Intercultural Communication, May 2011, University of Pennsylvania

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Dissertation directed by

Diana L. Burley
Professor of Human and Organizational Learning

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For the Greater Good: Network Citizenship Behavior and Innovative Behavior in
American Local Government

Jing Burgi-Tian

Dissertation Research Committee

Diana L. Burley, Professor of Human and Organizational Learning,
Dissertation Director

Maria Cseh, Associate Professor of Human and Organizational Learning,
Committee Member

Sharon Dannels, Associate Professor of Education Research,
Committee Member

Dedication

I dedicate this dissertation study to my wonderful and loving family:

To my husband, Constantin Bürgi, for always providing me with new ideas and perspectives and for always giving me the extra push when I am discouraged.

To my two beautiful boys, Augustus and Aurelian, who are inspirations for me every day to be the best I can be and serve as a good role model for them.

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Abstract of Dissertation

For the Greater Good: Network Citizenship Behavior and Innovative Behaviors in American Local Government

Innovation has an intrinsic relationship with social networks because it provides a mechanism for knowledge and information to travel freely between individuals.

Consequently, there is abundant research linking social networks to innovation. However, two critical gaps still exist. First, most research studies use a structural approach, and less attention is paid to the behavioral dimensions in a network. Secondly, the majority of empirical research focuses on egocentric networks or dyadic relationships between two organizations, instead of the network of organizations as a whole.

To bridge these gaps, the new concept of network citizenship behavior (NCB) was examined. NCB describes individual extra-role behavior exhibited by those who work in organizations that are part of a goal-directed multi-organizational network that supports the social and psychological environment in which the achievement of network-focused goals takes place. The research study set out to examine the relationship between NCB, measured by an adapted version of Autry, Skinner, and Lamb (2008)'s instrument, and individual innovative behaviors (IIB), measured by Kleysen and Street (2001)'s instrument. A second purpose was to explore the impact of network structural properties, measured by network centrality and tie strength, on NCB, IIB, and the relationship between them.

An Internet-based survey was distributed to 348 former participants of an interjurisdictional local government training program in a regional council of

governments in the Washington, D.C., area. A total number of 89 effective responses were received.

Both instruments were validated and modifications were made before conducting a series of multiple regression analyses to answer the research question. The results of the analyses suggest that NCB predicts 17.8% of IIB's variance. Among NCB's four subscales, loyalty explains the most variance in IIB (14.9%), followed by Compliance (11%) and Altruism (10.8%). Tolerance explains only a small fraction of variance in IIB (4.6%). Interestingly, one control variable, boundary spanning status, was found to be negatively significantly in impacting IIB, meaning that people without boundary spanning responsibilities exhibit higher levels of IIB. In addition, the findings of this research study also confirmed that closeness centrality and tie strength predict the exhibition of IIB. However, centrality and tie strength do not have any interaction effect with NCB to predict IIB.

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Chapter 1: Background and Introduction

Innovation in the Government

Innovation has become a widely discussed topic in both academia and among organizational development (OD) practitioners. Throughout history, innovation has been described as the driver of humankind's development (West & Farr, 1990), of the national economy (Black & Lynch, 2004), of organizational performance (Saunila & Ukko, 2012), and of individual growth (Janssen, van de Vliert, & West, 2004). It has long been taken as common knowledge that innovation is the key to success in the private sector (Sorensen & Torfing, 2012). On a firm level, research has demonstrated that innovation contributes to financial performance (Cho & Pucik, 2005; Gopalakrishnan, 2000), market performance (Tidd, 2001), and productivity (Klomp & Van Leeuwen, 2001).

Compared with the private sector, innovation in the government has been deemed difficult and challenging, because of its collective, democratic nature (McNabb, 2006) and its risk-averse culture (Bekkers, Edelenbox, & Steijn, 2011). In recent years, innovation in the government has been particularly complex because of the rapid changes in society. Many issues that confront governments in the late 2010s and early 2020s, such as environmental protection, terrorism, and mental health, have recently arisen as a demand from citizens, and are considered to be “wicked problems” that have no easy solutions using traditional governing methods. In addition, most government organizations are faced with massive fiscal pressure coupled with citizens' growing expectation of public services, which means government is tasked with “providing more with less” (Fuglsang, Ronning, & Enquist, 2014). At the same time, technology

advancement not only requires government agencies to catch up with the time, but also overwhelms them with constant opportunities for innovation and improvements.

Despite all the challenges, innovation in government institutions has been noticeably growing (Borins, 2014). The Innovation in the American Government Awards, created by the Harvard Kennedy School in 1985, is considered to be one of the most prestigious programs honoring government innovation. Since its inception, it has received over 27,000 applications and recognized over 500 government initiatives, with a combined grant funding of more than \$22 million (Harvard Kennedy School, Ash Center for Democratic Governance and Innovation, 2018). Even though it encourages innovation at all levels of government, the local level of government innovation initiative constitutes more than half of its finalists and winners. Examples of local innovation can be found in all aspects of government work, ranging from policing and community development to budgeting, education, and digital media. Through their application materials, it can be generalized that many of the applicants' innovation initiatives have a lasting impact on the constituents that they serve.

In the early 1990s, President Clinton and Vice-President Gore initiated a "reinventing government" program, which was designed to accelerate innovation in the government (McNabb, 2006). Before that, public servants rarely found the opportunity or incentive to innovate. Today, however, innovation has become so common that government agencies "find themselves in the position of either innovating or being forced to explain 'why not?'" (McNabb, 2006, p. 151). The growing innovation in government is also supported by the creation of innovation labs and other administrative innovation

units, the issuing of white papers on innovation, and the growing expectation that public managers and employees need to innovate (Sorensen & Torfing, 2012).

Innovation and Network in the Government

One way of growing innovation in the government is infusing new ideas through social networks and disseminating innovative practices across organizational boundaries (Rogers, 1983). Interorganizational collaboration in the government is becoming common in practice. Through studying the applications in the 2010 Innovation in American Government Awards presented by the Harvard Kennedy School, Borins (2014) found out that “the increase in the frequency of interorganizational collaboration as a characteristic of innovations is broad-based and widespread” (p. 181). Indeed, 81% of the 2010 award’s semifinalists used some form of intergovernment collaboration to achieve their innovations.

Innovation has an intrinsic relationship with social networks by its definition. Innovation was described by Schumpeter (1942) as a new combination of existing resources. For existing ideas to be able to take new forms in combination, it is necessary for knowledge to be able to flow freely among individuals. Social network theories can provide a theoretical lens to examine the structure and channel of the movement of knowledge. According to the knowledge-based view of innovation (Bland et al., 2010), knowledge and information cannot be separated from the specific social relations that produced them (Osterlund & Carlile, 2005). In other words, knowledge is embedded in social networks as it is produced, distributed, and reproduced.

As a result of the intrinsic link between innovation and social networks, there has been ample research connecting these two constructs. Interorganizational networks are a

particular focus of innovation and network research, because innovation is more likely to occur “at the interstices of collaborating groups and organizations” (Swan & Scarbrough, 2005, p. 914). However, two gaps still exist in the literature.

The first gap in the innovation and social network literature is its overemphasis on the structural approach. Social network analysts have examined extensively the relationship between innovation and network structural properties such as size, centrality, tie strength, density, and structural holes (Baer et al., 2015). However, there is a tendency among social network analysts to rely too much on social network structures, while downplaying actions and behaviors as merely the content or components of a social relationship (Nohria & Eccles, 1992). A few researchers have argued that focusing on structural elements alone does not sufficiently explain why actors share knowledge with one another, and called out to address the “action problem” in combination with structural properties in social network research (Dolfsma & van der Eijk, 2017; Obstfeld, 2005).

The second gap in the literature is that the majority of empirical research on interorganizational networks and innovation focuses on egocentric networks or dyadic relationships between two organizations, instead of focusing on a collection of organizations or the network as a whole (Najafian & Colabi, 2014). To call attention to this gap, Kilduff and Tsai (2003), Provan, Fish, and Sydow (2007), and Provan, Sydow, and Podsakoff (2014) proposed the notion of *whole network*. They defined whole network as a group of three or more organizations connected in ways that facilitate the achievement of a common goal. Whole networks comprise at least three legally autonomous organizations; involve multilateral ties among participating organizations;

are usually goal oriented; and are usually relatively clearly bounded by formal agreements (Provan et al., 2014). The difference between a “whole network” and an interorganizational network is that whole networks are usually legally bounded, which makes the network more stable and also provides a legitimate identity for employees in its membership organizations to relate to.

To fill this second gap, this current dissertation research focuses on whole networks in local government as the contextual background of the study. More specifically, the selected context is a whole network in the form of a regional local collaboration network in the United States.

This context is selected for several reasons. To begin with, the U.S. was selected because innovation played a pivotal role in American history. As Darrell West, Director of the Center for Technology Innovation at the Brookings Institute, pointed out in his testimony to the Hearing of the Joint Economic Committee of the U.S. Congress, “the success of the U.S. in technology innovation and building some of the world’s greatest Internet platforms has positioned the U.S. for global leadership” (West, 2018, p. 8). Second, the level of local government is selected because compared with the federal or state government, local is the level of government that works the closest with people and gains the most trust from people (Hendrix, 2019). Local governments must understand the changing needs of their constituents and catch up with the mindset of local business and residents to work effectively. Last, the regional local government network is targeted because competitive multi-jurisdiction regionalism is one of the key approaches to achieving collaborative economic growth (Wachsmuth, 2017). Across the U.S., it is becoming apparent that metropolitan regional networks are the backbone to economic

prosperity and social transformation in the U.S. (Katz & Bradley, 2013). After all, there are more than 800 regional councils in the country. The top 100 metropolitan areas in the U.S. occupy only 12% of the nation's land, but generate 75% of the U.S. GDP and accommodate about 66% of the country's population (Katz & Bradley, 2013).

To fill both empirical gaps, Provan et al. (2014) developed the concept of *network citizenship behavior* to describe the “individual behavior exhibited by those who work in organizations that are part of a goal-directed multiorganizational network that supports the social and psychological environment in which the achievement of network-focused goals takes place” (p. 9). The focus of network citizenship behavior (NCB) is on behaviors that promote the efficiency, well-being, and performance of a “whole network,” instead of being loyal to each individual's “home” organization. As a newly developed concept, NCB has received very little empirical analysis. The current study closely examines NCB and its contribution to innovation in the local government.

Statement of the Problem

NCB, as a newly developed concept, has not been examined very frequently, especially in the public sector. NCB has remained largely at a conceptual level (Provan et al., 2014), with practically no empirical research on it. A similar construct, interorganizational citizenship behavior (ICB), was developed in the supply-chain management field in 2008 (Autry, Skinner, & Lamb, 2008; Skinner, Autry, & Lamb (2009). However, after two papers that developed an instrument, no follow-up empirical research has been found in the literature. Thus, an empirical examination of NCB is a significant gap in the field that needs to be addressed.

In addition, as previously mentioned, even though there is abundant research on innovation and social networks, there are several gaps in the literature that have not been addressed by scholars. To begin with, government organizations find themselves in need of an innovative mindset to support the community they lead (Bekkers, Edelenbos, & Steijn, 2011). However, research on innovation has not been as adequate as in the private sector, so more scholarly examination is necessary to untangle the phenomenon of innovation in the government.

Moreover, innovation and social networks are closely connected to each other because social networks provide a mechanism for information and knowledge to freely travel among individuals. However, the existing research on social networks and innovation use an overwhelmingly structural approach, examining the relationships between structural properties and innovational outcomes (Obstfeld, 2005). A behavioral approach in combination with a structural approach in social network and innovation studies is largely missing, and has been called for in scholarly dialogues (Dolfsma & van der Eijk, 2017).

Finally, a large amount of social network and innovation research under the context of interorganizational collaboration network focuses on a dyadic relationship between two organizations (Kilduff & Tsai, 2003). The context of a whole network, or a semi-structured, goal-oriented arrangement that binds a consortium of autonomous organizations to work together on multiple levels, has not been looked at in depth (Provan et al., 2007).

Conceptual Framework

Individual innovative behaviors

Scholarly understanding of innovation has gone through several phases in the history of organizational studies. At first, from the economic perspective, innovation was seen as “creative destruction” that serves as the driver of a capitalist economy (Schumpeter, 1942, p. 83). After that, innovation has been viewed as a personality trait (Kirton, 1976), as a product (Zaltman, Duncan, & Holbek, 1973), or as a process (Anderson & King, 1991). From the 1990s until now, innovation has been perceived as a type of behavior (Scott & Bruce, 1994; West & Farr, 1990). Any organizational innovation is discovered, developed, endorsed, and implemented by its employees, who need to go above and beyond their established daily routines and job descriptions to search for new technologies, suggest new practices, apply new methods, and secure new resources. To implement innovation, people need to go through several stages, including idea generation, promotion, and implementation. Within each stage, employees exhibit certain behaviors for the innovation to be successful. Such behaviors are considered *individual innovative behaviors* (IIB) by researchers, and are defined as “intentional generation, promotion and realization of new ideas within a work role, work group or organization, in order to benefit role performance, the group or the organization” (Janssen, 2013, p. 348). The behavioral approach of innovation is particularly appropriate in the government, because other indicators of innovation, such as product, patent, or profit, are not necessarily relevant or available. Theoretical and operational definitions of IIB are provided in Table 1.

Network Citizenship Behavior

Under the context of a whole network, behaviors exhibited by individual employees of membership organizations are of critical importance to the free flow of

knowledge between organizations. The more interaction people from different organizations have with one another, the easier it is for knowledge and ideas to travel across organizational boundaries, and in turn it is easier to innovate (Nonaka, 1994). In the process of interacting with people from other organizations, different behaviors are exhibited that could support partnership organizations and the network as a whole. Drawing from the well-established construct of organizational citizenship behavior (OCB), NCB describes such behaviors as “individual extra-role behavior exhibited ... that supports the social and psychological environment in which the achievement of network-focused goals takes place” (Provan, Sydow, & Podsakoff, 2017, p. 162). A theoretical and an operational definition of NCB are provided in Table 1.

NCB’s theoretical foundation, OCB, includes seven types of discretionary behaviors that promote the effectiveness of the organization: helping/altruism, sportsmanship, loyalty, compliance, individual initiative, civic virtue, and self-development (Podsakoff et al., 2000). Before Provan et al. (2014) proposed NCB, in the field of supply-chain management and logistics, Skinner, Autry, and Lamb (2009) expanded OCB to an interorganizational level, and coined the term “interorganizational citizenship behavior” (ICB) to describe “interfirm behavioral tactics, generally enacted by boundary personnel, that are discretionary, not directly or explicitly included in informal agreements, and that in aggregate promote the functioning of the supply chain” (Autry et al., 2008, p. 54), which is a similar concept to NCB. They presented four subscales of ICB based on four of Podsakoff et al.’s (2000) seven categories of OCB:

1. Interorganizational tolerance: tolerance of inevitable inconveniences
2. Interorganizational altruism: helping a network partner in solving problems

3. Interorganizational loyalty: allegiance to partners and whole network, sometimes sacrificing self-interest.
4. Interorganizational compliance: orientation toward rules, policies, and processes of partners and whole network

In the current research, ICB and NCB are considered to be the same phenomenon. The two concepts were developed in different industrial contexts, but they both describe the same types of extra-role, goal-oriented behaviors toward partners in the same network of organizations. The term NCB is used because it is defined more clearly and specifically and within broader contexts, but research on ICB was also drawn upon to inform the study.

Although there is limited empirical evidence that OCB directly leads to innovation, it does predict innovation through the mediation of trust, social interaction, and shared vision (Gong, Chang, & Cheung, 2010; Molina-Morales & Martinez-Fernandez, 2010; Pearce & Ensley, 2004). When the OCB concept is applied to a network setting, it serves as a mechanism for firms to understand, acquire, and use external resources, which in turn leads to innovation (Gerke, Dickson, Desbordes, & Gates, 2017). Gerke et al. (2017) presented concrete examples of organizational NCB practices that take place in all phases of an innovation process.

Social Network Structural Properties

There are many structural properties in social network research, such as size, centrality, density, tie strength, and so on. Among them, centrality and tie strength are of particular interest as they relate to innovation. Network centrality refers to an actor's position in a network. Central positions are usually associated with power and influence

(Brass & Burkhardt, 1992). Research findings on the relationship between centrality and innovation have been relatively consistent, that a higher degree of centrality is likely to lead to a higher level of individual innovation (Bell, 2005; Casanueva, Castro, & Galan, 2013; Chiu, 2009; Ibarra, 1993; Perry-Smith, 2006; Tsai & Ghoshal, 1998; Tseng et al., 2016).

Tie strength refers to the nature of the relationship between two actors, which combines time investment, emotional intensity, intimacy, and reciprocal services (Granovetter, 1973). Strong ties entail intensive, frequent interactions with multiple functions, and weak ties involve simple, occasional relationships. In the literature, there is much debate about the effects of strong and weak ties on innovation. On one hand, strong ties are more beneficial for the exchange of knowledge and information by creating trust and the willingness to share that consequently facilitates innovation (Albrecht & Ropp, 1984; Bouty, 2000; Fritsch & Kauffeld-Monz, 2010; Krackhardt, 1992; Landry et al., 2002; Levin & Cross, 2004; Moran, 2005; 2002; Reagans & Zuckerman, 2001; Schulz, 2003; Smith et al., 2005; Uzzi & Lancaster, 2003;). On the other hand, the strength of weak ties is an often overlooked asset that brings access to disconnected partners and thus diverse information (Granovetter, 1973). To combine both perspectives, Leenders et al. (2003) discovered that it is an inverted U-shape relationship between tie strength inside a team and team creativity. They suggested that both too low and too high a level of interaction frequency impeded team creativity, whereas a moderate frequency produced the highest level of creativity.

It is difficult to hypothesize how a network's structural properties interact with NCB to impact IIB; however, exploring that interaction effect is needed to combine the

structural approach and the behavioral approach to better explain innovation in organizations.

Purpose and Research Questions

Thus, as Figure 1 demonstrates, the purpose of this research was to examine the relationship between IIB and NCB, and to explore the interaction effect between NCB, network structural properties, and IIB in the context of a whole network in local government in the United States. In this examination, the following research questions were asked:

1. What is the relationship between network citizenship behavior (and its four subscales) and individual innovative behaviors in a U.S. local government's whole network?

Hypothesis 1: the stronger the exhibition of network citizenship behavior, the stronger the exhibition of individual innovative behaviors.

2. How do social network structural properties of organizations, measured by network centrality and tie strength of organizations, impact the exhibition of individual innovative behavior, or the relationship between network citizenship behavior and individual innovative behavior?

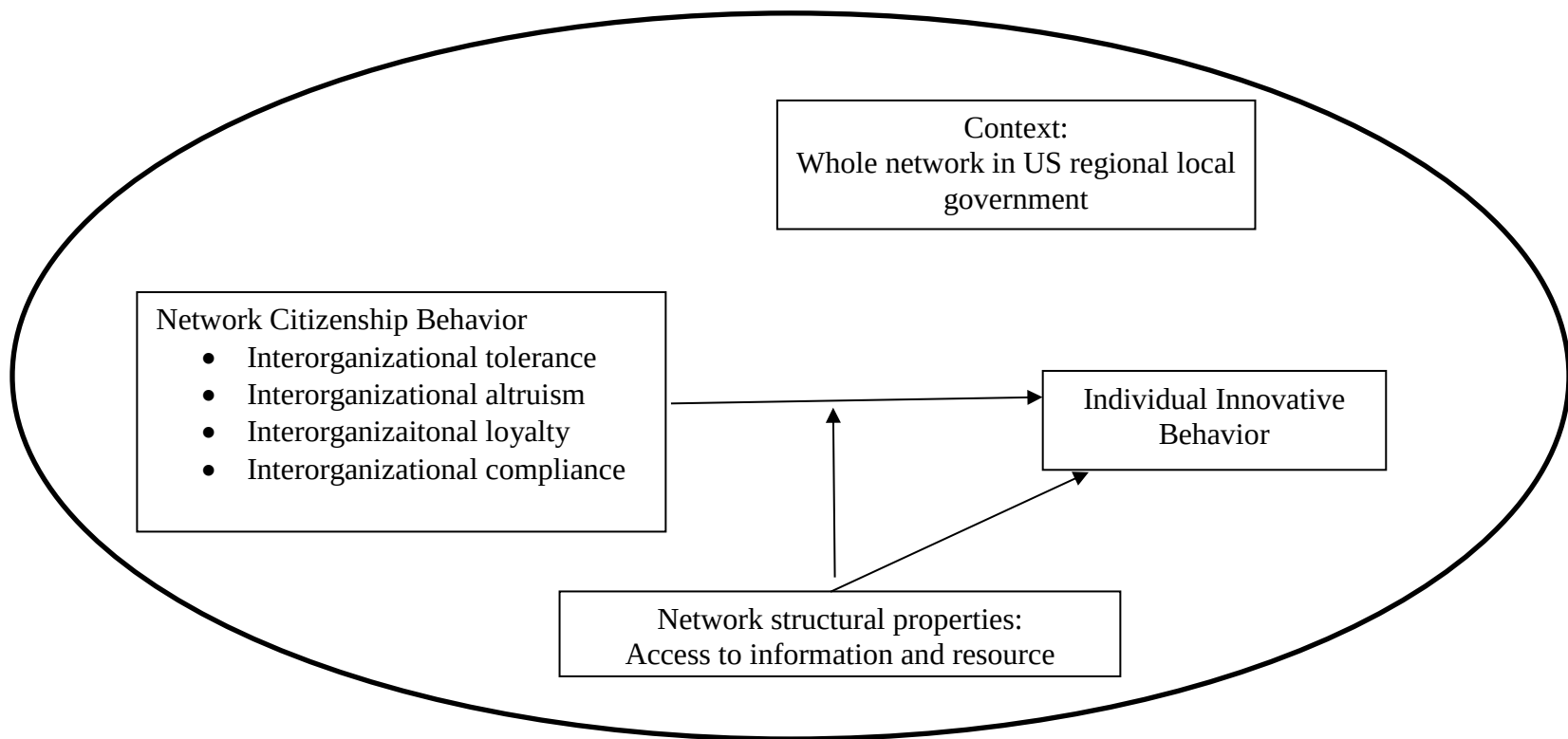


Figure 1: Conceptual Framework

Statement of Potential Significance

The current research contributes both academically and practically in several ways. From the theoretical perspective, and most importantly, it further examines and develops a new construct, NCB, in the local government whole-network context. Empirically, it applies the measurement of NCB to the new context of U.S. government. In addition, it brings a behavioral approach to an otherwise overwhelmingly structural field of research on social networks and innovation.

Practically speaking, the proposed study contributed to Human Resources (HR) professionals, organizational development (OD) practitioners, and organizational managers in several ways. Practitioners in similar regional networks gained insights into what type of behaviors in a whole network context were beneficial for innovation, and may use such knowledge to create incentives to encourage these behaviors, identify the right people for boundary-spanning positions, and design the right training to nurture such behaviors in those positions. It could also serve as a model that can be replicated in other industries, and lay the foundation that can be generalized to interorganizational collaboration networks in a broader context.

Summary of Methodology

Research Design

The current study sought to examine the empirical relationship between NCB and IIB and to explore the impact of social networks' structural properties under the context of a whole network in local government in the United States. To achieve this purpose, an empirical, cross-sectional study was designed and conducted using an Internet-based, self-administered survey approach. This method was chosen because of the nature of the

research questions, which focus on relationships among variables. Survey methodology allowed the identification of constructs and the relationships among them in a large population from a small group of individuals (Fowler, 2009). The Internet-based survey was used because of its low-cost, fast turnaround time in data collection, capacity for reaching widely dispersed participants, and direct data entry (Sue & Ritter, 2012).

Population and Sampling

The site selected for this study was a whole network in the form of a regional planning organization, the Metropolitan Washington Council of Governments (MWCOC), which is composed of 24 local government jurisdictions. The population of this study consists of employees who work for the local government jurisdictions in this network. It included employees who communicate with people from other jurisdictions or attend interjurisdictional committees, projects, legislative meetings, and other functions at MWCOC. A convenience sample was employed in this study because of the difficulty in obtaining contact information for all employees in this network due to privacy and confidentiality concerns in HR practices. The convenience sample was the participants of training programs delivered by the George Washington University Center for Excellence in Public Leadership (GWU-CEPL).

Measures

Data for this study were collected using previously established instruments for the two constructs: IIB and NCB.

IIB was measured using the 14-item instrument developed by Kleysen and Street (2001). Each item was measured on a 6-point frequency rating scale, anchored by “never” on one end and “always” on the other. Among all the measurements on

innovative behaviors in the literature, this particular instrument was chosen for its strong theoretical foundation, robust evidence of reliability, its appropriate length of 14 items, and its targeted participants of only individuals without their supervisors to keep it consistent with the independent variable, as well as its original language of English.

The measurement of NCB was adapted from the ICB instrument developed by Skinner et al. (2009). The original measurement was used in the supply-chain management context, which contained 11 items on a 1-7 frequency scale. This was the only measurement related to the construct and it reported high subscale reliability. A small selection of words was changed with the authors' permission to reflect the context of the government instead of the private sector.

Data Collection Procedure

The data were collected through Internet-based, self-administered surveys using the online survey tool Survey Monkey. An email explaining the purpose of the study was distributed together with the URL of the survey to all participants in the sampling frame. Participants agreed to an electronic informed consent form before accessing the survey.

Data Analysis

The data were collected and transformed into an Excel document. The Cronbach alpha was calculated for each subscale and the composite scores. An instrument validation was conducted for each measurement tool using both Cronbach's alpha in SPSS and confirmatory factor analysis (CFA) in LISREL. A series of regression analyses were conducted to answer the research questions using SPSS. To answer the first research question, a series of simple regression analyses were conducted to identify the

relationship between NCB and its four subscales and IIB. The regression model can be presented as

$$y_i = \alpha + \beta x_i + \varepsilon_i \quad (1)$$

where x_i is NCB and its four subscales, and y_i is IIB.

To answer the second research question, a series of multiple regression analyses were conducted to identify the relationship between structural properties, NCB and IIB. The regression model can be presented as

$$y_i = \alpha + \beta x_i + \varepsilon_i \quad (2)$$

where x_i is network structural properties centrality and tie strength, and y_i is IIB.

As an exploration of the impact of the interaction effect between network structural properties and NCB on IIB, a simple regression analysis was conducted, and the regression model can be presented as

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \varepsilon_i \quad (3)$$

where x_{1i} is network structural properties (centrality and tie strength), x_{2i} is NCB and, y_i is IIB.

Ethical Considerations

An electronic consent form was provided before the survey, and survey respondents could not proceed without agreeing to the informed consent form. Participants were assured that no information about the study would be shared with their employers, managers, or colleagues, and their employment and performance would not be impacted in any fashion by participating in the survey. All survey responses were anonymous. Permissions were sought and obtained from the original authors of each measurement before collecting data using their instruments.

Limitations and Delimitations

A full list of limitations and delimitations will be discussed in chapter 5, but in a short summary, the current study is subject to the following limitations. First of all, the sample size collected for the current study was not sufficiently large. Second, as a study that relies on a self-reported survey, personal bias, interpretation, and exhaustion might hinder the accuracy of the responses. Moreover, the current study is subject to a specific context and can hardly be generalized to broader contexts. In addition, the sampling frame is not a perfect representation of the population and might positively skew the exhibition of NCB. Finally, other factors that might influence the outcome variable were not accounted for in the current study.

To ensure that the study was manageable, several decisions were made intentionally to delimit the study. First, even though it is intended in the future to apply the findings to other sectors, the current research study *per se* focuses only on local government agencies in the U.S. Second, the current study only concerns the relationships among a limited number of constructs. There may be other factors contributing to the relationship, and additional research will be needed to include other factors into the model to further examine innovation or NCB.

Definition of Key Terms

Table 1

Definition of Key Terms

Construct	Theoretical Definition	Operational Definition
Innovative Behaviors	“Intentional generation, promotion and realization of new ideas within a work role, work group or organization, in order to benefit role performance, the group or the organization.” (Janssen, 2003, p. 348).	Kleyson & Street (2001), 14 items, Cronbach alpha = 0.945
Network Citizenship Behavior	“Individual extra role behavior exhibited by those who work in organizations that are part of a goal-directed multi-organizational network that supports the social and psychological environment in which the achievement of network-focused goals takes place” (Provan, Sydow & Podsakoff, 2017, p. 162).	Adapted from Skinner, Autry, & Lamb (2009)’s interorganizational citizenship behaviors, 11 items, 4 subscales. Subscale reliability alphas ranging from .702 to .916, using a sample of 225 managers in nine different organizations in a wide range of industrial and professional background.

Chapter 2: Literature Review

Purpose and Methodology of Literature Review

The purpose of this study was to examine the relationship between network citizenship behaviors (NCB) and individual innovative behaviors (IIB), and to explore the interactional effect between NCB and network structural properties on IIB. To serve this purpose, the purpose of this literature review was to achieve the following three goals: first, to examine past and present studies in the field to orient the proposed research study in the discourse of the existing literature; second, to provide a framework of argumentation that supports the research questions and hypotheses; and third, to examine methodologies used in previous studies to justify the decisions in research design.

The literature search went through two stages. In stage one, a comprehensive search was conducted using the following databases: IBM/Inform Complete Plus; Business Source Complete; ProQuest; JSTORE; ScienceDirect; and Google Scholar. These databases were recommended by librarians at the George Washington University Gelman Library as having rich resources in areas of the current study. Search terms included combinations of the following: “innovation,” “network,” “social network,” “network citizenship behaviors,” “interorganizational citizenship behavior,” “whole network,” “interorganizational collaboration,” and “interorganizational collaboration network.” Literature on or after the year 2000 was searched, because whole network–level research experienced a significant increase after 2000 (Provan, Fish, & Sydow, 2007).

The second stage of literature review was an iterative process as the researcher expanded the scope of literature and refined the research questions. Studies cited by

papers included in the first round of review were traced back in history if they related to the topic of the current study. There was no specific time-frame restriction in the second round of literature review. The goal was to provide enough context to position the current paper in the literature. Even though this was not an exhaustive review of all literature in the scholarly history, every effort was made to ensure a balanced argument, and no research was intentionally excluded for offering counterarguments to the proposed hypothesis.

The rest of the chapter is structured in the following way: In the first section the researcher sought to trace the historical definition of innovation and how IIB is defined today, and to present a list of instruments that have been developed to measure IIB to support the decision on instrument selection. In section two, the concept of social networks was brought in. After defining social networks and relating it to innovation, the researcher discussed two gaps in the current literature on innovation and networks, namely, too much emphasis on a structural approach and not enough focus on a behavioral approach; and too much work on ego-organizational or dyadic relationships between two organizations, and not enough research using whole networks as the context. Finally, in section three, social network properties, especially centrality and tie strengths, were discussed and their relationships with innovation were summarized.

Innovation

Definition of innovation. The scholarly investigation of the phenomenon of innovation can be traced back at least to Schumpeter, in the 1940s. In studying the economic development of capitalism, Schumpeter noticed innovation in the form of “creative destruction,” noting:

The fundamental impulse that sets and keeps the capitalist engine in motion comes from the new consumers' goods, the new methods of production or transportation, the new markets, the new forms of industrial organization that capitalist enterprise creates ... This process of 'Creative Destruction' is the essential fact about capitalism. (Schumpeter, 1942, p. 83)

Since then, scholars in economics, psychology, management, and organizational science have been studying innovation closely from various perspectives. The understanding of innovation has evolved significantly since Schumpeter's time and has gone through several stages.

In the 1960s and 1970s, innovation – interchangeably used with *creativity* at the time (Scott & Bruce, 1994; West & Farr, 1990) – was predominantly seen as a personality trait (West & Farr, 1990). Amabile (1988) pointed out that many researchers defined creativity as a characteristic of a person, or “a constellation of personality and intellectual traits shown by individuals who, when given a measure of free rein, spend significant amounts of time engaged in the creative process” (Findlay & Lumsden, 1988, p. 9).

From this perspective, the measurement of innovation was based on personality traits. Goldsmith (1986) compared four personality-based innovativeness scales in the 1970s. Among them, the most commonly used was Kirton's (1976) Kirton Adaptor-Innovator Inventory (KAI). KAI was developed based on the assumption that individuals' cognitive style might have a direct effect on their creativity (Shalley, Zhou, & Oldman, 2004). Kirton believed that people have an inherent tendency to adapt or to innovate. He proposed a continuum ranging from the ability to “do things better,” labeled as

“adaptive,” to “do things differently,” labeled as “innovative,” and everybody can be located on this continuum.

In addition to the “personality” perspective, innovation was also perceived as an “outcome.” For example, Zaltman, Duncan, and Holbek (1973) considered innovation to be “any idea, practice, or material artifact perceived to be new by the relevant unit of adoption” (p. 10). Amabile (1998) also agreed that definitions that focused on products were most common at that time. She defined organizational innovation as “the successful implementation of creative ideas within an organization” (p. 126). The “outcome” view of innovation also manifests itself in the literature when researchers discuss technological innovation, and the creation of each technological application is seen as “an” innovation. These innovations are countable, like products or items. When Tushman and Anderson (1989) discussed the management of strategic innovation, they linked technological advancement with business cycles and with the ways that new technological innovation serves as a punctuated equilibrium in the cycle of organizational change. Evidently they considered innovation as a product at a single point in time that inserted itself into the history of an organization.

During this period, innovation is measured by a countable output. Many research studies in the private sector used the amount of research and development (R&D) spending, the number of patents or new products, or the number of new ideas contributed to their organizations to measure the innovation of an individual or a firm (Cummings & Oldham, 1997; Monge, Cozzens, & Contractor, 1992; Sarooghi, Libaers, & Burkemper, 2015). Notably, in studying the relationship between organizational complexity and innovation, Damanpour (1996) identified 21 relevant articles, and among all of them,

“organizational innovation typically is measured by the rate of adoption of innovations, which is often operationalized by the number of innovations adopted within a given period of time” (p. 703).

Starting in the 1980s, scholars began to examine innovation as a process. Anderson (1990) described innovation as “the emergence, import or imposition of new ideas which are pursued towards implementation ... through interpersonal discussions and successive re-mouldings of the original proposal over time” (p. 3). In this case, a process in which innovation emerges, imports, and then becomes implemented was visibly presented, and more emphasis was placed upon the dynamics in which the innovation took place.

Kanter (1988) represented the emergence of the process view of innovation and laid the groundwork for the subsequent behavioral view of innovation. In her seminal work on innovation, Kanter (1988) emphasized that innovation was a process that was uncertain, knowledge-intensive, controversial, and boundary-crossing. She proposed that there were four major innovation tasks: idea generation, coalition building, idea realization, and transfer or diffusion. Even though she called them “tasks” instead of “stages,” and even though she did not hint at any sequential order of the events, this model does serve as a foundation of many process models that followed.

The subphases of the innovation process have been termed differently by various researchers, and there is no shared consensus in their definitions. However, generally speaking, the elements include two broad steps: idea generation and idea implementation. It is through this distinction in an innovation process that the concepts of innovation and creativity are clearly separated.

As previously mentioned, in the early years, innovation and creativity were used interchangeably (Carmeli et al., 2006; Scott & Bruce, 1994; Van Minh et al., 2017; West & Farr, 1990). As recently as 2007, in their proposed Climate for Creativity instrument, Hunter, Bedell, and Mumford (2007) still used the terms of creativity and innovation side by side. They cited both creativity and innovation research articles in their references to support their instrument. However, creativity and innovation can be clearly distinguished by the idea generation and idea implementation phases of an innovation process.

The English Oxford Living Dictionaries define creativity as “[the] use of imagination or original ideas to create something,” which focuses on the originality of the creation, its uniqueness, and the fact that something was brought from nonexistence to existence. Innovation, on the other hand, does not necessarily require an idea that did not exist before. West and Farr (1990) characterized innovation as only “new to the relevant unit of adoption, designed to significantly benefit the individual, the group organization or wider society” (p. 9).

In other words, *creativity* can be described as the idea-generation stage of innovation, and innovation encompasses both the proposal of new ideas (creativity) and their application (West & Farr, 1990). “Organizational innovation is the successful implementation of creative ideas within an organization” (Amabile, 1988, p. 126). In addition, innovation also has to be institutionalized into the current environment, whereas creativity does not necessarily need to be. Though there are many new ideas created by people, only a small percentage of them are expressed, discussed, and eventually implemented in organizations.

In addition to clearly distinguishing between creativity and innovation, the process view of innovation is also important because it serves as the theoretical foundation of innovative behaviors, which is the key construct in the current study.

Innovative Behaviors

Innovation, by its nature, is a social phenomenon. One way to acquire new knowledge requires interaction among people so ideas can be distributed, and for an innovative initiative to be implemented it requires influencing and mobilizing other people (Lukes & Stephan, 2017). Innovative behavior, on the other hand, is an individual-level phenomenon (de Jong & den Hartog, 2010). Innovative behavior is an explanatory construct that describes tasks and activities, performed by employees and required by innovation development (Messman & Mulder, 2012). It constitutes the micro-foundations of organizational innovation (Lukes & Stephan, 2017). Any organizational innovation is discovered, developed, endorsed, and implemented by an organization's employees, who need to go beyond their established daily routines and job descriptions to search for new technologies, suggest new practices, apply new methods, or secure new resources.

As a construct on the individual level, innovative behavior is impacted by both internal psychological factors and external environmental factors. Internally, employees' perceptions of effort-reward fairness (Janssen 2000), situation perceptions (Krause, 2004), individual intuitive problem-solving style, and systematic problem-solving style (Scott & Bruce, 1994) all influence the exhibition of innovative behaviors. Externally, the organizational climate (Scott & Bruce, 1994), leadership-member exchange, and

supervisory support (Messmann & Mulder, 2012) post a clear influence on the exhibition of innovative behaviors.

Scott and Bruce (1994) developed the first innovative work behavior inventory, which consisted of six items of innovative behaviors that they solicited from interviews with company executives. They specifically attributed the theoretical foundation of their identification of behaviors to Kanter's (1988) innovation process model. They viewed innovation as a three-stage process: idea generation, idea promotion, and idea implementation. In each stage, different behaviors are exhibited by employees. In the idea generation phase, employees actively search for new ideas. In the idea promotion phase, people look for champions and secure resources for the innovation. In the idea implementation phase, people develop plans and carry out the innovative change.

When many subsequent researchers were studying innovative behaviors, they generally also followed this process view, but looked for a wider range of behaviors that are exhibited at different stages of the innovation process. Janssen (2000), for example, while continuing Scott and Bruce's (1994) three-stage model, also solicited additional and more specific behavioral items and expanded the instruments from six items to nine.

Following Janssen (2000), additional expansions were made by several researchers. De Jong and Den Hartog (2010) added a new phase of "idea exploration" before idea generation. Based on creativity research, they saw idea exploration and idea generation as distinctive concepts: idea exploration entails identifying a problem to be solved as an opportunity; idea generation, on the other hand, is the reorganization of existing information. Idea exploration precedes idea generation and serves as a target or goal for new ideas to be generated.

Subsequently, Messman and Mulder (2012) added another phase of reflection at the end of the process. They see innovation as dynamic and context-bound. Innovation is dynamic because employees' previous performance and knowledge impact their future innovative capacities as employees' professional careers develop. Innovation is context-bound because innovation does not have to be brand new, but only needs to be new in its current context. As a result, a phase of reflection is necessary for employees to learn from previous experiences to improve their performance and professional capability to bring in more innovation in the future.

Alongside the development of this series of innovative behavior identification, Kleysen and Street (2001) conducted a separate line of literature search, through which they identified 289 creative and innovation-related activities and coded them into five broad stages that are completely different from what other researchers had discussed. They see innovative behaviors consist of five phases: opportunity exploration (look extensively for innovative opportunities); generativity (direct behaviors toward generating beneficial change); formative investigation (flesh out ideas, solutions, and opinions and try them out through investigation); championing (bring creative ideas to life); and application (make innovations a regular part of business). Among these five phases, opportunity exploration and generativity can be considered to be the idea generation phase, and formative investigation, championing, and application can be considered to be the idea implementation. The validity of this 5-stage model found its theoretical roots through a thorough and comprehensive literature review on creativity and innovation as well as three rounds of coding, as described by the authors.

Innovative behavior has largely been a quantitatively operationalized concept, with many instruments developed to measure it. Between the 1990s and 2010s, a number of innovative behavior measurements were developed. Researchers critiqued, learned from, and built upon each other's work with regard to its behavior subscales, wording, and structure. A list of measurement instruments of innovative workplace behaviors is described and compared in Table 2.

Table 2

Individual Innovative Behavior Measurements

Name of assessment	Author and year	# of items	Sample	Cronbach alpha	Validity evidence	Comments
Innovative Behavior Measure	Scott & Bruce (1994)	6	172 R&D scientists and engineers	0.89	Significant correlation between ratings and objective invention rate obtained through archival data	Pioneering attempt to measure the construct, but overly simplistic Completed by both respondents and their supervisors; items created based on Kanter (1988)
Employee Creativity Items	Tierney, Farmer, & Graen (1999)	9	191 R&D non-clerical employees in a chemical company	0.95	Adequate discriminant validity tested by comparison of nested models	Only asking supervisors; Too generic Measuring “creativity” instead of “innovation”
Innovative Work Behavior	Janssen (2000)	9	170 non-management employees in a Dutch food company	0.95 for self & 0.96 for manager	Significant correlation between self and supervisor ratings	Drawing from Scott & Bruce (1994) Reported by self and supervisor
Individual Innovative Behaviors	Kleyson & Street (2001)	14	225 from nine different organizations	0.945	Not supported extensively, poor model fit	Set out to create a multidimensional instrument, but validity test did not support it; combined as a single dimensional instrument
Creativity Scale	Zhou & George (2001)	13	149 employees from a manufacturing company	0.96	Not reported	Does not differentiate among innovation processes; it is called creativity, but partly measures innovation
Innovative Behaviors and Innovation-	Krause (2004)	16	339 middle managers from German	>0.8	Exploratory factor analysis showed that	Originally in German; First assessment to include innovation-blocking behaviors;

Blocking Behaviors			companies of various sizes and sectors		two dimensions are distinct	two dimensional; aimed for process-innovation
Innovative Work Behavior	De Jong & Den Hartog (2010)	10	703 knowledge workers from 94 small knowledge-intensive service firms in The Netherlands	>0.7	Significant correlation between innovative work behavior and innovative output	Supervisor rated Set out to measure four dimensions, but failed to distinguish between the four dimensions
Innovative Work Behaviors Scale	Messman & Mulder (2012)	37	335 employees from a German auto supply company	0.76-0.87	Construct validity tested by exploratory factor analysis and confirmatory factor analysis	Too many items; Seeks to integrate three previous scales; Originally in German, awkward translation five dimensions
Innovative Behavior Inventory	Lukes and Stephen (2016)	23	Total of 2,812 employees of Czech, German, Italian, and Swiss companies	0.60-0.88	Most comprehensive validity study: Significant factor loadings Significant correlations with innovative output Construct validity showed by significant correlations for each of the six dimensions with similar constructs Cross-culture validation with samples from four countries	Too many items, but overall comprehensive and nicely worded Six dimensions Less satisfactory Cronbach Alpha reliability score

In this current study, IIB was used as the dependent variable. It is an individual level instead of an organizational-level phenomenon, which is consistent with the level of analysis of the independent variable, NCB. Not only has it come a long way from the historical understanding of innovation to reflect the most current behavioral view of innovation, but it is also the most appropriate in the government sector. In the government, there are usually no patents or products that can be counted as innovative output, and very few local governments could afford to provide funding for R&D or innovation. IIB is one of the most identifiable and observable variables that can be used to measure innovation.

With that said, compared with earlier measurement of innovation outcomes, using innovative behavior measurements to indicate “innovation” does pose several limitations, but these limitations can be overcome. First of all, the exhibition of innovation behaviors does not necessarily equate to generating innovative output, such as the implementation of practices or the creation of new policies, and is linked even less to organizational outcomes such as profit or improved quality (Kanter, 1988). This limitation can be overcome by selecting a theoretically sound instrument with a strong validity, so its predictive power from a behavior to an outcome can be supported through theoretical argument or empirical testing. Secondly, even though, theoretically speaking, innovation can be perceived as a multistep process and each step entails different behaviors, so far, there have been limited satisfactory measurements that reflect the multidimensional nature of innovation. Several attempts have been made to create a multidimensional measurement of innovative behaviors; however, the correlations between the dimensions are always too high to distinguish them as different subscales (De Jong & Den Hartog,

2010; Janssen, 2000; Kleysen & Street, 2001). The only truly multidimensional instrument (Lukes & Stephan, 2016) did not provide sufficiently high-reliability evidence (.60-.88). However, the multidimensionality of innovative behavior is not at the core of this study. The main goal of the current study was to examine NCB as a novel construct, and IIB merely serves as its outcome variable to test its predictive validity. A reliable and valid unidimensional instrument could be sufficient to serve as a sound dependent variable.

For this current research, the 14-item measurement developed by Kleysen and Street (2001) was used for several reasons. First, Kleysen and Street (2001) have a robust theoretical foundation. The innovation stages and the behaviors identified for each stage were based on a thorough and comprehensive literature review and multiple rounds of coding and synthesizing. The theoretical foundations of some other measurement instruments might be problematic. For example, Lukes and Stephan (2017) proposed a six-step process model of innovation: idea generation, idea search, idea communication, implementation starting activities, involving others, and overcoming obstacles. However, some of the phases clearly overlap with each other, and there is no clear sequencing of these phases. For example, the “idea communication” phase needs to “involve others,” and “overcoming obstacles” is unavoidable in every stage. Individual innovative behaviors is theoretically strongly rooted in the stage theories of innovation (Kanter, 1988), and if the specific behavior and actions identified in the instrument cannot be used to distinguish stages, its theoretical foundation is weak and flawed. Another example of theoretically weak instrument is Zhou and George (2001)’s measurement, in which they

cited Scott and Bruce (1994)'s three items, and added 10 additional items without explanation or theoretical grounding.

Second, the selected measurement instrument has one of the highest reliability scores among the instruments in the literature. Its Cronbach alpha value is .945 (N=225 from nine organizations), which is higher than almost all other instruments, with the exceptions of Tierney, Farmer, and Graen (1999) (Cronbach alpha value = .95, N=191 R&D nonclerical employees in a chemical company) and Janssen (2000) (Cronbach alpha value of .95 for self-reported items and .96 for manager responses, N=170 nonmanagement employees in a Dutch food company). Other instruments that are more recent and integrative of previous measurements had much lower reliability – for example, Lukes and Stephan (2016), the Cronbach alpha value of which is only .60-.88 (N= 2,812 employees of Czech, German, Italian, and Swiss companies).

Moreover, the instrument developed by Kleysen and Street (2001) seeks responses only from employees, not from supervisors or other colleagues. Although innovative behaviors could be observed by supervisors, the independent variable of this study, NCB, usually takes place outside of employees' organizational boundaries (Provan et al., 2017) and might not be directly observed by supervisors or colleagues. Selecting an instrument that solicits only self-reported data would not only significantly simplify the data collection process and reduce confusion, but would also report consistently between dependent and independent constructs. Thus, many other instruments that seek responses from supervisors, including Scott and Bruce (1994), Tierney, Farmer, and Graen (1999), Janssen (2000), and De Jong and Den Hartog (2010), were not selected for this study.

In addition, the length of a questionnaire has a significant impact on response rate (Cook, Heath, & Thompson, 2000). Multiple empirical studies have demonstrated that there is a negative relationship between the questionnaire length and the response rate (Deutskens, De Ruyter, Wetzels, & Oosterveld, 2004). Several textbooks and papers have suggested that Internet-based surveys should not take more than 10-15 minutes to complete (Czaja & Blair, 2005; de Leeuw, Hox, & Dillman, 2008; Fan & Yan, 2010). Given there were two additional constructs to measure, many innovation instruments had too many items to include in the questionnaire, which was likely to generate a low response rate. For example, even though Messman and Mulder's (2012) instrument reported an acceptable reliability indicator (.76-.87) and was more recent, it contained as many as 37 items. Combined with 11 items of NCB, 6 items for demographic information, and 1 complex item for the network, the total length of such a survey would be too long to be effective.

Finally, the Kleysen and Street (2001) instrument was developed in Canada, an English-speaking country that is similar to the U.S.. Many other more recent and popular measurements were developed in European countries in languages other than English [de Jong & den Hartog (2010) and Janssen (2000) in the Netherlands; Krause (2004) in Germany]. To avoid subtle, nuanced meanings lost in translation and to select a measurement with more accurate, targeted wording without cultural barriers, Kleysen and Street (2001) was the ideal instrument for the current research study.

However, despite its advantages, the Kleysen and Street (2001) measurement does have its limitations. Even though it intended to create a multidimensional instrument, and identified five stages in the innovation process – opportunity exploration, generativity,

formative investigation, championing, and application – the instrument failed to distinguish between the five dimensions, and in the end had to present a simple one-dimensional instrument.

In addition, one critical shortcoming of the Kleysen and Street (2001) instrument was that the study did not provide enough evidence for validity. The authors described their effort to ensure the construct validity in the process of developing the instrument. Through a literature search, they identified 28 articles on creativity and innovation for examination and coding. From these 28 articles, 289 creativity and innovation-related activities were identified. During the first round of coding, these 289 activities were condensed into 24 groups of activities. In the next round of coding, 17 behavioral processes emerged from the 24 groups of activities. Lastly, the 17 behaviors are categorized into 5 factors. Such a process ensured that the items were good representations of the construct of IIB. However, the authors did not provide evidence for other types of validity. They did not work with experts or stakeholders to review the survey items, and they did not test its criterion validity by relating it to other constructs.

In short, so far we have defined innovation and innovative behavior, and compared and selected measurements of innovative behaviors. As noted by Anderson, de Dreu, and Nijstad (2004), innovation research has been routinized, and individual and environmental factors that lead to innovation have been studied thoroughly. One way to continue to contribute to the study of innovation is by developing new constructs that have not been examined extensively yet and to explain how these new constructs could impact innovation. The next section introduces social networks, which has a close

relationship with innovation and serves as both a critical influencing factor and an analytical tool to empirically examine innovation.

Social Networks

The social network perspective. The term “social network” has gained considerable attention in the scholarly world in recent decades, and the research on social networks has grown exponentially (Borgatti & Foster, 2003). The social network perspective views social relationships in the form of nodes and ties, wherein nodes represent individual actors and ties represent the relationships between the actors (Wasserman & Faust, 1994). Network research differs from traditional perspectives in organizational studies, which focus on characteristics of individuals in isolation. A network perspective seeks to explain social phenomena through the relationships among social actors and the structure of these relationships (Brass et al., 2004). The relationships can be directed, dichotomous, or valued (Wasserman & Faust, 1994). There are many different types of relationship, such as friendship, information sharing, or advisory, which assume different functionalities (Borgatti & Foster, 2003).

Network research could find its roots in a number of theoretical foundations, such as social capital (Burt, 1992), the resource-based view (Gulati, 1999), and the resource dependence theory (Pfeffer & Salancik, 2003). Most notably it draws from the social exchange theory (SET). SET is conceived as “actions that are contingent on rewarding reactions from others and cease when these expected reactions are not forthcoming.” (Blau, 1964, p. 6). SET implies a “two-sided, mutually contingent, and mutually rewarding process involving ‘transactions’ or simply ‘exchange’” (Emerson, 1976, p. 336). Using a network perspective, the relationship that two actors have with each other

contains an exchange of content, either tangible materials such as funding or resources, or intangible materials such as information or emotion. SET helps us to understand the mechanism through which “specific transactions take form and how relationships emerge” (Oparaocha, 2015, p. 537).

So far, this stream of research has been referred to as the social network perspective, instead of the social network theory, because it has been inconclusive whether network research was able to form its own theories. Some researchers believe that the network perspective does not by itself constitute a theoretical lens (Zaheer, Gozubuyuk, & Milanov, 2010). Others have argued that it was too early to call “network research” a paradigm but rather an “ongoing research program” (Kilduff, Tsai, & Hanke, 2006, p. 1032) with a set of shared core assumptions that are fundamental in informing this study. These assumptions are—

1. Social relations: views an organization as a system of objects linked by relationship among its internal members and with its external environment.
2. Embeddedness: posits that actors prefer to interact with other network members instead of actors outside of the community.
3. Structural patterning: assumes that the complexity of social life can be explained by the pattern of connectivity, clustering, and centralization.
4. Utility of network connections: states that social networks constrain and facilitate outcomes of importance to individuals and groups.

It is important to distinguish between social network analysis as a methodology or an analytical tool, versus research with a social network perspective. Wasserman and Faust (1994) defined social network analysis (SNA) as a set of analytical methods to

demonstrate, dissect, and examine patterns of relationships among connected network actors. This set of studies take a structural view (Kilduff et al., 2006) and a quantitative approach, and examine such structural measures as density, centrality, structural holes, and cliques to analyze the relationships between network locations and other factors. On the other hand, studies from a network perspective “encompass theories, models, and applications that are expressed in terms of relational concepts or processes” (p. 4). Using this approach, researchers adopt the relational view, but the study itself need not involve network structures. The two different perspectives present a gap in the literature, which will be discussed later.

Innovation and Network

Innovation has an intrinsic relationship with social networks, particularly through the transfer of knowledge, information, and ideas. Thorgren, Wincent, and Örtqvist (2009a) found a direct, positive relationship between knowledge transfer and corporate entrepreneurship that stimulates innovation. Innovation was described by Schumpeter (1942) as a recombination of existing resources and knowledge, and by Kanter (1988) as a knowledge-intensive process. Knowledge, particularly tacit knowledge, is intangible, ambiguous, path-dependent, and difficult to replicate (Nonaka, 1994). It is cumulative based on experiences (Alguezaui & Filieri, 2010), and it exists on a collective level.

This collective, (quasi-) public nature of tacit knowledge makes it nonexclusive and nonrivalrous: it is not consumed by its use, but rather grows when it is used and shared (Bland et al., 2010; Groff & Jones, 2003). Tacit knowledge may reside in individuals, but it is through networks that knowledge is exchanged, put to use, and developed further (Dolfsma & van der Eijk, 2017). An individual’s knowledge and

information cannot be separated from the specific social relations through which they are produced and reproduced (Osterlund & Carlile, 2005). Through this reproduction, knowledge is no longer just “transferred,” but “transformed,” whereby the final knowledge is different from all its previous owners and transferors. The transfer of knowledge is an active process of “grafting and transplanting” instead of a passive process of copying and pasting (Hartley & Benington, 2006, p.104). As knowledge is taken into a new setting or is rediscovered in relation to a new purpose, knowledge is continuously reviewed and recreated alongside existing “old” knowledge. In a way, this transformation of knowledge is innovation (Rice et al., 2017). In other words, innovation is the display of how a bundle of knowledge is brought together into a new configuration (Gubbins & Dooley, 2014).

Because of this intrinsic relationship between social networks and innovation, there has been abundant research linking the two constructs together. However, there have been two significant gaps in the literature. Briefly, first, most social network and innovation research uses social network analysis as a methodology and an analytical tool to study the relationships between structural properties and innovation, but little emphasis is placed upon actions and behaviors in the network context. Secondly, most innovation studies have focused on either a single organization or dyadic relationships between two organizations in the private sector. Relatively less focus has been placed upon a network of organizations (“whole network”) in the public sector. The following sections will summarize and critique previous literature on social network and innovation to demonstrate the existence of these two gaps and the significance of bridging them.

Gap 1: Structural property versus a behavioral approach. There has been abundant research that links social networks to innovation from the structural perspective, mostly studying structural properties such as size, density, structural holes, and strengths and how they are related to innovation. Table 3 describes major structural properties of network research and their relationships to innovation demonstrated in empirical research.

Table 3

Network Structural Properties and Innovation Outcome

Structural property	Definition	Generalized conclusion	Citation	Detailed (caveat) conclusion	Citation
Size	Number of ties (direct or indirect)	Positive	Ahuja, 2000; Baer et al., 2015; Shan et al., 1994; Smith et al., 2000;	Inverted U-shape as a result of interaction with cost	Cantner, Conti & Meder, 2010; Delgado-Marquez et al., 2018; Mcfadyen & Cannella, 2004;
Centrality	An actor's position in a network, associated with power and influence (Brass & Burkhardt, 1992)	Positive	Bell, 2005; Casanueva, Castro & Galan, 2013; Chiu, 2008; Tsai & Ghoshal, 1998; Ibarra, 1993; Perry-Smith, 2006; Tseng et al., 2016	Interaction effect with size: Benefit of centralized location can be replaced by size of outside ties.	Perry-Smith (2006)
Tie Strength	The nature of the relationship between two actors which combines the invested time, emotional intensity, intimacy, and reciprocal services. Strong ties entail intensive, frequent interactions with multiple functions, while weak ties involve simple, occasional relationships (Granovetter, 1973).	Inverted U-shape: moderate level interaction frequency produces the highest level of creativity	Leenders et al. (2003); McFadyen & Cannella (2004)	Strong ties create trust and willingness to share knowledge	Albrecht & Ropp, 1984; Bouty, 2000; Fritsch & Kauffeld-Monz, 2010; Krackhardt, 1992; Landry et al., 2002; Levin & Cross, 2004; Moran, 2005; Reagans & Zuckerman, 2001; Schulz, 2003; Smith et al., 2005; Uzzi & Lancaster, 2003
				Weak ties impede transfer of complex knowledge	Hansen, 1999

				Weak ties bring access to unique knowledge	Granovetter, 1973; Hansen, 1999; Levin & Cross (2004); Perry-Smith (2006);
				Strong ties reduce innovation by “locking” people in existing relationships at the expense of gaining new partner and access	Yli-Renko, Autio & Sapienza, 2001
Density	Connectedness of the contacts that an actor or a network has	Process view: Idea generation phase: large network rich in structural holes and weak ties Implementation phase: dense, cohesive closed network with strong ties.	Gubbins & Dooley, 2014; Kijkuit & van den Ende, 2010; Macri, Tagliaventi, & Bertolotti (2001); Perry-Smith & Mannucci, 2017	Positive: trust and social cohesion Redundancies that ensure full penetration of information Governance mechanism to enhance information flow	Abrahamson & Rosenkopf, 1997; Ahuja, 2000; Casanueva, Castro & Galan, 2013; Krackhardt, 1992; Montanari & Saberi, 2010; Morgan & Soerensen, 1999; Morrison, 2002; Obstfeld, 2005; Tseng et al., 2016; Uzzi, 1996
Structural hole	The separation between nonredundant contacts (Burt, 1994).			Positive: bridges mutually unconnected actors and diverse sources of knowledge	Burt, 2004; Fleming et al., 2007, McFadyen, Semadeni & Cannella, 2009).

This structural approach described in detail in Table 3 has made up the majority of network innovation studies. For example, Levin and Cross (2004) studied the debate between strong and weak ties and their impact on effective knowledge transfer, one critical building block of innovation (Gubbins & Dooley, 2014). They started off acknowledging that in the literature, there is a disagreement in regards to which type of network structure is more conducive to knowledge transfer, strong or weak ties. Through surveying 127 midlevel managers in three different companies that belong to three different industries in three different cities, they identified 508 relationships that these respondents have, half of which were strong ties, and the other half of which were weak ties. They concluded that strong ties, in general, lead to the receipt of useful knowledge. However, when the concepts of trust, particularly benevolence-based and competence-based trust, were introduced, the positive effect of strong ties on the knowledge transfer disappeared once trust is controlled. In other words, trust mediates the relationship between strong ties, and after controlling for trust, weak ties have a stronger impact on knowledge transfer than strong ties. Here Levin and Cross (2004) clearly realized that there is something in knowledge transfer that network structure alone cannot explain. The content, characteristics, or behavior of these dyadic relationships interact with network structural properties to impact the outcome variables. Their research suggested that trust is the construct that made the difference. However, a disadvantage of trust, as a characteristic of relationships, does not fully answer the practical question of “how do I do it?” It provided critical evidence in regards to the psychological state of mind that is most important to the relationship between network structural properties and knowledge transfer, but it did not go a step further to explain how this state of mind could be

achieved. In fact, toward the end, the authors already hinted at behavior as the natural next step, when they discussed “strong ties encourage exploration of new ideas and techniques” (p. 1487); however, the article itself was not able to address the action problem.

The study conducted by Levin and Cross (2004) is a good example of the early realization that network structure cannot convey everything. However, among social network analysts, this is not the opinion of the majority. In the field of social network analysis, very little focus has been placed upon behaviors. Network researchers tend to perceive social networks as encompassing everything, and the content of relationships is “limited only by a researcher’s imagination” (Brass et al., 2004, p. 795). Behaviors and other individual characteristics are merely one type of content of relationship, and can be conceptualized as a part of the social network. For example, Nohria and Eccles (1992), in introducing a book collecting cutting-edge network papers, suggested that underlying all network research is the premise that “the actions (attitudes and behaviors) of actors in organizations can be best explained in terms of their position in networks of relationships” (p. 6). In so arguing, he simply incorporated behaviors as a content or object that is exchanged in the social network, without studying the specifics of the behaviors.

However, such an assumption is not necessarily accurate. It limits our ability to explain phenomena in organizations. Network structural properties alone do not explain why, when, and how actors share knowledge with one another, or how the relationship is initiated, developed, and sustained. A similar network structure could yield different results because of differences in actors’ behaviors. For example, even Burt (2004), a

well-respected structuralist network researcher, admitted that even though structural holes could lead to access to good ideas, they do not necessarily lead to effort of implementation, let alone success of implementation. A combination of the structural approach and the behavioral approach could provide us with a broader lens to uncover innovation in social networks. As suggested by Dolfsma and van der Eijk (2017)'s theoretical paper, the structural approach alone is particularly problematic in knowledge transfer and innovation research because these activities are usually extra-role and discretionary. Most employees, even within whole networks, are not required within their job responsibility to innovate or to share knowledge with people outside or even inside their organizations, so sharing knowledge with others or not is a personal choice. Studying these behaviors together with structural properties under the social network context is critical in understanding the success and failure of innovation.

So far in the literature that has been identified, there are only two published peer-reviewed papers that attempted to address the behavioral gap in social networks and innovation research, but neither of them addressed the gap sufficiently. The first one is Obstfeld (2005), who proposed the term “*tertius iungens*” (or the strategic, behavioral orientation toward connecting people together) as a behavioral dimension critical to knowledge-sharing across structural holes. Through studying the automotive industry's production of new cars and collecting extensive quantitative and qualitative data, Obstfeld pointed out that even though structural holes provide access to unique, novel knowledge, the dispersed and unconnected actors usually possess only weak ties that are harder to mobilize and act. The information advantage that structural holes can provide is usually offset by the lack of motivation to act for people with weak ties. He proposed

tertius iungens as a behavior orientation that partially supplements the structural approach to explain why knowledge is shared between certain actors but not others.

Obstfeld (2005) made a critical contribution to the literature by openly calling to address “the action problem” in network and innovation research; however, his examination of behaviors did not go very far from the structural approach. Tertius iungens, by definition, describes a strategic, behavioral orientation toward connecting people together, which, in its essence, is still about the establishment of connections. It could partially explain why relationships are initiated but does not explain the behavioral differences after the relationship has become stabilized.

Following Obstfeld’s (2005) call to address the “action problem,” the second study that looked into behaviors in network and innovation research was Dolfsma and van der Eijk’s (2017) theoretical paper on gift-exchange behavior. They argued that because of the cumulative nature and inherent uncertainty of knowledge, the structural approach of the social network is insufficient to address the diffusion and development of knowledge, let alone innovation. They proposed gift-exchange behavior – give, receive, and reciprocate informal obligations between actors – as a supplement to social network structures to explain why knowledge is shared with certain connections but not with others.

Dolfsman and vander Eijk’s (2017) theoretical proposal was an important step further than Obstfeld (2005) because it really addressed one type of core behavioral dimensions that do not relate to social structure directly. It is also an extra-role behavior in which people have the flexibility to decide whether they want to engage or not. However, gift-exchange behavior has two limitations with regard to the study of

innovation. First of all, the authors acknowledged that gift-exchange behavior cannot be enforced and is often found to be used when the official alternative is not preferable, which subjects the knowledge transfer to the vulnerability of abuse. Second and more importantly, gift-exchange behavior is a transactional behavior: a person reciprocates a favor when he/she is in a social debt. However, it does not explain how gift exchange is initiated to trigger the reciprocity process.

In short, both Obsteld (2005) and Dolfsma and van der Eijk (2016) clearly saw the gap in the literature and proposed their own behavioral constructs; however, studies of the behavioral foundation that underlies the relationship between social network and innovation are far from adequate. Additional research, particularly research on behaviors in combination with structural properties, are needed to further examine the gap.

Gap 2: Whole networks in the public sector. Brass et al. (2004) defined interorganizational networks as “long-term cooperative relationships between organizations ... in which organizations retain control over their own resources but jointly decide on their use” (p. 802). The developmental dynamics of an interorganizational network are highly complex. There are usually “multiple and interacting subprocesses; they are recursive; subject to external shocks, and the ability of participating actors to anticipate and then forestall their complete unfolding is highly confounding” (Bardach, 2001, p. 149).

Rich empirical research has suggested that innovation is more likely to occur in an interorganizational context. Swan and Scarbrough (2005), for example, through comparative studies of three cases of networked innovation, pointed out that innovation “increasingly needs to occur at the interstices of collaborating groups and organizations”

(p. 913). Powell, Koput, and Smith-Doerr (1996) pointed out a trend in innovation research in which the locus of innovation no longer lies in individuals or firms, but in the network in which a firm is embedded. Burt (2004) believed that compared with people with only intraorganizational ties, brokers across structural holes between heterogeneous groups (inter-organizations) are more likely to express ideas, less likely to have ideas dismissed, and more likely to have ideas evaluated as valuable. Pouwels and Koster (2017) showed a strong positive relationship between interorganizational cooperation and organizational innovation in 32 European countries and six different sectors.

In interorganizational network research, another gap in the literature is the level of focus and relationship among participating organizations. In network research, there are two complementary perspectives: one from the individual organization (actor level) and one from the network level (Provan et al., 2007). Many studies focused on specific organizations or dyadic relationships instead of focusing on the collection of organizations or the network as a whole (Najafian & Colabi, 2014). By focusing only on individual organizations themselves and their interaction with others, the needs, desires, and outcomes of those individual organizations tend to be exaggerated and the importance of network-level outcomes is underemphasized (Provan et al., 2007).

Kilduff and Tsai (2003) made the important distinction between the egocentric network versus the “whole network.” A whole network is defined as “a group of three or more organizations connected in ways that facilitate achievement of a common goal” (Provan et al., 2007, p. 482). A closer look characterized “whole networks” as comprising at least three legally autonomous organizations; involving multilateral ties among

participating organizations; being goal-oriented; and relatively clearly bounded by formal agreements (Provan et al., 2017).

To point out the need to understand “whole networks” in more depth, Provan et al. (2007) conducted a literature review of interorganizational network research from the whole network perspective. They divided interorganizational network research into four categories based on organizational-level and network-level variables serving as dependent and independent variables. Table 4 represents a typology of interorganizational network research developed by Provan et al. (2007).

Table 4
A Typology of Interorganizational Network Research

Independent Variable or Input Focus	Dependent Variable or Outcome Focus	
	Individual Organizations	Collectivities of Organizations
Organizational variables	Impact of organizations on other organizations through dyadic interactions	Impact of individual organizations on a network
Relational or network variables	Impact of a network on individual organizations	Whole networks or network-level interactions

Based on Provan et al. (2007), only when both the dependent variable and the independent variable are on a network level could a study be considered a whole network study. Such empirical studies are very limited in number. An extensive search of the literature provided only 26 papers that fit the criteria of studying “whole networks” between 1985 and 2005 (Provan et al., 2007). Most whole network research studies were conducted in the private sector. Another big concentration is on the public sector, but only in the health and human services area. This is clearly explained by the authors: given the complexity and scope of whole networks, sound research on whole networks takes a huge amount of resources. Many seminal works on whole networks occupied a large team of researchers for years. Only the private sector and the health industry could

provide enough funding to sustain a research initiative of this magnitude. Studies on whole networks among government agencies other than health and human services is practically nonexistent.

The current study uses a regional planning organization as the site, which is composed of 24 independent local government jurisdictions. It is bounded by a formal agreement and has a clearly articulated goal of strengthening the well-being of the region. It obviously satisfies all the requirements in the definition of a whole network as described by Provan, Sydow, and Podsakoff (2014). However, technically, the current study does not fit their description of whole network research. The level of analysis of the current study is individual, and no whole network–level dependent variable is studied. It merely used a “whole network” as the background and context of the study. Even though conducting a real whole network study was the first preference of the researcher, as has been pointed out, the magnitude of such a study was not feasible for one single researcher within the time frame and with the resources available for a dissertation study.

Nevertheless, this is not to say that the concept of whole network is not important or relevant in the current study. Regional local government collaboration networks have never been studied through the lens of a whole network, but conclusions made in the whole network literature have significant implications in this context.

On whole networks in the public sector, Keith Provan is one of the most critical and prolific researchers in the field (Galaskiewicz, 2014). He extensively studied interorganizational collaboration networks, its governance modes, social network structures, and network effectiveness, particularly in mental health systems. Drawing on his own prior research and working together with Lemaire (Provan & Lemaire, 2012),

Provan identified five characteristics of an effective whole network. In addition to “involvement at multiple levels” and “the importance of network design,” three other characteristics are particularly important to the regional local government setting: appropriate governance, building and maintaining legitimacy, and stability.

Appropriate governance. Provan and Kenis (2008) proposed three basic models of network governance, focusing on their distinct structures: participant-governed networks (or bottom-up approach), lead organization-governed networks (or top-down approach), or network administrative organizations (NAO). The governance structure of the regional collaboration network in the current study fit the description of an NAO. Even though the most appropriate mode of governance depends on a variety of contingency factors, and even though the network governance structure is not the single deciding factor of network effectiveness, there was evidence that NAO is an effective governance mode. For example, through a detailed comparative case study, Provan and Milward (1995) studied four networks of mental health organizations, in Tucson, Akron, Albuquerque, and Providence respectively, and concluded that Providence, which was managed by a strong core organization (NAO), provided the most effective network outcome, measured by the satisfaction of client, families, stakeholders, and case managers.

Building and maintaining legitimacy. Legitimacy is often referred to as credibility to others, or an externally conferred status (Suchman, 1995). Human and Provan (2000) pointed out that there is internal legitimacy (developing trust, demonstrating shared value, and building sustainable network governance) and external legitimacy (promoting the network and its activities to outsiders, seeking new members

and providing outside resources to meet network goals). Perceiving the regional local government collaboration network from the whole network perspective is particularly important because of legitimacy: a legally bounded network provides a legitimate identity for employees of its membership organizations to relate to. Internally, it provides a platform to develop trust and shared value, and externally it gives members of the network a tangible entity to promote and defend in front of others.

Stability. In general, when other factors are controlled, network effectiveness can be enhanced under system stability; however, stability alone does not serve as a sufficient condition for effectiveness (Provan & Milward, 1995). For example, Provan and Milward (1995), through studying the comparison of four networks of mental health organizations in four different cities, discovered that in general, cities with relatively higher system stability (Albuquerque and Providence) outperformed the ones with relatively lower system stability (Tuscon and Akron). However, the case in Albuquerque demonstrated that too much stability, on the other hand, would push participating organizations to work around system shortcomings instead overcoming them (Provan & Milward, 1995).

Network Citizenship Behavior

To address the two gaps in network and innovation research mentioned above – namely, lack of attention on behavioral dimensions and lack of focus on whole network level in the public sector as the context – the construct of **network citizenship behavior (NCB)** was identified from the literature. NCB is a relatively newly formed construct, first presented at the American Management Association conference in 2014 by Provan, Sydow, and Podsakoff, and published for the first time in 2017.

NCB might be new, but its theoretical roots in organizational citizenship behavior (OCB) have been comprehensively researched. OCB was first proposed by Organ and his colleagues in the 1980s (Bateman & Organ, 1983; Organ, 1988; Smith, Organ, & Near, 1983), and its definition has remained relatively consistent as “individual behavior that is discretionary, not directly or explicitly recognized by the formal reward system, and [which] in the aggregate promotes the efficient and effective functioning of the organization” (Organ, Podsakoff, & Mackenzie, 2006, p. 3).

The key word in OCB’s definition is *discretionary*, which means not reflected in the job description or performance evaluations. The behavior is based purely on the employees’ own willingness and intrinsic motivation, without external requirements or pressure. Over the years, there are many different forms of OCB that have been identified in literature. Indeed, Podsakoff, MacKenzie, Paine, and Bachrach (2000) identified more than 30 different forms of OCB, and LePine, Erez, and Johnson (2002) identified more than 40. In general, OCB is categorized into seven dimensions: helping/altruism, sportsmanship, organizational loyalty, organizational compliance, individual initiative, civic virtue, and self-development (Podsakoff et al., 2000).

NCB was defined by Provan, Sydow, and Podsakoff (2017) as “individual extra-role behavior exhibited by those who work in organizations that are part of a goal-directed multiorganizational network that supports the social and psychological environment in which the achievement of network-focused goals takes place” (p. 162). Provan et al. (2017) proposed that NCB is led by eight antecedent factors: four on the organizational level – goal congruity, reward system, organizational culture, organization’s network position; and four on the network level – domain characteristics,

type of network governance, network legitimacy, and network configuration.

Interestingly, even though NCB is an individual-level construct, none of the antecedents identified by Provan et al. (2017) was on the individual level.

Before NCB was conceptualized, a similar construct was developed in a separate line of research in the field of supply-chain management. Also based on the conceptualization of OCB, Autry, Skinner, and Lamb (2008) developed the concept of interorganizational citizenship behavior (ICB), and defined it as “interfirm behavioral tactics, generally enacted by boundary personnel, that are discretionary, not directly or explicitly included in formal agreements, and that in the aggregate promote the effective functioning of the supply chain” (p. 54) on the network level. They presented four subscales of ICB based on four of Podsakoff et al.’s (2000) seven categories of OCB:

1. Interorganizational tolerance: tolerance of inevitable inconveniences.
2. Interorganizational altruism: helping a network partner in solving problems.
3. Interorganizational loyalty: allegiance to partners and whole network, sometimes sacrificing self-interest.
4. Interorganizational compliance: orientation toward rules, policies, and processes of partners and whole network.

Even though NCB and ICB were developed in completely different streams of research and there is no direct inheritance between the two conceptualizations, the two concepts are quite similar. Both constructs are individual-level, behavioral concepts: NCB was very specific and accurate when it stressed in its definition that it is about “individual behaviors.” ICB, on the other hand, mostly was discussed as an individual-level construct, but at times appeared to be confused with an organizational-level

construct. For example, its instrument used “we” instead of “I” in every item. However, in its definition, ICB is described as “generally enacted by boundary personnel,” which is definitely individual. The reason “we” was used instead of “I” in the construction of its items could be that boundary personnel, in a way, represent the entire firm when interacting with its suppliers or clients. Moreover, even though both are individual behaviors, both concepts clearly adopt a network view, and consider the behavior to be embedded in the social relationships of networks. NCB talks about how “this behavior is socially embedded in the system in which it is exhibited,” whereas ICB takes a “relationist” view on partner firms. In addition, both constructs stress the fact that it is an extra-role, discretionary behavior, which means employees are not required to perform these behaviors, and the exhibition of such behavior is completely voluntary by employees. Lastly, both behaviors are targeted toward achieving a commonly shared goal. NCB specifically pointed out the “goal-directed” (p. 9) nature of the whole network, whereas ICB stated that it aimed at “promoting the effective function of the supply chain” (p. 54).

In addition to their similarities, the two concepts showed subtle differences in two ways: first, ICB was developed in the supply-chain management field, and was never intended to be extended beyond this background. This is evident in the fact that they specifically included the word “our suppliers” in all of their items. NCB, on the other hand, came out of the research on public sector health and human services organizations, but it aspires to be expanded into other sectors. In the examples that they provided, Provan et al. (2017) pointed out that such phenomena could be found in public, nonprofit, or private sectors, and in addressing issues such as job training and placement, regional

economic development, R&D consortia, and small and medium enterprises within a regional industry. Secondly, NCB specifically targets closed, legally bounded entities of a “whole network” as the background, whereas ICB is open to being exhibited toward any supplier or client without confining all the participating organizations to a single association. However, ICB’s definition of “common goal” clearly hinted at a loosely connected identity. Given that the publication of ICB predates the development of whole network, this difference between ICB and NCB could be merely a matter of concept maturity.

For the purpose of this current research, NCB and ICB are considered to represent the same phenomenon. The term NCB will be used to echo the contextual background of this research as “whole networks,” partly because NCB is defined more clearly and specifically, within broader contexts.

So far there has been no instrument developed to measure NCB in the public sector. Provan et al.’s (2017) work was theoretical, and clearly did not imply any measurement work to follow. However, an instrument for ICB has been developed by two consecutive research projects, those of Autry et al. (2008) and Skinner, Autry, and Lamb (2009). It contains 11 items, including four subscales, those being interorganizational tolerance, interorganizational altruism, interorganizational loyalty, and interorganizational compliance, which corresponds to Podsakoff et al.’s (2000) sportsmanship, helping/altruism, loyalty, and compliance, respectively. It reported high subscale reliability alphas ranging from .702 to .916. Even though the studies were limited in scope and not much follow-up research has been conducted, the two research studies did provide satisfactory validity evidence. The items were selected and adapted

from existing OCB scales, and a panel of executives and experts discussed the fit of these items to the constructs. Before the instrument was used, a pilot study was conducted using 208 online panel respondents from a variety of industries and organizations of varying sizes. Exploratory factor analysis was conducted to delete items with bad loadings. Afterwards, a confirmatory factor analysis was conducted with the same data set to confirm that the items were good representatives of each subscale. A second, separate data set with 318 procurement professionals was collected to evaluate the convergent validity of the instrument, and provided statistically significant results with outcome variables such as market performance, financial performance, and relationship quality.

Linking NCB to innovation. As the theoretical roots of NCB, in previous research, OCB was not related to innovation very extensively. The majority of research studies on OCB used some forms of financial indicators as the dependent variable (Mackenzi, Podsakoff, & Podsakoff, 2018). Organ, Podsakoff, and KacKenzie (2006) called for researching OCB's organizational outcomes other than financial performance, such as firms' innovativeness and other improvements in business processes. So far, only a handful of studies used innovation as an outcome of OCB. For example, Detert et al. (2013) studied how "speaking up" – one type of OCB – could lead to expressed ideas and actions generated as a result. Zhang, Wan, and Jia's (2008) research demonstrated that OCB mediates the relationship between HR practices and corporate entrepreneurship. Similarly, Yoon (2009) discovered that OCB leads to intention to innovate in an IT context.

Among these three examples, Zhang, Wan, and Jia (2008) was the only one that was able to empirically establish a relationship between OCB and innovation. They set out to examine the relationship between HR practices, OCB, and corporate entrepreneurship. Based on their definition, corporate entrepreneurship incorporates three components: innovation, venturing, and strategic renewing activities. They surveyed 139 small- to medium-sized biotechnology companies in China and 695 frontline supervisors in these companies, and concluded that OCB mediates the relationship between HR practices and corporate entrepreneurship. As a necessary condition of the mediation relationship, OCB was found to be positively related to corporate entrepreneurship, which encompasses innovation. Zhang, Wan, and Jia (2008) provided a sound quantitative research study, using a large sample size, reliable instruments, and appropriate statistical modeling. However, the research did present a few limitations. Even though the OCB instrument they chose to use presented a high internal consistency (Cronbach's alpha value at .87), its construct validity is not as satisfactory. The instrument was developed in 1990 (Podsakoff et al., 1990), relatively obsolete for an article published in 2008, and because of how outdated the instrument was, it did not fully describe OCB as a construct. The instrument described OCB as containing five behaviors: help, courtesy, sportsmanship, civic virtue, and conscientiousness. However, as the construct matured, OCB also included organizational compliance and self-development (Podsakoff et al., 2000).

Even though there are only a few direct empirical research studies, OCB relates to innovative outputs on the theoretical level, for two reasons. To begin with, OCB shares similarities with several other behaviors that have been found to relate to innovation. As

stated earlier, to bridge the behavioral gap in social network research, Obstfeld (2005) proposed the *tertius iungens* strategic orientation (or “third who joins”) and Dolfisma and van der Eijk (2017) used “gift exchange behavior” to explain why and how knowledge-exchange relationships are established and maintained to produce innovation. Similarly to these two constructs, OCB is also an extra-role behavior that is not mandated by job description or requirements. The tendency of introducing unconnected people together (*tertius iungens*) and the obligation to give, receive, and reciprocate (gift-exchange behavior) can be described as altruistic and helpful for the purpose of enhancing organizational efficiency, as OCB is defined. In a way, *tertius iungens* and gift-exchange behavior can be considered as examples or subdimensions of OCB. In previous research, through studying a sample of 124 professionals in a new vehicle design process at an auto-making company, Obstfeld (2005) suggested *tertius iungens* behavior leads to innovation. Dolfisma and van der Eijk (2017), in their theoretical arguments, proposed that gift-exchange behavior will lead to innovation. As a result, OCB, as an overarching concept, might lead to innovation as well.

In addition, even though OCB does not predict innovation directly, in empirical research it does lead to innovation indirectly. OCB is found to lead to organizational efficiency and financial well-being through several mediator factors, including team spirit (Podsakoff et al., 2000), development of team norms (Ehrhart & Naumann, 2004; Raver, Ehrhart, & Chadwick, 2012), trust (Gong, Chang, & Cheung, 2010), and team-shared vision (Pearce & Ensley, 2004). In turn, in the innovation literature, a number of environmental factors on the team level have been found to lead to innovation, including trust (Akhavan et al., 2015; Levin & Cross, 2004; Molina-Morales & Martinez-

Fernandez, 2010; Zheng, 2010), effective communication (Monge, Cozzens, & Contractor, 1992; Odoardi et al., 2015; Ortega-Egea, Moreno, & Dominguez, 2014), social interaction (Molina-Morales & Martinez-Fernandez, 2010), team-shared vision (Molina-Morales & Martinez-Fernandez, 2010; Zheng, 2010), and norms (Russel & Russel, 1992; Zheng, 2010). Thus, a logic chain can be established from OCB to trust, social interaction, and shared vision, which consequently leads to innovation.

When citizenship behavior is expanded to the network level, its connection to innovation is expected to be even greater, because the locus of innovation is increasingly moving toward interorganizational networks (Powell, Koput, & Smith-Doerr, 1996). However, because of the novelty of the construct, so far there is only one research study that explicitly set out to study the relationship between NCB and innovation. Gerke et al. (2017) conducted a single-case study in the New Zealand marine racing industry in Auckland, and collected extensive qualitative data, including 27 interviews, 4 observations, 12 organizational documents, and 1 piece of archival data. Through data coding analysis, they identified 16 practices that belong to the seven types of ICB, and based on a literature review, they identified three subphases of innovation: ideation, invention, and exploitation. Next, they mapped these 16 practices onto these three subphases of innovation. Their findings presented that advancement and altruism are important in all three phases of innovation. Other than that, conscientiousness is most prominent during the ideation phase; advancement, loyalty, and tolerance are more important during the invention phase; and constructiveness is more critical in the exploitation phase. The study was the first clear attempt to connect ICB with innovation,

and it provided concrete examples in natural organizational settings. However, it was not able to establish a strong, quantitative link between ICB and innovation.

Despite the lack of empirical research, theoretically, NCB is expected to relate to innovation. A related construct, even though it is not a behavior, is *network competence*, developed by Ritter, Wilkinson, and Johnston (2002). Network competence includes two dimensions:

1. Network management qualification, such as social and specialist qualification.
2. The execution of network management tasks, such as planning, coordination, adaptation, and controlling activities (Chiu, 2009).

Even though the first dimension is usually in-role, the second dimension shares much similarity with NCB. Because network competence has been discovered to have a significant relationship to innovation (Chiu, 2009; Ritter, Wilkinson, & Johnston, 2002), this suggests that NCB can predict innovation as well.

Therefore, the following relationship is hypothesized:

Hypothesis: The stronger the exhibition of network citizenship behavior, the stronger the exhibition of individual innovative behavior.

Social Network Structural Properties

Among the social network structural properties discussed earlier, centrality and tie strength have a particularly strong impact on innovation. Network centrality refers to an actor's position in a network. Central positions are usually associated with power and influence (Brass & Burkhardt, 1992). There are three types of centrality that have been operationalized in the research: *degree centrality*, which measures the extent of network ties of an actor among the rest of the network; *closeness centrality*, which focuses on how

close an actor is to all the other actors in the network; and *betweenness centrality*, measuring the degree to which an actor “sits” in between all other pairs of random actors (Wasserman & Faust, 1994). All three types of centrality measure the structural importance of an actor, but they might or might not be the same for a given actor in a given network. Research findings on the relationship between centrality and innovation have been relatively consistent, that a higher *degree* of centrality is likely to lead to a higher level of innovation (Bell, 2005; Casanueva, Castro, & Galan, 2013; Chiu, 2009; Ibarra, 1993; Perry-Smith, 2006; Tsai & Ghoshal, 1998; Tseng et al., 2016). The only caveat lies in the interaction effect of centrality and size: Perry-Smith (2006)’s study found that the advantage of a centralized location can be replaced by the size of outside ties. For example, for researchers who are on the periphery of their lab’s network (low centrality), creativity was higher when they had more outside connections (Perry-Smith, 2006).

Tie strength refers to the nature of the relationship between two actors, which combines time investment, emotional intensity, intimacy, and reciprocal services (Granovetter, 1973). Strong ties entail intensive, frequent interactions with multiple functions, whereas weak ties involve simple, occasional relationships. Many may confuse tie strength with density because people with strong ties are more likely to be similar to each other and have redundant connections (Granovetter, 1973; Perry-Smith, 2006; Zheng, 2010). However, these two are distinct concepts. Density is a structural concept that captures the connectedness of the contacts that an actor has, whereas tie strength describes the characteristics of the connections, and it is technically not a structural property. A strong tie can exist in a sparse network and a weak tie can also exist in a

dense network. The two constructs are correlated but not interchangeable (Reagans & McEvily, 2003; Zheng, 2010).

In the literature, there is much debate about the effects of strong and weak ties on innovation. On one hand, strong ties are more beneficial for the exchange of knowledge and information by creating trust and a willingness to share that consequently facilitates innovation (Albrecht & Ropp, 1984; Bouty, 2000; Fritsch & Kauffeld-Monz, 2010; Krackhardt, 1992; Landry et al., 2002; Levin & Cross, 2004; Moran, 2005; Reagans & Zuckerman, 2001; Schulz, 2003; Smith et al., 2005; Uzzi & Lancaster, 2003). Strong ties are created by multiple layers of interpersonal exchange that facilitate the discussion of innovation in organizations (Albrecht & Ropp, 1984). Weak ties can impede the transfer of complex knowledge, which is more needed to generate innovation (Hansen, 1999). On the other hand, the strength of weak ties is an often overlooked asset that brings actors access to disconnected partners and thus diverse information (Granovetter, 1973). Hansen (1999) showed that weak inter-unit ties help a project team search for useful knowledge. Perry-Smith (2006) also found that a person's weak ties had a positive effect on creativity. Strong ties increase trust levels, which can reduce levels of innovation by locking them into existing relationships at the expense of gaining access to diverse knowledge from new partners (Yli-Renko, Autio, & Sapienza, 2001). Levin and Cross (2004)'s research also showed that when trust is controlled at the same level, weak ties have a stronger prediction of innovation than strong ties.

To combine both perspectives, Leenders et al. (2003) discovered that it is an inverted U-shape relationship between tie strength inside a team and team creativity. They suggested that both too low and too high a level of interaction frequency impeded

team creativity, whereas moderate frequency produced the highest level of creativity. McFadyen and Cannella (2004) had a similar finding, that after a certain point, more interactions seem to bring diminishing benefits.

In this research, the interactional relationship between structural properties, particularly centrality and tie strength, and behavioral dimensions were explored to examine how they impact the exhibition of individual innovative behaviors in combination. Specifically, the following research question is explored:

How do social network structural properties, measured by network centrality and tie strength, impact the exhibition of NCB, IIB, or the relationship between NCB and IIB?

Chapter 3: Methodology

Overview

As previously discussed, the purpose of this research was to examine the relationship between individual innovative behaviors (IIB) and network citizenship behavior (NCB) in the context of a whole network in the local government in the United States. For this examination, the following research questions were asked:

1. What is the relationship between network citizenship behavior (and its four subscales) and individual innovative behaviors in a U.S. local government's whole network?

Hypothesis 1: the stronger the exhibition of network citizenship behavior, the stronger the exhibition of individual innovative behaviors.

2. How do social network structural properties of organizations, measured by network centrality and tie strength of organizations, impact the exhibition of individual innovative behavior, or the relationship between network citizenship behavior and individual innovative behavior?

To answer these research questions, an empirical, cross-sectional, quantitative research study was conducted using an Internet-based, self-administered survey method and multiple regression analysis. This research design was selected for a number of reasons:

To begin with, the quantitative design was selected because the researcher holds a worldview of postpositivism, which puts forth that there are laws that govern the world, and truth exists independent of individuals' minds. A postpositivist believes in "a deterministic philosophy in which causes (probably) determine effects or outcomes"

(Creswell, 2014, p. 7). The postpositivistic worldview is best manifested in quantitative research design, which seeks to examine a conjectured relationship among variables, usually operationalized as numbers using instruments (Creswell, 2014). The nature of the current research questions fall under the umbrella of untangling relationships among variables, thus making quantitative research design the natural choice for the current study.

Secondly, the survey method was chosen based on the nature of the research question, which sought to understand relationships among variables. Using a survey method allowed an understanding of constructs and the relationships among them in a large population from a small group of individuals (Fowler, 2009). A nonexperimental, cross-sectional survey method was used because the research questions sought to understand phenomena in organizations in a natural setting, at one point in time, and did not seek to impose any intervention or change. Internet-based surveys were used because of their low cost, fast turnaround time in data collection, capacity for reaching widely dispersed participants, and allowance for direct data entry (Sue & Ritter, 2012).

Participants

Population and research site. The site selected for this study is the Metropolitan Washington Council of Governments (MWCOC), a regional planning organization for the greater Washington, D.C., area that is composed of 24 local government jurisdictions, featuring urban, suburban, and rural communities that range in size from about 10,000 to more than one million residents. Figure 2 illustrates a map that covers the area that MWCOC represents.



Figure 2: Jurisdictions of Metropolitan Washington Council of Governments

As it is described on its website:

Metropolitan Washington is a diverse and dynamic region home to more than five million people and one of the nation's largest economies. More than a million new residents and jobs are forecast between now [2016] and 2045. Managing this growth – and enhancing the region's quality of life – requires partnership. And that's why the region relies on the Metropolitan Washington Council of Governments, also known as COG. (Metropolitan Washington Council of Governments, 2016a)

Established in 1957, MWCOG is an independent, nonprofit association, whose memberships consist of 300 elected officials from those 24 local governments, the Maryland and Virginia state legislatures, and the U.S. Congress. It also invites private businesses and other quasi-governmental organizations that have a significant impact on

the well-being of the region, such as Washington Metropolitan Area Transit Authority (WMATA), to be nonvoting members of the network. The goal of MWCOG is to help tackle the biggest challenges that metropolitan Washington faces as a region, ones that no single jurisdiction alone would be able to address without the support of the others. Such challenges include “revitalizing the Potomac River, ensuring the Metro system was fully built, and strengthening emergency preparedness after 9/11,” to name a few (Metropolitan Washington Council of Governments, 2016b, para 1).

Such a site was a strong manifestation of Provan, Sydow, and Podsakoff’s (2014) “whole network” concept, and was an ideal setting in which to study NCB. Using MWCOG as the institution, the local governments of the Washington Metropolitan Region make up a large “whole network.” As its definition describes, it

1. Contains more than three legally autonomous organizations, because each of the 24 jurisdictions is a legally independent entity.
2. Involves multilateral ties among participating organizations, because jurisdictions work with each other on individual, team, departmental, and jurisdictional levels in various ways.
3. Is goal-oriented, which can be demonstrated by MWCOG’s Region Forward Vision that “focuses on creating a more prosperous, accessible, livable and sustainable metropolitan Washington.” (Metropolitan Washington Council of Governments, 2017, para 1).
4. Is clearly bounded by formal agreements, which can be presented by the existence of MWCOG’s bylaws. All jurisdictions in the Washington, DC, area must pay dues to become a member of MWCOG. The Board of

Directors, consists of 34 members appointed by local jurisdictions, serves as the official governing body of the MWCOG network.

The population of this study was employees who work for local government jurisdictions in this network who communicate with people across jurisdictional boundaries and attend committees, projects, legislative meetings, and other functions at MWCOG. It was difficult to estimate the full size of the population due to the fluid nature of interjurisdictional collaboration, but roughly it could be estimated at around a few thousand people.

Sampling frame. The sampling frame of this study was a full list of individuals who had participated in cross-jurisdictional training programs, the purpose of which was to build a collaborative network among regional jurisdictions. The training program, known as the Regional Executive Development Program (REDP), “assists local government managers in metropolitan Washington area in dealing with the challenges of providing services and meeting the needs or residents in a growing metropolitan area” (Institute of Regional Excellence, 2019, para 4). It targets mid- to senior-level managers in local governments in metropolitan Washington area. Participants were first selected by their respective jurisdictions, and then jurisdictions submit their applicants to the program. Selection criterion include managerial experience and personal perspective on regional challenges. Experience or responsibility to work across jurisdictions as a part of their job is not a requirement. There were approximately 350 participants in this sampling frame.

The proposed sampling frame represented the population to be studied satisfactorily. Given the unpredictable nature of interorganizational collaboration work,

as well as the sensitivity of certain interorganizational work committees, it was impossible to obtain a large sampling frame that represented the population perfectly. The proposed sampling frame was a convenience sample. It represented the population to be studied well because of the participants' exposure and sensitivity toward the importance of interorganizational collaboration work through the interjurisdictional training, and it was easily accessible. It is important to note that before joining the program, some of the participants may already have had a boundary spanning role on their jobs, thus exhibiting frequent NCB. However, the majority of the participants did not have interjurisdictional work experience, and NCB would be relatively new to them. The number of people from each jurisdiction in the sampling frame roughly represented their population and resources, as larger jurisdictions usually had more resources to support more people to take the training, and smaller jurisdictions usually faced a fiscal constraint therefore supporting fewer participants. The sampling frame represented 19 out of the 24 jurisdictions in the network. The remaining jurisdictions were missing from the sampling frame either because of their late participation in the network, or because of a small workforce.

However, this sampling frame does present one limitation. Given that the training program was interjurisdictional in nature, and cross-boundary collaboration is one key component of the program, participants received significant exposure to NCB and had a deeper understanding of the benefit of it. In a way, the sampling frame might be more positively skewed toward the exhibition of NCB than the population it represents. Nevertheless, given the difficulty in obtaining permission to survey or sample the whole population, this is the best framework available to the current study.

Sample, sample size, and selection criteria. A convenience sample was used in this study because of the difficulty in obtaining contact information for all employees in this network due to privacy and confidentiality concerns in HR practices. All individuals who met the selection criteria were contacted to participate in the study. The selection criteria included—

1. Respondent must be a current employee of participating organizations of MWCOG (people who had retired, moved out of the region, or left their employer were excluded, as indicated by the accessibility of their work email address).
2. Respondent must currently participate in or have previously participated in interjurisdictional training programs through MWCOG.

A power analysis was conducted to calculate the minimal required sample size to conduct rigorous regression analyses. Consistent with practice in the field, the power for this study was set at .8, and the alpha value was set as .05. There is one independent variable, NCB, with four subscales ($m = 4$). According to Green (1991), who cited Cohen (1988), “a typical study in the behavioral sciences would have a medium effect size” (Green, 1991, p. 503), which equals to $R^2 = .13$ or $f^2 = .15$. Based on Green (1991):

$$L = 6.4 + 1.65 * m - .05 * m^2$$

As a result, in this case:

$$L = 6.4 + 1.65 * 4 - .05 * 4^2 = 12.2$$

$$N \geq L/f^2 = 12.2 / .15 = 82$$

Thus, a sample of at least 82 respondents was desired in data collection to conduct regression analyses to answer the two research questions.

Survey Instruments

Two instruments were used to conduct the current study: IICB and NCB. Network data were collected through name generator questions, and five different types of control variables were collected to explore their effects.

Individual Innovative Behavior (IIB). Based on a thorough review of the literature, comparing a number of instruments of innovative behaviors, the Kleysen and Street (2001) instrument was selected to be employed in this study. Consistent with the literature, Kleysen and Street (2001) followed the understanding of innovative behaviors that originated with West and Farr (1989), “all individual actions directed at the generation, introduction and/or application of beneficial novelty at any organizational level” (Kleysen & Street, 2001, p. 285). After the authors removed items without a good factor loading, the instrument contained 14 six-point frequency rating scale items that measure the frequency of exhibition of certain behaviors, from “never” to “always.” The full instrument of Kleysen and Street’s (2001) IIB measurement is included in the Appendix. Permission has been obtained from one of the original authors to use this instrument in the current study.

The Kleysen and Street (2001)’s IIB instrument had satisfactory reliability and validity. In its original study, a sample of 225 responses from managers in nine different organizations was used, covering a wide range of industrial backgrounds and professional backgrounds. The instrument started with 34 items, and after taking out items with inadequate factor loadings, 14 items were left. Its internal consistency reliability, the Cronbach’s alpha, was reported at .945. With regard to its construct validity, the researchers originally set out to create a multidimensional measurement of individual

innovative behaviors. However, given that the model did not support the multidimensional structure, only one overall scale was presented. Nevertheless, because all the measurement items were developed based on the literature and represent a rich domain of observable variables associated with the construct of IIB, the authors were confident in its construct validity.

After its original development, several studies adopted the instrument in various contexts. As the examples in Table 5 show, several studies that used Kleysen and Street's (2001) IIB instrument presented their reliability and validity evidence. All studies treated Kleysen and Street (2001) as a one-dimensional instrument. The multidimensionality of the instrument was never supported empirically.

Table 5

Reliability and Validity for Kleysen & Street (2001) Instrument

Paper	Sample	Items	Reliability	Validity
Kleyses & Street (2001)	225 employees from 9 different organizations covering a wide range of industries and professions	14 items	.945	Based on the authors, construct validity was achieved through literature review which solicited 289 creativity and innovation-related activities and was coded into 17 behaviors in 5 dimensions
Kim & Park (2015)	347 nurses in 6 general hospitals in central South Korea	14 items	.95	Good model fit Goodness-of-fit>.90, root mean square residual <.08 Significant correlations with independent variables, including creative self-efficacy, innovative organizational culture, self-leadership; individual knowledge sharing; and organizational knowledge share (indirectly through innovative organizational culture)
Wojtczuk-Turek & Turek (2015)	166 respondents from the database of a service company + 70 postgraduate students of Warsaw School of Economics, data collected from January to May 2012	12 items (did not specify which two items were dropped)	.94	Confirmatory factor analysis indicated good fit ($X^2 = 47.950$; $df = 37$; $p = .107$; $RMSEA = .042$; $CFI = .993$; $GFI = .954$; $NFI = .966$; $TLI = .987$. Factor loadings range between .669 and .833 Significant correlations with psychological capital as independent variable
Wojtczuk-Turek & Turek (2016)	246 employees from 76 companies in Poland, managerial and non-managerial, wide range of industry and profession. Data from Feb 2013 to Jan 2014	14 items	.95	Confirmatory factor analysis indicates good fit of the model; significant correlation with all independent variables: perceived social-organizational climate; organizational citizenship behaviors of other employees; mediated by person-organization fit
Mishra et al. (2017)	398 service sector employees in India	6 items (did not explain which 6 or why these 6)	Composite reliability: .93	Average variance explained (AVE): .69 Average shared squared variance (ASV): .29 Convergent validity: $CR > AVE$ Discriminant validity: $ASV < AVE$

Network Citizenship Behavior. “Network Citizenship Behavior” was defined by Provan et al. (2014) as the “individual behavior exhibited by those who work in organizations that are part of a goal-directed multiorganizational network that supports the social and psychological environment in which the achievement of network-focused goals takes place” (p. 9). Because of the novelty of the construct, there is no existing instrument that measures network citizenship behavior. A related but different concept, “Inter-organizational Citizenship Behavior,” was developed by Skinner, Autry, and Lamb in 2009. They defined it as “interfirm behavioral tactics, generally enacted by boundary personnel, that are discretionary, not directly or explicitly included in informal agreements, and that in the aggregate promote the functioning of the supply chain” (p. 229).

NCB and ICB are similar in the way that they are both theoretically grounded in the concept of organizational citizenship behavior (OCB) (Organ, 1988; Podsakoff, Mackenzie, & Podsakoff, 2018), which is a discrete, extra-role behavior that promotes the effectiveness of the organization without seeking immediate returns. They are different in that the context of ICB is in the private sector, and more specifically in the supply-chain management area, whereas NCB describes a broader range of contexts, and stresses the importance of goal congruence.

Because of the similarities between the construct to be measured (NCB) and the existing scale of ICB, an adapted version of Skinner et al.’s (2009) ICB scale was used. The original scale included four subdimensions: interorganizational tolerance, interorganizational altruism, interorganizational loyalty, and interorganizational compliance. The authors developed the items through literature reviews and through

interviews with a panel of 25 company executives. The survey was sent to an Internet panel of 800 business owners identified as having experience in interorganizational exchange; 195 of the 800 potential participants, or 27.6%, returned complete data for analysis. After an exploratory factor analysis (EFA) and a confirmatory factor analysis (CFA), the authors deleted 7 items with poor factor loadings, and a total of 11 items are left, measured on a 1-7 frequency scale. With the exception of interorganizational tolerance, which had two items, the other three subscales contained three items each. Interorganizational compliance has the highest Cronbach's alpha score, .916, whereas interorganizational altruism has the lowest, .702. Interorganizational tolerance and interorganizational loyalty have Cronbach's alpha scores of .847 and .734, respectively.

The instrument went through a rigorous validation process. The items were selected and adapted from existing OCB scales, and a panel of executives and experts discussed the fit of these items to the constructs. Before the instrument was used, a pilot study was conducted using 208 online panel respondents from a variety of industries and organizations of varying sizes. EFA was conducted to delete items with bad loadings. Afterwards, CFA was conducted with the same data set to confirm the items were good representatives of each subscale. A second, separate data set with 318 procurement professionals was collected to evaluate the convergent validity of the instrument, and provided statistically significant results with outcome variables such as market performance, financial performance, and relationship quality (Skinner et al., 2009).

After the initial development and validation, the instrument was rarely used in other studies. It has been cited in a number of papers on the subject, and has been used in

the sports industry in a qualitative study (Gerke, Desbordes, & Dickson, 2014), but no additional reliability studies have been reported.

As a result of the different contextual background in the current research, the instrument was modified before distributing it to the participants. Wording of the original instrument was adjusted based on the new contextual background. Because the original scale was used in a supply-chain management context, verbiage such as “supplier” was used in almost every item, and words such as “product” or “business” were frequently referred to as well. To apply the scale in a government context, all usage of “supplier” was changed to “partner” (for all 11 items), “business” or “company” to “organizations” (1 item for each), and the word “products” was deleted (for 1 item). Permission to use and adapt the survey was obtained from one of the original authors. After the instrument was updated, a panel of three experts who have firsthand observation of the NCB phenomenon were invited by the researcher to review the measurement. The results of the panel review indicated that the change of wording carried minimal impact on the validity of the scale, and no additional changes were recommended.

Network data. The network data of this study was collected through a series of questions asking, “Thinking about the top five people from other jurisdictions with whom you usually communicate regarding work issues, which jurisdictions are they from, in the order of frequency of communication?”

Control variables exploration. A total number of six demographic variables were collected in the survey, including age, tenure in organization, boundary spanning status, managerial level, jurisdictional background, and professional background. In addition, jurisdictional budgets for the Fiscal Year 2019 were also collected from online

archival data to indicate available resources as a control variable. The impacts of these control variables on IIB were explored in the study.

Data Collection Procedure

Survey administration. Prior to any effort in data collection, the study underwent a rigorous review at The George Washington University's Institutional Review Board (IRB) to ensure it had minimal impact on the human participants. After the IRB approval, data collection took place in the summer of 2019.

The data for this study were collected through email with the URL link to the survey embedded in the body of the email. In the online survey, before the demographic information and the content of the instrument, an informed consent form was presented to the respondents that reiterated the purpose of the study and its confidentiality provisions. Additional reminder emails to all nonresponsive participants were distributed approximately one week, two weeks, and three weeks after the initial distribution. The full survey is presented in the Appendix.

Strategies for maximizing the Response Rate.

Ensuring accurate email information. After the initial survey invitation email was distributed, a list of 35 bounced-back and unavailable email addresses was generated. Additional methods were used to seek the current email addresses of the respondents, including, but not limited to, double-checking hardcopies of documentations, Internet research, and inquiring the respondents' peers and colleagues. Unfortunately no additional correct email addresses were obtained.

Follow-up with respondents. Three emails were sent to nonresponsive participants to readdress the importance of the survey and to politely encourage the

recipients to participate in the survey. The online survey website allowed for contacting nonresponsive participants without disclosing their identity to the researcher or anybody else.

Data Handling and Preparation

The survey was sent to the entire proposed sampling frame, which included a total of 348 participants after deleting undeliverable email addresses. During the data collection process, data were collected and stored in a password-protected account on a secure online platform. After the data collection process, data were downloaded and saved in a password-protected computer.

Between July and September of 2019, 109 responses were received, generating a response rate of 31.32%. After initial screening, 20 responses were deleted due to missing data. Missing data were defined as missing 5% or more from the combination of NCB and IIB instrument items. If demographic data were missing, it was not counted toward the 5%. All 20 responses in question had significant amounts of missing data, leaving more than half of the survey blank. As a result, computational scores were not possible to replace missing data. Thus, listwise deletion was used. In total, 89 responses were left for analysis, yielding an effective response rate of 25.57%.

All instrument data were transformed from frequency or Likert scales to numerical scales. NCB variables were measured on a 7-point Likert scale, and IIB variables were measured on a 6-point frequency scale. Even though they are ordinal in nature, it is suggested that such variables with more than five categories can be treated as continuous (Dimitrov, 2012). Two levels of composite scores were calculated: the first level was subscales – a composite score adding scores of all items for each of the four

subscales of NCB and each of the five subscales of IIB were calculated; the second level was the latent variables – the dependent variable of IIB, and independent variables of NCB. Once data were cleaned, they were imported into SPSS 25 and LISREAL 10.20 (student) for further analysis.

For network data, survey respondents were asked which jurisdictions their top five professional contacts were from. Only responses from jurisdictions within the whole network territory were considered. A total of 19 jurisdictions were identified through the survey, and a 19*19 matrix was created. A tie from Jurisdiction A to Jurisdiction B was counted when a respondent from Jurisdiction B identified someone in Jurisdiction A as a frequent professional contact, because information flows from the contacts to the survey respondents. Ties were weighted and the most frequent contact was assigned a weight of 5, the second most frequent contact a weight of 4, the third a weight of 3, and so on.

Data Analysis

Demographic data were presented to describe response rate and respondents of this study. A list of demographic questions can be found in the survey in the Appendix. After preliminary handling, data were transferred from Excel to SPSS to conduct data analysis. As a reliability measure, Cronbach's alpha was calculated for each composite score and was compared with the Cronbach's alpha obtained in the literature. Cronbach's alpha value is set at .7 for this study.

Before conducting data analysis, the following assumptions were tested to examine whether the dataset met the prerequisite of regression analysis: outliers, normal distribution of dependent and independent variables; normal distribution of residuals; non-multicollinearity among independent variables; linear relationship between dependent

variables and independent variables; linear relationship between independent variables and the standard residual of dependent variables.

To answer the first research question, what is the predictive relationship between NCB (and its four subscales) and IIB in a U.S. local municipal government's whole network, a simple regression analysis was conducted. The regression model can be presented as

$$y_i = \alpha + \beta x_i + \varepsilon_i \quad (1)$$

where x_i is NCB, and where y_i is IIB.

To answer the second research question, a series of exploratory multiple regression analyses were conducted to explore the impact of structural properties, measured by centrality and tie strength, on IIB. The regression model can be presented as

$$y_i = \alpha + \beta x_i + \varepsilon_i \quad (2)$$

where x_i is network structural properties (centrality and tie strength) and y_i is IIB.

As an exploration of the impact of the interaction effect of network structural properties and NCB on IIB, a multiple regression analysis was conducted, and the regression model can be presented as

$$y_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i} x_{2i} + \varepsilon_i \quad (3)$$

where x_{1i} is network structural properties (centrality and tie strength), x_{2i} is NCB, and y_i is IIB.

Human Participants and Ethics Precautions

Each participant was assured that the survey responses would be kept confidential because no identifiable information was collected. Their employers and peers would not be aware that they participated in the study. An informed consent form was included at

the beginning of the Internet-based survey, stating the purpose and the scope of the study. The survey could not be taken without the participants' agreement to the consent form. Participants were assured that only the study's researcher had access to the data collected. Data collected using Internet-based survey tools were deleted from the Internet survey site as soon as the data collection procedure ended. Participants were also informed that participation in this study would lead to minimal risks. No individual information was shared with their employer or any other individual other than the researcher.

Chapter 4: Data Analysis

As previously discussed, the current research sought to answer the following two research questions:

1. What is the predictive relationship between network citizenship behavior (and its four subscales) and individual innovative behaviors in a U.S. local municipal government's whole network?

Hypothesis 1: the stronger the exhibition of network citizenship behavior, the stronger the exhibition of individual innovative behaviors.

2. How do social network structural properties of organizations, measured by network centrality and tie strength of organizations, impact the exhibition of individual innovative behavior, or the relationship between network citizenship behavior and individual innovative behavior?

The primary purpose of this chapter is to outline the process and results of the statistical analysis of the current research. The chapter is organized as follows: first, the demographic information of the data were presented; then an extensive validation was conducted for both NCB and IIB instruments. Lastly, the analysis of the two research questions are described, with both tests of the assumptions and tests of the hypotheses.

Demographic Information

Tables 6, 7, and 8 summarize the demographic information, jurisdictional origins, and professional backgrounds of the 89 effective responses. One thing to note in particular is that 19 out of 89 respondents (or 21.35%) identified themselves as nonboundary spanning personnel. At first glance, it might seem counterintuitive, because all respondents in the sampling frame participated in a training program that was

designed for the purpose of enhancing interjurisdictional collaboration. However, having a boundary-spanning role as a component of their job responsibility was not a requirement to participate in the program. In fact, people without boundary-spanning experience were preferred when recruiting participants, so that the importance of interjurisdictional collaboration would be transferred to front-line employees who normally are not required to perform boundary-spanning tasks.

Table 6
Demographic Characteristics

Characteristics	Frequency	Frequency Accumulation	Percentage	Percentage Accumulation
Age (What is your age?)				
Below 30	2	2	2.25%	
30-40	15	17	16.85%	19.10%
40-50	35	52	39.33%	58.43%
50-60	25	77	28.09%	86.52%
60 and above	10	87	11.24%	97.76%
Unidentified	2	89	2.25%	100%
Tenure (How long have you been working with local governments?)				
Less than 5 years	6	6	6.74%	
5-10 years	14	20	15.73%	22.47%
10-20 years	38	58	42.70%	65.17%
20-30 years	16	74	18.00%	83.15%
30 years and above	14	88	15.73%	98.88%
Unidentified	1	89	1.12%	100%
Boundary Spanning Status (In your role, are you required to work with people from other jurisdictions?)				
Yes	70	70	78.65%	
No	19	89	21.35%	100%
Managerial Level				
Entry Level	4	4	4.49%	
Front Line	11	15	12.36%	16.85%
Supervisory				
Middle Management	43	58	48.31%	65.16%
Strategic Leadership	31	89	34.83%	100%

Table 7
Jurisdictional Background

Jurisdiction	Jurisdiction Budget	Frequency	Frequency Accumulation	Percentage	Percentage Accumulation
Arlington County	\$1.4 billion	4	4	4.49%	
Charles County	\$425,097,500	10	14	11.24%	15.73%
City of Alexandria	\$761,500,000	6	20	6.74%	22.47%
City of Fairfax	\$145,333,489	1	21	1.12%	23.59%
City of Falls Church	\$92,801,742	2	23	2.25%	25.84%
City of Gaithersburg	\$68,300,000	1	24	1.12%	26.96%
City of Greenbelt	\$31,139,100	1	25	1.12%	28.08%
City of Laurel	\$34,228,093	1	26	1.12%	29.21%
City of Rockville	\$138,900,000	1	27	1.12%	30.34%
District of Columbia	\$15.4 billion	4	31	4.49%	34.83%
Fairfax County	\$4,449,429,561	19	50	21.35%	56.18%
Frederick County	\$637,700,000	2	52	1.12%	57.3%
Loudoun County	\$2,600,000,000	1	53	2.25%	59.55%
Montgomery County	\$5,800,000,000	12	65	13.48%	73.03%
Metropolitan Washington Council of Governments	\$31,766,500	6	71	6.74%	79.77%
City of Virginia Beach*	\$2,077,926,082	1	72	1.12%	80.90%
Prince George's County	\$3,806,113,603	2	74	2.25%	83.15%
Prince William County	\$1,180,000,000	9	83	10.11%	93.26%
Washington Metro Area Transit Authority	\$3,500,000,000	3	86	3.37%	96.63%
Unidentified		3	89	2.25%	100%

*Note: Even though City of Virginia Beach is not officially a voting member of the MWCOG network, it does have a close relationship with the national capital region in general, and with several jurisdictions in the region in particular, such as Arlington County. Both Virginia Beach and Arlington have a large military presence, and employees from both jurisdictions have frequent communication.

Table 8
Professional Background

Profession	Frequency	Frequency Accumulation	Percentage	Percentage Accumulation
Administration	9	9	10.11%	
Budgeting and Finance	6	15	6.74%	16.85%
Human Resources	9	24	10.11%	26.97%
IT and Information System	8	32	8.99%	35.96%
Parks and Recreation	1	33	1.12%	37.08%
Public Safety	13	46	14.6%	51.69%
	(including one identified as other)			
Public Works	5	51	5.62%	57.30%
Social Services	8 (including on identified as other)	59	8.99%	66.29%
Other				
Communication	6	65	6.74%	73.03%
Planning	10	75	11.24%	84.27%
Environment	3	78	3.37%	87.64%
Law	2	80	2.25%	89.89%
Libraries	2	82	2.25%	92.13%
Other (such as applied science, real estate appraisal, occupational safety, risk management, interdisciplinary, etc.)	7	89	7.87%	100%

Instrument Validation for NCB

As discussed in Chapter 3, in previous research, the instrument used in this study to measure NCB, also known as ICB, developed by Skinner, Autry, and Lamb (2009), had undergone very limited empirical testing. When the instrument was first developed, the authors conducted exploratory factor analysis (EFA), confirmatory factor analysis (CFA), and Cronbach's alpha reliability test to refine the instrument with 298 procurement professionals. The overall model fit statistics of the final CFA was chi-

square (384.22) at degree of freedom of 183, RMSEA (.066), CFI (.895), and NFI (.870). Cutoff values of these indices are included in Table 11. Cronbach's alphas for each of the four subscales were calculated at .846, .702, .734, and .916 for Tolerance, Altruism, Loyalty, and Compliance respectively. No overall Cronbach's alpha was reported.

Because no additional published validation has been reported in the literature since then, the current study first sought to validate the instrument before conducting analysis to answer the research questions. Following the methodology of the original study, both CFA and Cronbach's alpha were tested.

The purpose of factor analysis is to determine "the extent to which measurement overlap – that is, shared variance – exists among a set of variables" (Mertler & Vannatta, 2005, p. 249). In other words, it attempts to determine if different observed variables measure the same phenomenon. There are two types of factor analysis: CFA and EFA. CFA is a theory-driven approach, where the observed variables and the latent variables that they measure are already proposed by existing theories, and the purpose of CFA analysis is to confirm or disconfirm the proposed hypothesis. EFA, on the other hand, is typically used when there is not enough theoretical information to provide a model framework. The goal of EFA is to explore whether any observed variables can be grouped together to measure an unidentified latent variable (Dimitrov, 2005). In this current study, CFA was chosen because the existing NCB measurement serves as theory that provides a model to confirm.

Cronbach's alpha, or stratified alpha established by Cronbach, Shonenmann, and McKie in 1965, was developed to estimate the reliability of multiple items that are grouped into scales on the basis of content (Dimitrov, 2012). It is an estimate of the

internal consistency reliability of observed variables. Kline (2016) recommended that a Cronbach's alpha score of .90 and above is considered to be excellent; .80 and above is considered to be good, and .70 and above is considered to be adequate. In this current study, Cronbach's alpha score was set at .70 to be considered acceptable.

Admittedly, the current sample size was calculated based on power analysis for the purpose of multiple regression analyses, instead of CFA, and CFA normally does require a larger sample size. However, the current research only set out to study the relationships between NCB, IIB, and network structural properties. The validation of the instruments was not an original purpose, but rather an unexpectedly added contribution. As a result, even though the small sample size does posit a limitation to the current study and the results of CFA should be interpreted with caution, the conclusion drawn from CFA using the current sample was still a valuable first step toward a more rigorous validation and improvement of the instrument.

In addition, determining sample size required by CFA is a complicated matter (Kline, 2016). A number of aspects could impact the determination of the recommended sample size, such as the number of factors, the number of indicators, and the magnitude of loadings (Wolf et al., 2013). Among other features, the magnitudes of loading of both the NCB and IIB instruments are quite high. Based on original research, loadings of NCB are mostly over .80 and loadings of IIB are mostly over .70. The higher expected loadings decrease the required sample size, and thus increase the value of CFA conducted using the sample collected in the current study.

Testing assumptions. Before conducting instrument validation, the following statistical assumptions were tested: univariate and multivariate outliers, normality, linearity, homoscedasticity, and multicollinearity (Kline, 2016).

Univariate outlier. A univariate outlier is defined as “a score that is extreme on a single variable” (Kline, 2016, p. 72). Univariate outliers can be detected through transforming the data to z-scores. For a small- to medium-sized sample with less than 100 responses, a z score larger than 3 or smaller than -3 is considered to be an outlier (Mertler & Vannatta, 2005).

Z scores for all individual NCB items were calculated, and one response (response #52) was deleted because it contain five items with a z score lower than -4 (items 3, 4, 9, 10, and 11). Thus, 88 responses were left for NCB instrument validation analysis.

Multivariate outlier. A multivariate outlier is defined as a case that “has extreme scores on two or more variables, or a pattern of scores that is atypical” (Kline, 2016, p. 73). The Mahalanobis distance test is usually used to detect multivariate outliers. A response is considered to be a multivariate outlier if the Mahalanobis distance is significant at the $p < .001$, by comparing the Mahalanobis distance value to the chi-square critical value (Mertler & Vannatta, 2005). In this study, no multivariate outlier was identified for NCB items.

Normality. To use the maximum likelihood estimation, the multivariate normality assumption needs to be met. To determine the multivariate normality of a sample, univariate normality needs to be calculated first. Univariate normality is defined as the degree to which all observations for a given variable are distributed normally (Mertler & Vannatta, 2005). The univariate normality assumption can be tested by the skewness and

the kurtosis tests. The skewness measures the degree of symmetry of a distribution around the mean, whereas the kurtosis measures the degree of peakedness of a distribution (Mertler & Vannatta, 2005). Dimitrov (2012) suggested that a skewness value of 2.0 or less and kurtosis value of 7.0 or less would meet the normality assumption.

In this current study, skewness and kurtosis tests were conducted for all 11 observed variables of NCB, and no skewness score higher than 2.0 and no kurtosis score higher than 7.0 were detected. Thus, the univariate normality assumption for the NCB instrument is met.

Multivariate normality refers to “the extent to which all observations in the sample for all combinations of variables are distributed normally” (Mertler & Vannatta, 2005). Univariate normality is a necessary but not sufficient condition for multivariate normality (Mertler & Vannatta, 2005), and it is impractical to examine all possible bivariate normality distribution (Kline, 2016). As a result, in the current study, it is assumed that the multivariate normality assumption is met.

Linearity. The assumption of linearity proposes that a linear relationship exists between two variables. The linearity assumption can be tested by observing the scatterplots between observed variables. For the current study, a scatterplot of the relationship matrix of all possible combinations of NCB items was created, and it was concluded that the linearity assumption is met.

Homoscedasticity. Homoscedasticity is the assumption that “the variability in scores for one continuous variable is roughly the same at all values of another continuous variable” (Mertler & Vannatta, 2005, p. 34). The homoscedasticity assumption can be

tested by observing the residual scatterplot. A residual scatterplot was conducted for each observed variable against all other variables, and it was concluded that the homoscedasticity assumption was met.

Multicollinearity. Multicollinearity can occur when independent variables are highly correlated with each other. Multicollinearity can be tested by tolerance and variance inflation factor (VIF). A tolerance value $< .10$ or a VIF value > 10.0 indicates multicollinearity (Kline, 2016). In the current study, no tolerance value smaller than .10 and no VIF value larger than 6 was detected, so it is assumed that the multicollinearity assumption was met.

CFA for NCB instrument validation: First iteration. Based on the original measurement developed by Skinner, Autry, and Lamb (2009), a model is identified as demonstrated by Figure 3. The current CFA model did not yield a satisfactory result, with the Chi-square estimated at 98.006 with a degree of freedom of 38, RSMEA (.133), CFI (.836), GIF (.849), NNFI (.762), and SRMR (.088). Figure 4 outlines factor loadings and standard errors of the first CFA model for NCB instrument validation, and Table 10 provides a list of all model fit indices and their comparisons to cutoff values. The cutoff values were identified based on Dimitrov (2012) and Hancock and Mueller (2010). As a result of the less than satisfactory model fit, modification recommendations were sought to improve the quality of the NCB instrument.

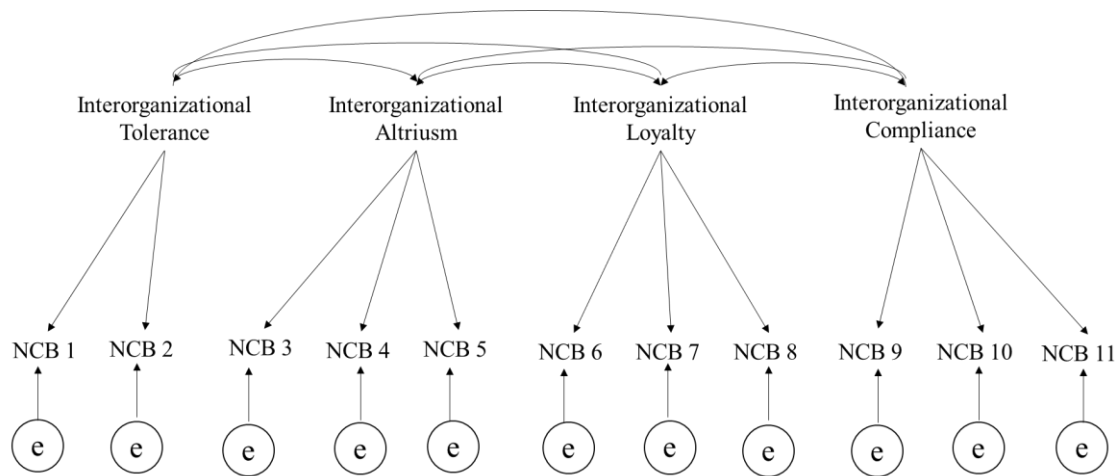


Figure 3: First CFA Model for NCB Instrument Validation based on Skinner, Autry, and Lamb (2009)

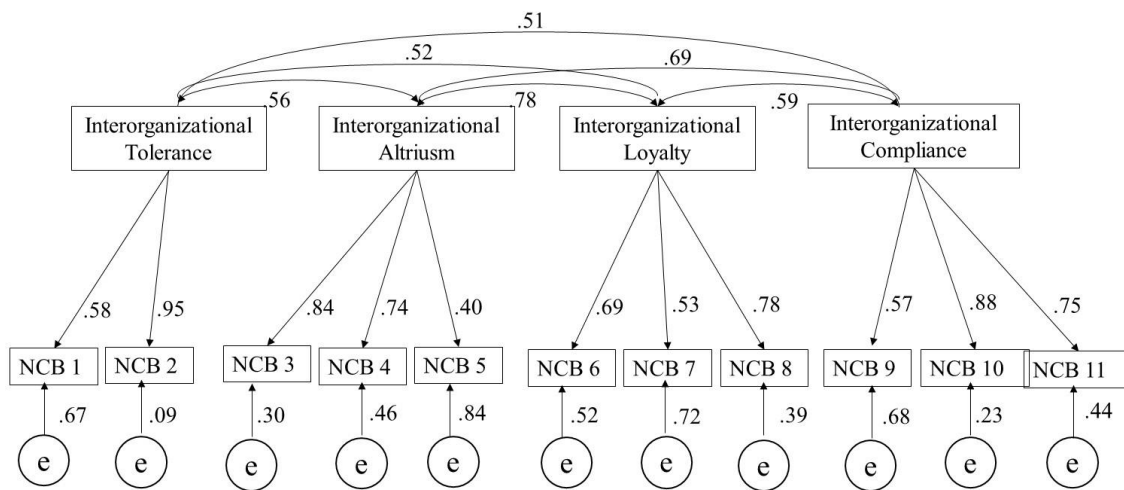


Figure 4: Factor Loadings of the first CFA Model for NCB Instrument Validation

As Figure 4 demonstrated, a few factor loadings are less than .60, namely items 1, 5, 7, and 9. The first iteration of the CFA model also provided a few recommendations, which are demonstrated in Figure 5. Specifically, item 7 loads onto the latent variable of Tolerance instead of Loyalty; and item 9 loads onto Altruism instead of Compliance.

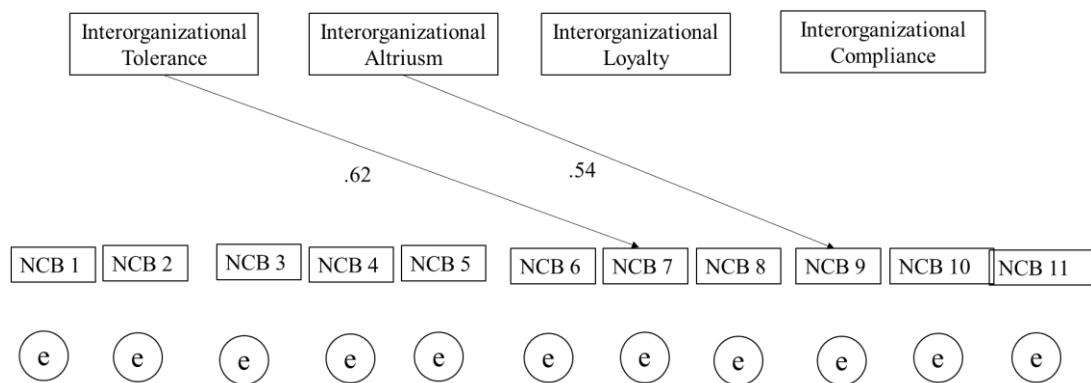


Figure 5: Model Modification Recommendations for NCB Instrument Validation

Cronbach's alpha reliability tests. In addition to factor analysis, Cronbach's alpha reliability tests were also used to determine the internal consistency of the instrument and to improve its overall quality. A Cronbach's alpha reliability test was conducted for all 11 items of NCB, which was calculated at .828. Among all items, if item 1 were deleted, the Cronbach's alpha score would increase to .832, and if item 5 were deleted, the Cronbach's alpha value would increase to .834. As a result, items 1 and 5 were deleted from subsequent analysis.

Next, a Cronbach's alpha score was calculated for each of the four latent variables, and additional exploratory tests were conducted based on the modification recommendations of CFA and the overall NCB Cronbach's alpha. Specifically—

Tolerance. In the original instrument, Tolerance was measured by items 1 and 2, and the original Cronbach's alpha of the Tolerance subscale was calculated at .681. First, item 1 was deleted because the overall NCB's Cronbach's alpha score would significantly increase from .828 to .832 if item 1 is deleted. Next, based on the modification recommendation of the CFA analysis, item 7, which originally belonged to the Loyalty subscale, was added to the tolerance subscale (factor loading .62). Item 2 and

item 7 together generated a new Cronbach's alpha score of .639. This was lower than the original Tolerance subscale Cronbach's alpha, and also lower than the usually recommended score of .70 (Kline, 2016). Thus, item 7 was deleted from the Tolerance subscale. As a result, only item 2 was left to measure Tolerance.

Having only one item to measure a particular subscale does pose a limitation, because the reliability of the Tolerance scale cannot be tested. However, this one item was kept to measure Tolerance, because it is a critical component of NCB, and the current research is interested in investigating the relationship between NCB as well as all four of its subscales and IIB.

Altruism. In the original instrument, Altruism was measured by items 3, 4, and 5, and the original Cronbach's alpha for the Altruism subscale was .612. First, item 5 was deleted from the Altruism subscale because the Cronbach's alpha score for the overall NCB instrument would increase to .834 from .828 without item 5. Without item 5, items 3 and 4 provided a new Cronbach's alpha score of .765, higher than the original number, which confirmed that deleting item 5 was the right decision. Next, item 9, which originally belonged to the Compliance subscale, was added to the Altruism subscale based on the modification recommendation of the CFA analysis (factor loading .54). Updated Cronbach's alpha score for items 3, 4, and 9 was calculated at .768, higher than the original score for the Altruism subscale. As a result, items 3, 4, and 9 were left to measure Altruism.

Loyalty. In the original instrument, Loyalty was measured by items 6, 7, and 8, and the original Cronbach's alpha for the Loyalty subscale was .669. Item 7 was deleted from the Loyalty subscale because it was suggested to be reloaded to the Tolerance

subscale based on modification recommendations that the CFA analysis provided.

Deleting item 7 increased the subscale Cronbach's alpha to .726. As a result, items 6 and 8 are left to measure Loyalty.

Compliance. In the original instrument, Compliance was measured by items 9, 10, and 11, and the original Cronbach's alpha of the Compliance subscale was calculated at .744. Based on the modification recommendation of the CFA analysis, item 9 should be reloaded to the Altruism subscale, thus being deleted from the Compliance scale. The new Cronbach's alpha score for item 10 and item 11 increased to .790. As a result, items 10 and 11 were left to measure Compliance.

Table 9 describes the original NCB instrument, the modifications that were done to all the modified items, the newly updated NCB instrument version 2 (NCB 2.0), and the Cronbach's Alpha comparisons of all subscales between the original NCB and NCB 2.0. After all the exploration and items reloading described above, the NCB measurement is left with 8 items. The new Cronbach's alpha score for the updated instrument increased to .840.

Table 9

Item and Cronbach's Alpha Scores Comparison of Original and Updated NCB Instruments

Original NCB Instrument (Cronbach's alpha: .828)	Updated Instrument (NCB v. 2.0) (Cronbach's alpha: .840)
Tolerance (Cronbach's alpha: .681) 1. We accept the inevitable mistakes that our partner sometimes makes without complaining. (deleted) 2. When our partner is unable to fulfill certain responsibilities, we try to be understanding.	Tolerance (Cronbach's alpha: N/A) 2. When our partner is unable to fulfill certain responsibilities, we try to be understanding.
Altruism (Cronbach's alpha: .612) 3. We go out of our way to help our partner with business-related issues if we sense that they are in need. 4. We provide assistance to our partner if they ever have a problem in an area where we have expertise. 5. We sometimes volunteer our resources in order to lighten our partner's workload. (deleted)	Altruism (Cronbach's alpha: .768) 3. We go out of our way to help our partner with business-related issues if we sense that they are in need. 4. We provide assistance to our partner if they ever have a problem in an area where we have expertise. 9. We attempt to meet or exceed any standards that our partner is also required or expected to meet.
Loyalty (Cronbach's alpha: .669) 6. We tell others that our partner is a good organization with whom to do business. 7. We sometimes sacrifice our own best interests in order to provide additional benefits to our partner. (deleted) 8. We promote our partner's services to others.	Loyalty (Cronbach's alpha: .726) 6. We tell others that our partner is a good organization with whom to do business. 8. We promote our partner's services to others.
Compliance (Cronbach's alpha: .744) 9. We attempt to meet or exceed any standards that our partner is also required or expected to meet. (reloaded to Altruism) 10. When dealing with our partner, we comply with expected standards of social behavior. 11. We show respect for our partner's organizational policies and try to abide by them ourselves.	Compliance (Cronbach's alpha: .790) 10. When dealing with our partner, we comply with expected standards of social behavior. 11. We show respect for our partner's organizational policies and try to abide by them ourselves.

New CFA. Based on the updated NCB measurement, a new model was identified to conduct a second CFA. Figure 6 demonstrates the loading relationship of the new model.

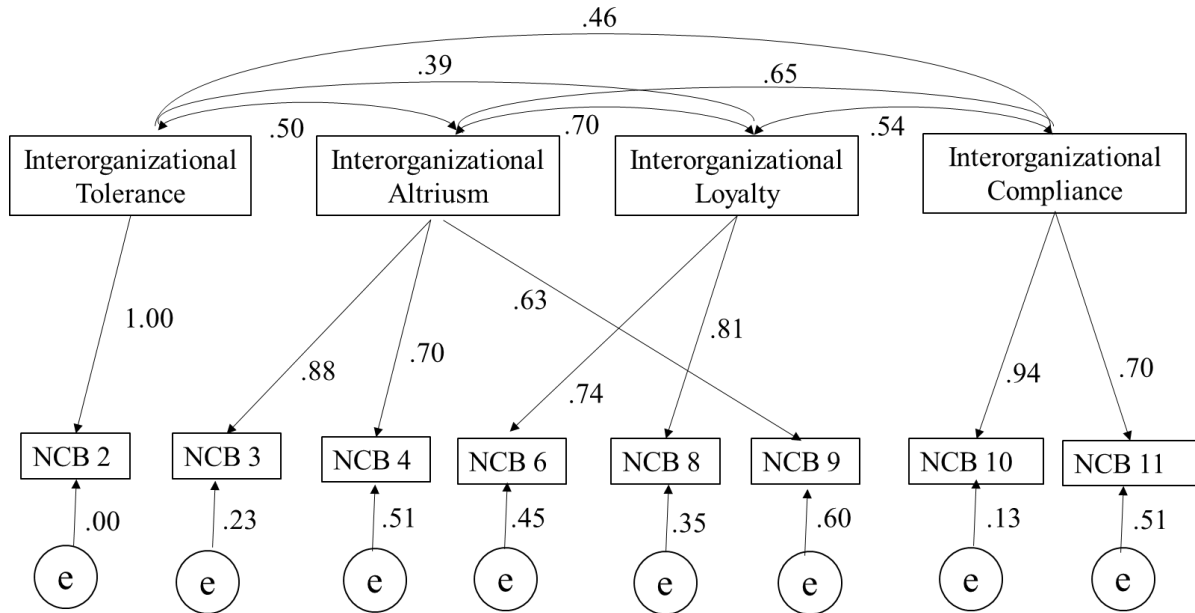


Figure 6: New CFA for NCB 2.0

The new model provided much better results for the updated, NCB 2.0 measurement. Table 10 provides model fit indices and its comparisons to the first CFA model as well as the cutoff values. The cutoff values were identified based on the recommendations of Dimitrov (2012) and Hancock and Mueller (2010). All model fit indices have significantly increased from the original NCB to NCB 2.0. NCB 2.0 provided satisfactory indices for CFI (.962), GIF (.945), SRMR (.0466), and Chi-square to degree of freedom (1.644), which were not satisfactory in the original model. The other indices, NNFI (.929) and RMSEA (.0861), even though they are still out of the range of a good fit, are much closer to the cutoff value than the original model. In particular, the still poor RMSEA score can be explained by the fact that RMSEA could be

misleading when the degrees of freedom are too small and the sample size is too small (n<100) (Kline, 2016).

In conclusion, the newly updated NCB 2.0 measurement with eight items was used to test the hypotheses and to respond to the research questions.

Table 10
CFA Model Fit Indices Comparison

Indicators	Cutoff values	NCB 1 (original)	NCB 2 (updated)
GIF	>.90	0.849	.945
AGIF	>.95	0.739	.868
Chi-square		98.006	24.663
df		38	15
Chi-square to df	<2	2.579	1.644
p value	<.05	p=.0001	p=.0547
CFI	>.95	0.836	0.962
NNFI	>.95	0.762	0.929
SRMR	<.08	0.0880	0.0466
RMSEA	<.08	0.133	0.0861
RMSEA	[0, .08]	[0.101, 0.166]	[0.0, 0.145]
Confidence intervals			

Note: Cutoff points identified based on Dimitrov (2012) and Hancock & Mueller (2010). GIF: Goodness of Fit; AGIF: Adjusted Goodness of Fit; CFI: Comparative Fit Index; NNFI: Non-Normed Fit Index, aka Tucker-Lewis Index (TLI); SRMR: Standardized Root Mean Square Residual; RMSEA: Root Mean Square Error of Approximation.

Instrument Validation for IIB

In addition to NCB as the independent variable, the dependent variable, IIB, was also measured by an existing instrument, and required another rigorous validation process. Similar to NCB, the instrument used in this study to measure IIB, developed by Kleysen and Street (2001), has received a limited number of validations. When the instrument was first introduced, the authors conducted EFA to eliminate items and create latent variables, as well as CFA to test its reliability. A revised CFA reported chi-squared = 715.1 (p<.001), chi-square/df = 10.07, CFI = .718, GFI = .724, AGFI = .592, and RSMEA = -.201. All of the model fit indicators were not satisfactory, suggesting the

items did not indicate latent variables as the theory suggested. The authors also conducted a Cronbach's alpha coefficient test and produced a score of .945 for IIB as a unidimensional scale. Thus, Kleysen and Street (2001) concluded that evidence for latent factors was not sufficient in the model and the instrument should be used as a unidimensional measurement.

Subsequently, a limited number of additional studies also used Kleysen and Street (2001)'s IIB measurement (Kim & Park, 2015; Mishra et al., 2017; Wojtczuk-Turek & Kurek, 2015; Wojtczuk-Turek & Turek, 2016), and all the research confirmed IIB to be a unidimensional variable. Mishra et al. (2017) reported a Cronbach's alpha of .88 (n = 398 service organization employees), and Kim and Park (2015) reported a Cronbach's alpha value of .94 (N = 347) registered nurses in six hospitals in South Korea), but no CFA was conducted for either study. Sojtczuk-Turek and Turek used a sample of 166 service company employees in Poland and 70 graduate students in Warsaw School of Business. They reported a Cronbach's alpha of .94, and a CFA confirmed satisfactory result for a single-factor IIB model (Chi Square = 47.950; df = 37; p = .107; RMSEA = .042; CFI = .993; GFI = .954; NFI = .966; TLI = .987; Factor loadings ranged between .669 and .833).

As a result of the less than satisfactory reporting of the instrument's subscales' reliability, a similar validation process was conducted for IIB using CFA and Cronbach's alpha, in the same manner as NCB.

Testing assumptions. One univariate outlier was identified based on z scores of all IIB items (response #5 with z scores lower than -4 for items 2, 3, and 9), and one multivariate outlier was identified based on the Mahalanobis distance test (response 40,

$p < .001$). A total number of $N = 87$ cases were used for analysis of IIB's reliability and validity. The normality assumption was met based on skewness and kurtosis tests for each observed variable. The linearity assumption was met based on scatterplot graphs of each possible pair of observed variables. The homoscedasticity assumption was met based on residual scatterplot graphs for each possible pair of observed variables. All tolerance values were larger than .10 and all VIF values were smaller than 6, confirming that the multicollinearity assumption was met.

CFA for IIB instrument validation: First iteration. Based on the framework of the original measurement developed by Kleysen and Street (2001), a model was identified, demonstrated by Figure 7. The current CFA model did not yield a satisfactory result, with Chi-Square estimated at 125.299 with a degree of freedom of 67, $p = .00002$, RSMEA at .100, GFI at .856, CFI at .925 and NNFI = .898. Therefore, modifications to the model was sought attempting to improve the overall quality of the IIB instrument.

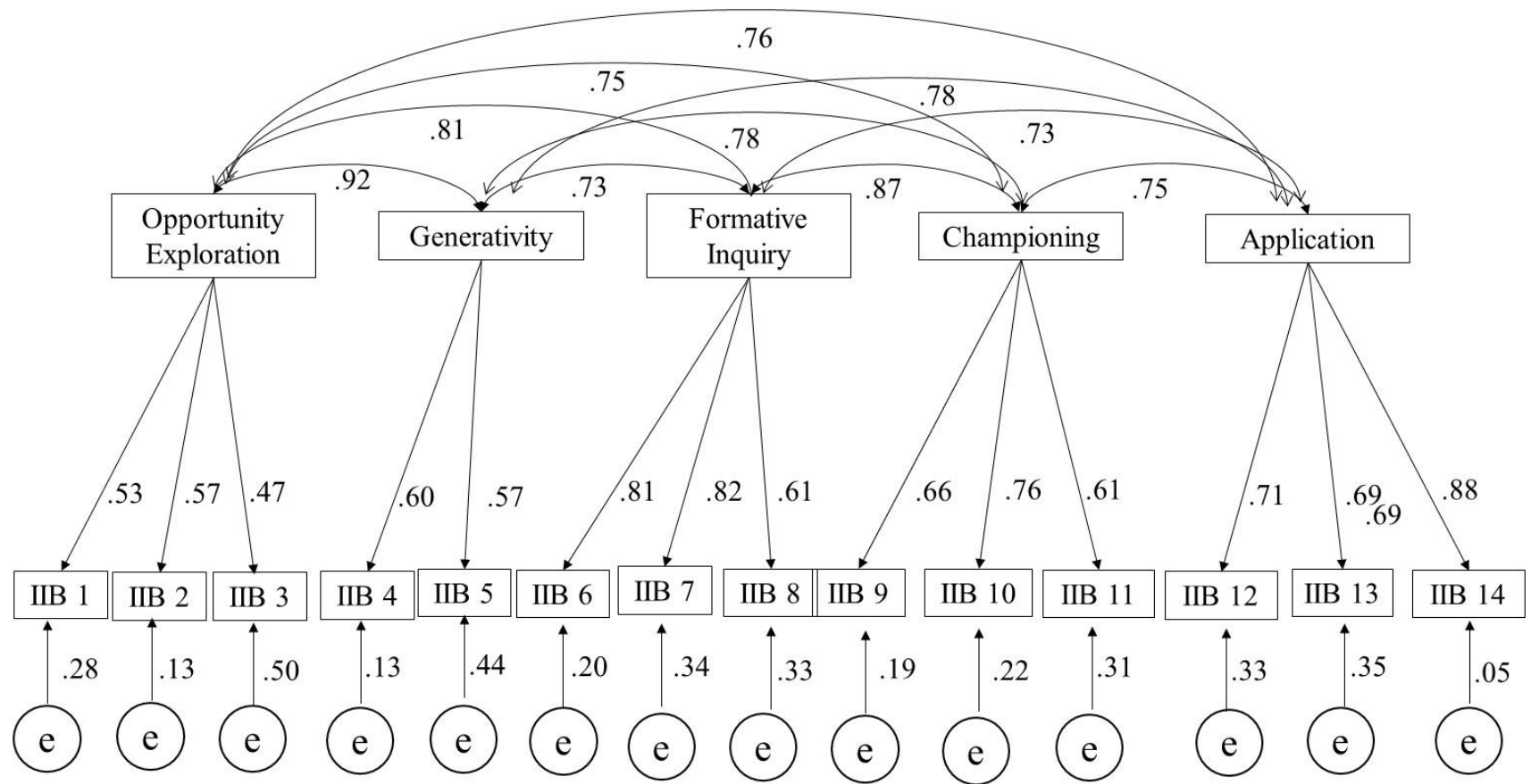


Figure 7: First CFA Model for IIB Instrument

Cronbach's alpha reliability test. The IIB instrument's overall Cronbach's alpha coefficient score is calculated at .932. The only change that IIB's overall Cronbach's alpha score suggested was the elimination of item 3. When item 3 was deleted, the Cronbach's alpha coefficient value increased to .933. As a result, a slightly modified IIB instrument without item 3 was used for subsequent analysis.

Subscale Cronbach's alpha.

Opportunity exploration. In the original instrument, Opportunity Exploration was measured by items 1, 2, and 3, and the original Cronbach's alpha for the Opportunity Exploration subscale was calculated at .719. Item 3 was deleted from the subscale because doing so increased the Cronbach's alpha value of this subscale to .776. As a result, only items 1 and 2 were left to measure Opportunity Exploration.

Generativity. In the original instrument, Generativity was measured by items 4 and 5, and the original Cronbach's alpha score for the generativity subscale was calculated at .705. Based on the modification recommendation of the CFA analysis, item 11, which originally belonged to the Championing subscale, was reloaded to Generativity subscale. However, the updated Cronbach's alpha score for items 4, 5, and 11 decreased to .664, lower than the .70 cutoff value (Kline, 2006). As a result, item 11 was not reloaded to generativity, and only items 4 and 5 were left to measure Generativity.

Formative inquiry. In the original instrument, Formative Inquiry was measured by items 6, 7, and 8, and the original Cronbach's alpha is .846. Adding item 12 increased the Cronbach's alpha to .854.

Championing. In the original instrument, Championing was measured by items 9, 10, and 11, and the original Cronbach's alpha is .844.

Application. In the original instrument, Application was measured by items 12, 13, and 14, and the original Cronbach's alpha is .861.

CFA: Model modifications. The CFA estimation provided several modification recommendations, as demonstrated by Figure 8. Particularly, item 11 could load onto Application and Formative Inquiry instead of Championing, and item 12 could load onto Formative Inquiry and Championing instead of Application.

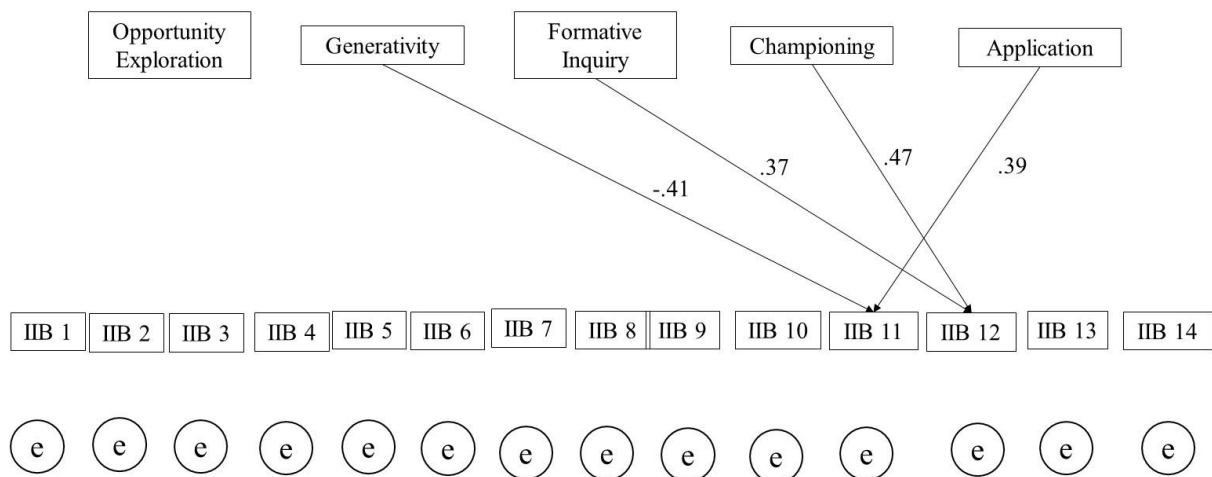


Figure 8: Model Modification Recommendations for CFA for IIB Instrument Validation

Depending on how items 11 and 12 were loaded, five additional models could be identified, in addition to loadings of the original model:

1. Both items 11 and 12 are loaded onto Championing
2. Both items 11 and 12 are loaded onto Application
3. Item 11 loads onto Application and item 12 loads onto Championing
4. Item 11 loads onto Application and item 12 loads onto Formative Inquiry
5. Item 11 loads onto Championing and item 12 loads onto Formative Inquiry

A CFA model for each of these five scenarios was conducted, and Table 11 presents the model fit indices of the original model as well as all five additional models

based on modification recommendations, and their comparisons to cutoff values. The cutoff values were identified based on Dimitrov (2012) and Hancock and Mueller (2010).

As Table 11 showed, even after several iterations of CFA, almost none of the goodness-of-fit indices were satisfactory. Model 1 and Model 2 had slightly better model fit results, providing satisfactory SRMR and Chi-square to degree of freedom values, but all other indicators (GIF, CFI, NNFI, and RMSEA) of these two models were still not acceptable. This is consistent with prior literature and confirms that IIB is a unidimensional instrument.

In conclusion, the validation process confirmed that IIB is a single factor, unidimensional instrument, and no subscales were identified. The only modification that was confirmed in the validation process was the deletion of item 3, to increase the Cronbach's alpha score. As a result, item 3 was deleted, leaving IIB as a 13-item unidimensional measurement to be used in answering the research questions.

Table 11

IIB Confirmatory Factor Analysis Model Fit Indices Comparison

Indicators	Cutoff values	Original Model	Model 1	Model 2	Model 3	Model 4	Model 5
GIF	>.95	Not available	.856	.859	.841	.843	.844
Chi-square		125.299	100.132	96.253	113.447	115.360	109.411
df		67	55	55	55	55	55
Chi-square to df	<2	Not available	1.82*	1.75*	2.06	2.10	2.00
p value	<.05	.0002	.0002	.0005	.0000	.0000	.0000
CFI	>.95	.925	.939	.944	.921	.918	.926
NNFI	>.95	.898	.913	.921	.888	.884	.895
SRMR	<.08	.0676	.0571*	.0563*	.0614	.0653	.0589
RMSEA	<.06	.100	.0971	.0929	.111	.112	.107
RMSEA Confidence intervals	[0, .08]	[.0725, .127]	[.0661, .127]	[.0610, .123]	[.0815, .139]	[.0835, .141]	[.0771, .136]

Note: Cutoff points identified based on Dimitrov (2012) and Hancock and Mueller (2010); GIF: Goodness of Fit; CFI: Comparative Fit Index; NNFI: Non-Normed Fit Index, aka Tucker-Lewis Index (TLI); SRMR: Standardized Root Mean Square Residual; RMSEA: Root Mean Square Error of Approximation; Model 1: both items load onto Championing; Model 2: both items load onto Application; Model 3: item 11 loads onto Application, item 12 loads onto Championing; Model 4: item 11 loads onto Application, item 12 loads onto Formative Inquiry; Model 5: item 11 loads onto Championing; item 12 loads onto Formative Inquiry. *: satisfactory model fit indices

Research Question 1: NCB and IIB

Research question 1 asks if a relationship exists between NCB as well as its four subscales and IIB, and hypothesis 1 stated “the stronger the exhibition of NCB and its four subscales, the stronger the exhibition of IIB.” To answer this research question and to test this hypothesis, the dependent variable was identified as the updated 13-item measurement of IIB, and the independent variable of the model was identified as the updated 8-item instrument NCB 2.0 and four subdimensions.

Among the seven demographic variables collected through survey or archival data, five of them were used as control variables. Jurisdictional background and professional background contained too many categories and would not be able to provide meaningful results as control variables, thus were excluded from further analysis. Among the five control variables, boundary spanning is considered to be a categorical variable with two categories, and the others are considered to be continuous variables.

Therefore, a total of 10 independent variables were identified to conduct assumption and hypothesis testing: age, tenure, boundary spanning status, managerial level, organizational budget, NCB-total, NCB-tolerance, NCB-altruism, NCB-loyalty, and NCB-compliance.

Testing of assumptions. Z scores were calculated for each independent variable, and two responses were deleted as univariate outliers (Response 52 has a z score of -5.555 for NCB, and response 5 has a z score of -4.020 for IIB, both lower than the cutoff value of -4). Response 40, although it was considered to be an outlier when conducting CFA for IIB, met the z score requirement for the regression analysis of research question 1, thus it was kept in the remainder of this analysis. No multivariate outlier was identified

from the Mahalanobis distance test. A total number of $n = 87$ responses were left for additional analysis, larger than the required sample size of 82 based on power analysis.

Based on skewness and kurtosis tests, all independent variables were normally distributed. Based on scatterplot graphs, all independent variables had a linear relationship with each other. The homoscedasticity assumption was met based on residual scatterplots for each possible independent variable pair, and multicollinearity assumption was met because all tolerance values were larger than .10 and all VIF values were smaller than 6.

Regression analysis. First, a simple regression model was conducted to test if NCB significantly predicted IIB. The results of the regression indicated that NCB explained 17.8% of the variance in IIB ($R^2 = 17.8$, $F(1, 85) = 18.436$, $p < .05$).

Next, a multiple regression analysis was conducted to test if NCB significantly predicted IIB after five demographic variables are controlled. The results of the multiple regression indicated that together with the four demographic variables, NCB predicted 30.5% of the variance in IIB ($R^2 = .305$, $F(6, 75) = 5.485$, $p < .001$). However, only boundary spanning was a significant predictor of IIB ($\beta = -.237$, $t = -2.339$, $p = .022$), whereas age ($\beta = -.114$, $t = -.938$, $p = .351$), tenure ($\beta = .186$, $t = 1.525$, $p = .131$), management level ($\beta = .013$, $t = .123$, $p = .902$), and budget ($\beta = -.190$, $t = -1.915$, $p = .059$) were all not significantly predicting IIB.

Thus, nonsignificant control variables were deleted, and only boundary-spanning status was left as a control variable. After controlling boundary-spanning status, it was discovered that the NCB explains even less variance in IIB, only 11.56% ($\beta = -.221$, t

= -2.194, $p = .031$, partial $r^2 = .340$). Boundary-spanning status had a negative impact on the exhibition of IIB.

A separate series of simple regression analysis was conducted for each of the subscales for NCB. Specifically, Tolerance was found to significantly predict IIB, but explained only a small fraction of the IIB variance ($R^2 = .046$, $F(1, 85) = 4.080$, $P = .047$). Altruism explained about 10.8% of the total IIB variance ($R^2 = .1108$, $F(1, 85) = 10.249$, $P = .002$). Loyalty explained about 14.9% of total IIB variance ($R^2 = .149$, $F(1, 85) = 14.903$, $P < .05$). Compliance explained about 11% of total IIB variance ($R^2 = .110$, $F(1, 85) = 10.475$, $P = .002$). Thus, hypothesis 1 was supported.

Research Question 2: Network Data

The second research question explored how social network structural properties, measured by network centrality and tie strength, impact the exhibition of NCB, IIB, and the relationship between the two variables. To answer the second research question, network data were first prepared and transformed, and then a series of multiple correlation analyses and regression analyses were conducted.

Network data calculation. Three responses were deleted because they did not specify which jurisdictions they were from (responses 27, 55, and 56), so a total of $n = 84$ was used for subsequent analysis related to network. A weighted network matrix was established and a communication network map among these 19 jurisdictions is presented in Figure 9.

Seven types of network structural properties were identified and calculated for each jurisdiction. Consistent with the literature, each of the structural properties was defined and calculated as follows:

Degree centrality: Degree-centrality is defined as the extent of network ties of an actor among the rest of the network. Degree centrality is directional: in-degree centrality means the extent of other network nodes that are linked to a given actor, whereas out-degree centrality means the extent of a given actor that links to other network nodes. Based on Wasserman and Faust (1994), degree centrality is calculated as

$$C_D(n_i) = \frac{\sum_j x_{ij}}{n - 1}$$

where x_j represents the number of links node j has.

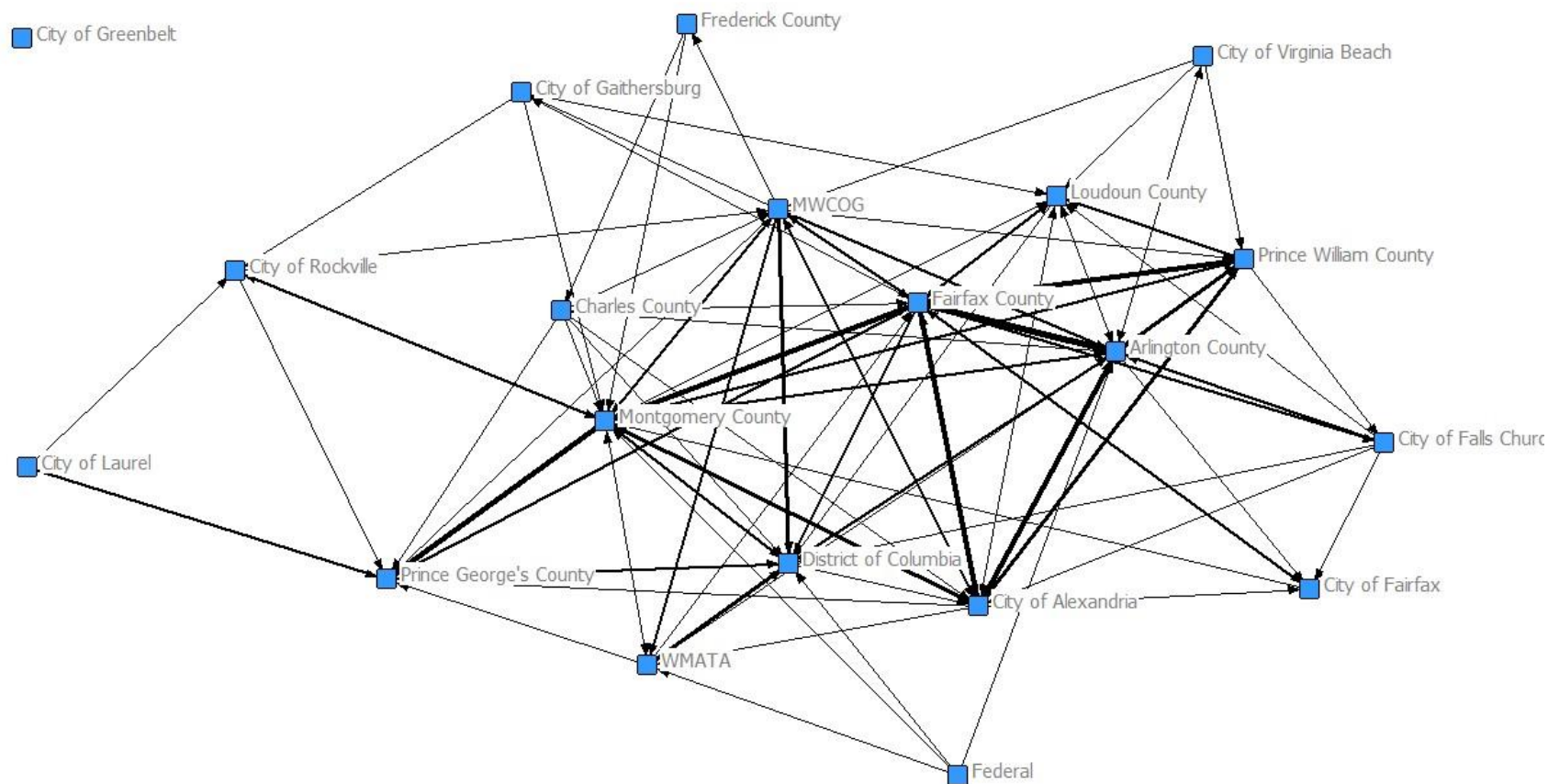
Betweenness Centrality is defined as the degree that an actor “sits” in between all other pairs of random actors. Betweenness centrality is nondirectional. Based on Wasserman and Faust (1994), betweenness centrality is calculated as:

$$C_B(n_i) = \sum_{j < k} g_{jk}(n_i) / g_{jk}$$

where g_{jk} represents the number of geodesic steps linking the two actors.

Closeness centrality focuses on how close an actor is to all the other actors in the network. Closeness centrality is directional, where in-closeness centrality measures how easily all other nodes in a network can reach a given actor, and out-closeness centrality

Figure 9
Network Map



measures how easily an actor can reach all other nodes in the network. Based on Wasserman and Faust (1994), closeness centrality is calculated as

$$C_c(n_i) = \left[\sum_{j=i}^g d(n_i, n_j) \right]^{-1}$$

where $d(n_i, n_j)$ means the number of lines in the geodesic linking actors i and j . The standard closeness centrality calculation only applies to connected networks. For unconnected nodes, undefined distance is set to be the number of nodes in the network.

Tie strength refers to the nature of the relationship between two actors, which combines time investment, emotional intensity, intimacy, and reciprocal services (Granovetter, 1973). In this current research, tie strength is operationalized through ranking of communication frequency. Tie strength is directional: in-degree tie strength is calculated as the accumulation of all relationship rankings that an actor receives, and out-degree tie strength is calculated as the accumulation of all relationship rankings that an actor provides.

Three types of software were used to conduct analyses related to social networks. UCINET is the most widely known and commonly used statistical analysis software for conducting social network analysis, developed by social network researchers Freeman, Everett, and Borgatti. UCINET was used to calculate all the centrality measures. NetDraw is a network visualization tool developed by Borgatti that transforms network data into network configuration. In addition, Microsoft Excel was used to calculate tie strength. A full list of these network structural properties and their values for each of the 19 jurisdictions is presented in Table 12.

Table 12

Network Structural Properties of Jurisdictions

	Betweenness Centrality	Out-Degree Centrality	In- Degree Centrality	In- Closeness Centrality	Out- Closeness Centrality	In-Degree Tie strength	Out-Degree Tie strength
Arlington County	34.033	7.681	18.919	42.222	21.839	54	133
Charles County	4.833	3.841	.284	22.892	22.353	27	2
City of Alexandria	22.375	10.811	11.38	39.583	22.093	76	80
City of Fairfax	.658	.711	2.418	32.759	20.213	5	17
City of Falls Church	0.5	4.267	1.565	30.159	21.111	30	11
City of Gaithersburg	2.083	1.565	1.138	31.667	20	11	8
City of Greenbelt	0	0	0	5	5	0	0
City of Laurel	0	1.991	0	5	21.591	14	0
City of Rockville	14.75	1.849	0.996	32.759	20.43	13	7
City of Virginia Beach	.75	1.991	0.142	30.645	21.111	15	2
District of Columbia	18.367	2.134	11.238	41.304	20.43	23	87
Fairfax County	40.117	21.053	17.212	39.583	23.171	148	121
Frederick County	17	.996	.427	28.358	20.213	7	3
Loudoun County	6.408	2.134	6.543	38	21.111	15	46
Montgomery County	65.192	12.091	10.242	43.182	22.892	97	84
MWCOG	51.875	9.673	3.698	37.255	23.171	68	26
Prince George's County	2.75	1.138	8.962	40.426	17.593	8	63
Prince William County	6.733	11.522	3.983	36.538	20.897	81	28
WMATA	1.575	3.841	1.991	35.185	21.111	27	14

Correlation analysis. When structural properties were considered on the organizational level, given the small sample size ($n = 19$), Spearman's rho and Pearson's correlation analyses were used instead of a regression analysis. The average scores of the following six variables were calculated for each jurisdiction: NCB total, NCB-Tolerance, NCB-Altruism, NCB-Loyalty, NCB-Compliance, and IIB. A Spearman's rho correlation test and a Pearson's correlation test were conducted for the seven network structure properties and the six NCB and IIB measurements. Table 13 and 14 demonstrate the results of the two correlation analyses. No statistically significant findings between network structural properties and the behaviors were found.

Multiple regression. In its essence, the current research study examines phenomena on an individual level instead of on organizational or multiorganizational levels. As a result, even though network structural properties are calculated for each jurisdiction, it is necessary to transform the data back to individual responses to study the role that network structural properties play on NCB, IIB, and their relationships. Therefore, the network structural properties for each jurisdiction were matched back to the original individual responses. Respondents from the same jurisdiction have the same values for their network structural properties. These network structural properties were identified as the independent variable, and the exhibition of IIB was identified as the dependent variable. Organizational budget was added as a control variable.

Table 13

Spearman's Rho Correlations

		Spearman's Rho Correlations						
		Betweenness Centrality	Out- Degree Centrality	In- Degree Centrality	In- Closeness Centrality	Out- Closeness Centrality	In-Degree Tie Strength	Out- Degree Tie Strength
NCB-T	Correlation Coefficient	-.163	-.124	-.215	-.153	-.141	-.096	-.189
	Sig. (2-tailed)	.504	.613	.376	.532	.566	.695	.439
NCB-A	Correlation Coefficient	.084	.049	.242	.259	-.208	.076	.253
	Sig. (2-tailed)	.733	.841	.318	.283	.392	.756	.296
NCB-L	Correlation Coefficient	-.123	-.319	-.195	-.187	-.419	-.283	-.167
	Sig. (2-tailed)	.616	.183	.423	.444	.074	.240	.494
NCB-C	Correlation Coefficient	.189	-.024	.193	.101	-.168	-.018	.199
	Sig. (2-tailed)	.439	.923	.428	.681	.492	.943	.415
NCB- TTN157	Correlation Coefficient	.022	-.043	.052	.076	-.236	-.004	.075
	Sig. (2-tailed)	.929	.862	.833	.756	.331	.989	.761
IIB-N3	Correlation Coefficient	.417	.224	.324	.291	.131	.232	.295
	Sig. (2-tailed)	.076	.357	.176	.227	.594	.340	.220

No statistically significant findings were identified.

Table 14
Pearson's Correlations

		Correlations						
		Betweenness Centrality	Out-Degree Centrality	In-Degree Centrality	In-Closeness Centrality	Out-Closeness Centrality	In-Degree Tie Strength	Out-Degree Tie Strength
NCB-T	Pearson Correlation	0.007	-0.028	-0.007	-0.152	-0.136	-0.018	0.001
	Sig. (2-tailed)	0.979	0.910	0.977	0.536	0.578	0.940	0.997
NCB-A	Pearson Correlation	-0.015	0.055	0.150	.511*	0.181	0.039	0.133
	Sig. (2-tailed)	0.951	0.823	0.539	0.025	0.459	0.873	0.587
NCB-L	Pearson Correlation	-0.183	-0.101	-0.111	0.066	-0.246	-0.105	-0.113
	Sig. (2-tailed)	0.452	0.680	0.651	0.788	0.311	0.669	0.645
NCB-C	Pearson Correlation	0.099	0.059	0.182	0.065	-0.027	0.045	0.166
	Sig. (2-tailed)	0.685	0.812	0.457	0.791	0.914	0.855	0.498
NCB- Total	Pearson Correlation	-0.054	-0.010	0.065	0.228	-0.064	-0.019	0.055
	Sig. (2-tailed)	0.825	0.969	0.790	0.348	0.795	0.938	0.823
IIB- Total	Pearson Correlation	0.210	0.356	0.353	0.266	0.326	0.340	0.335
	Sig. (2-tailed)	0.389	0.135	0.138	0.272	0.173	0.155	0.160

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

Two responses were deleted for being univariate outliers (responses 16 and 20) for Out-Degree Centrality (response 20 also for In-Degree Centrality), leaving a total number of $n = 82$ responses. No additional multivariate outlier was identified. The normality assumption was met. The linearity assumption was met. The multicollinearity assumption was not met because several network structural properties (In- and Out-Closeness Centrality, and In- and Out-Degree Tie Strength) significantly correlate with each other.

A series of multiple regressions was conducted to test whether these network structural properties predict IIB. Table 15 summarizes the results of the multiple regressions. As the table demonstrates, closeness centrality and tie strength were significant predictors of innovative behaviors, after controlling for jurisdictional budget. Specifically, after controlling for budget, In-Closeness Centrality predicts 7.24% of the variance in innovative behaviors ($R^2 = .108$, $F(2,79) = 4.799$, $p = .0013$, partial $r = .269$); Out-Closeness Centrality predicts 7.34% of the variance in innovative behaviors ($R^2 = .109$, $F(2,79) = 4.853$, $p = .012$, partial $r = .271$); In-Degree Tie Strength explains 6.66% of the variance in innovative behaviors ($R^2 = .102$, $F(2, 79) = 4.493$, $p = .018$, partial $r = .258$); and Out-Degree Tie Strength explains 6.76% of variance in innovative behaviors ($R^2 = .103$, $F(2,79) = 4.541$, $p = .017$, partial $r = .26$).

Moderator effects. The research question was interested in exploring whether NCB and network structural properties interact with each other when predicting IIB, thus, a series of multiple regression analyses were conducted to see if network structural properties were statistically significantly moderating the relationship between NCB and

IIB. However, as Table 16 demonstrates, none of the moderating relationships were found to be significant.

Table 15

Network Structural Properties and IIB

	Predicting IIB			Controlling Budget Predicting IIB				
	R2	F (1,80)	p	R2	F (2,79)	Model p value	Predictor p value	Partial r
Betweenness Centrality	0.004	0.345	0.558	0.036	1.468	0.237	0.931	-0.01
In-Closeness Centrality	0.06	5.104	0.027 *	0.108	4.799	0.011*	0.013*	0.269
Out-Closeness Centrality	0.027	2.207	0.141	0.109	4.853	0.01*	0.012*	0.271
In-Degree Centrality	0.001	0.093	0.761	0.037	1.5	0.229	0.793	0.029
Out Degree Centrality	0	0.014	0.906	0.052	2.149	0.123	0.254	0.126
In-Degree Tie Strength	0.049	4.105	0.046*	0.102	4.493	0.014	0.018*	0.258
Out-Degree Tie Strength	0.019	1.53	0.22	0.103	4.541	0.014	0.017*	0.26

* All regressions are individual simple regressions

Table 16

Network Structural Properties as Moderators

Controlling budget, Interaction effect with NCB Predicting IIB					
	Adjusted R2	F (4,77)	Model p value	interaction effect p value	partial of interaction effect
Betweenness Centrality	0.242	7.475	0	0.102	0.16
In-Closeness Centrality	0.259	8.066	0	0.982	0.002
Out-Closeness Centrality	0.257	7.992	0	0.691	0.038
In-Degree Centrality	0.222	6.78	0	0.575	0.055
Out Degree Centrality	0.235	7.212	0	0.258	0.111
In-Degree Tie Strength	0.257	8.016	0	0.845	0.019
Out-Degree Tie Strength	0.257	7.996	0	0.572	0.054

* All regressions are individual simple regressions

Chapter 5: Interpretations and Recommendations

Working across boundaries in a multiorganizational network context has become a common practice, and more research on the behavioral dimension in this context has been called for (Provan, Sydow, & Podsakoff, 2014). Thus, the researcher sought to understand if a novel concept, network citizenship behavior (NCB), in addition to network structural properties, impacts innovation in the public sector. This chapter summarizes the conclusions drawn from the results of the data analysis, provides implications for both theory and practice, discusses limitations and delimitations, and finally offers direction for future research.

Conclusion 1: Does NCB Predict IIB?

As shown in Figure 10, the result of the current research suggested that NCB is statistically significant in predicting IIB, explaining at least 17.8% of the variance in IIB. Even though 17.8% might not make NCB a large, dominant predictor of IIB, considering how complex a phenomenon like innovation is and how many other factors impact IIB, this R squared is quite significant and should not be ignored.

Previous research on innovation has been comprehensive, and Anderson, de Dreu, and Nijstad (2004) suggested that innovation research has become “routinized.” There are many, many factors, both internally to the employees (Afsar & Badir, 2016; Akhavan et al., 2015; Carmeli, Weitar, & Weisberg, 2006; Cummings & Oldman, 1997; Janssen, 2000, 2005; Kirton, 1976; Knol & van Linge, 2008; Messman & Mulder, 2011, 2013, 2015; Odoardi, et al., 2015; Shalley, Gilson, & Blum, 2009; Tierney & Farmer, 2002; Tierney, Farmer, & Graen, 1999; Yuan & Woodman, 2010), as well as externally related to job (Cummings & Oldman, 1997; Dorenbosch, van Engen, & Verhagen, 2005; Janssen

2000; Oldman & Cummings, 1996; Ramamoorthy et al., 2005; Shalley & Gilson, 2004; Shalley, Gilson, & Blum, 2009; Spiegelaere, et al., 2014; Tierney & Farmer, 2002), leaders (Basu & Green, 1997; Mishra et al., 2017, Mumford, Scott, Gaddis, & Strange, 2002; Reuvers et al., 2008; Scott & Bruce, 1993; Tierney, Farmer, & Graen, 1999; Van Minh, Badir, Quang, & Afsar, 2017), and the organizational culture in general (Afsar & Badir, 2016; Amar & Juneja, 2008; Damanpour, 1996; Erez & Nouri, 2010; Miron et al., 2004; Shalley & Gilson, 2004; Shalley, Gilson, & Blum, 2009; Shalley, Zhou & Oldham, 2004; Wojtczuk-Turek & Turek, 2015; Sarooghi, Libaers, & Burkemper, 2015) that have been found to predict innovation.

Based on prior quantitative research, many factors have been found to predict innovation. On an individual level, psychological capital contributes 30.69% of the variance in innovation (Wojtczuk-Turek & Turek, 2015), psychological empowerment contributes about 36% (Odoardi et al., 2015), positive affect contributes about 15.21% (Messman & Mulder, 2011), 13.69% is from work engagement (Messman & Mulder, 2011), and 10.2% from work locus of control (Messman & Mulder, 2011). Externally, job autonomy contributes up to 29.7% to innovation (Spiegelaere et al., 2014), whereas job complexity contributes about 5% of variance in innovation. Related to leadership, leader-member exchange predicts about 3% of the variance in innovation (Tierney, Farmer, & Graen, 1999), and leadership support predicts about 4% (Basu & Green, 1997).

For example, Wojtczuk-Turek & Turek (2015) established a model to describe the relationships among HR flexibility, individual flexibility, psychological capital, and IIB. They conducted the study with a sample of $N = 237$, composed of employees of service

industry organizations and postgraduate students at the Warsaw School of Economics. Their findings suggested that HR flexibility and individual flexibility predicted psychological capital, and psychological capital predicted IIB ($r = .554$, $p < .01$).

Admittedly, an organization is a complex system with intertwined phenomena, and these factors are not mutually exclusive. For example, psychological capital in some ways encompasses psychological empowerment and positive affect, and work engagement and work locus of control mutually influence each other. Good leadership support contributes to effective leader-member exchange. As a result, cumulatively, the predictors of innovation from various previous research exceed 100%.

As demonstrated above, the most significant predictors of innovation are psychological capital and psychological empowerment, both of which are internal individual motivators. Other than these two factors, all other factors' contributions to innovation are relatively small, ranging from 3% to 15%. The fact that NCB, as a newly developed concept, could explain as much as 17.8% of the variance in IIB, which is more than most of the other factors that have been studied previously, is a significant finding that should be emphasized.

Another interesting finding of the analysis is that one of the controlled variables, boundary-spanning status, serves as a negative controlled variable, meaning people who did not have interorganizational responsibilities in their job description have a higher level of innovative behavior. This is a particularly interesting and counterintuitive finding, because traditionally in the research the emphasis has been placed upon boundary-spanning personnel. However, this finding suggests it is employees that are not required to work across organizational boundaries who bring in more innovation to their

organizations if they choose to interact with colleagues from other organizations. There is no previous literature on which to base this finding, thus it is difficult to explain the reason behind this finding. One possible conjecture would be that boundary-spanning personnel might not be involved in work delivery as deeply as non-boundary spanning personnel, and non-boundary spanning personnel have more power to implement something innovative in the work that they do. However, this requires further examination in the future.

Among the four types of NCB measured in this current research, Loyalty explains the most variance in IIB (14.9%), followed by Compliance (11%) and Altruism (10.8%). Tolerance explains only a small fraction of variance in IIB (4.6%). The low predictive power of Tolerance could partially be explained by the quality of the Tolerance measurement: after deleting items with bad loadings, only one item was left to measure Tolerance, which makes it difficult to assess its reliability. In addition, it is also not surprising that Tolerance does not relate to innovation very much: Interorganizational Tolerance describes employees' tendency to forgive unavoidable mistakes made by partner organizations. It is an important type of behavior to create trust, but it does not necessarily lead to knowledge exchange.

What is interesting among the four subscales of NCB is that Loyalty, instead of Altruism, provides the strongest prediction of IIB, counter to what the researcher expected. One would think that Altruism usually takes place in the process of problem solving and conducting actual work deliverables, and ideas are most likely to be exchanged during this process. This work-orientation of Altruism was manifested in its items through verbiage such as "help our partner with business-related issues" or

“provide assistance to our partner if they have a problem.” One possible explanation could be that when solving problems, exchanges are most likely to be focused exclusively on the problem at hand, and very little free conversations take place to allow people to exchange ideas with each other without a sense of urgency. However, no prior research was found to explain Altruism’s lack of predictive power on IIB, and further research is needed to confirm the assumption. Loyalty, on the other hand, describes how much employees would recommend their partnership organizations to others. It may be through these introductory and explanatory conversations free of restrictions that people notice the best practices of partnership organizations, and thus think about how to bring innovation to their home organizations.

Another explanation of the importance of Loyalty could be found in prior research on whole-network’s legitimacy. Human and Provan (2000) proposed that legitimacy, both internal legitimacy (developing trust, demonstrating shared value, and building sustainable network governance) and external legitimacy (promoting network and its activities to outsiders, seeking new members, and providing outside resources to meet network goals), are significant to the effectiveness of a whole network. Loyalty, by its definition, describes a person’s allegiance to the whole network and the partnership organizations within this whole network, which is a demonstration of a network’s external legitimacy. There are two items that measure Loyalty in the instrument. Item 6 states, “We tell others that our partner is a good organization with whom to do business,” and item 8 states as “We promote our partner’s services to others.” Both of these items discuss how to promote activities of partners within the network to outsiders, which aligns with the definition of external legitimacy.

Conclusion 2: What Is the Role of Network?

The discussion of the network is constrained by the limited scope that the sample was able to provide. Network research is sensitive to the sample size and the response rate. People and jurisdictions in the network are limited to the respondents of the survey and the contacts that they have identified, which might not be a perfect representation of the sampling frame or the population. Admittedly, because participants in the sampling frame already have experience in interorganizational work, their exhibition of NCB might be more positively skewed than the general population.

Network map. The current research generated a network map that provided interesting insights into how information flows between these organizations in the Metropolitan Washington region. A few observations could be made about the network structure of the respondents of the survey and their main professional contacts based on Figure 9 in Chapter 4. First of all, a few large jurisdictions serve as hubs of information flow, such as Arlington County, the City of Alexandria, the District of Columbia, Montgomery County, and Fairfax County. Interestingly, these are the jurisdictions that are closer to the center of the region geographically (based on Figure 2), and they are also the larger jurisdictions with larger budgets and more resources (based on Table 7). Smaller jurisdictions, such as the Cities of Rockville, Gaithersburg, or Fairfax, and jurisdictions that are on the outskirts of the region, such as Frederick County and Charles County, occupy peripheral positions and have a relatively smaller role in the information flow dynamics of the region.

In addition, jurisdictions that are closer to each other geographically (based on Figure 2) and are similar in size (based on Table 7) seem to have more interaction with

each other. This is true to central organizations as well as peripheral organizations. For example, Fairfax County, Arlington County, City of Alexandria, and Prince William County are all jurisdictions in northern Virginia that are adjacent to each other, and their information exchanges are frequent and intense, represented by the thicker lines in between those actors. The same relationship exists between Montgomery County and Prince George's County, both large jurisdictions in southern Maryland that border each other. For smaller jurisdictions, even though their interaction is not as intensive, the organizations that they tend to speak to are mostly nearby similar jurisdictions, such as between the City of Rockville and City of Gaithersburg, and between City of Falls Church and City of Fairfax, etc.

Finally, the role of a network administrative organization (NAO, Provan & Kenis, 2008) – one that is dedicated to coordinating the whole network – is instrumental in the information flow. For example, the Metropolitan Washington Council of Governments (MWWOG) occupies a very centralized location in the network. Even though its strength with each jurisdiction might not be the strongest, it definitely serves as a hub for the information flow, with many contacts speaking with employees of MWWOG. The information from different jurisdictions becomes centralized here, and then is distributed to others. This confirms findings in previous research (Span, Luijkx, & Schols, 2012) that even though it might not be the single deciding factor for network effectiveness, an NAO has been found to be the most successful governance structure in managing a network of organizations.

Network structural property and IIB. The findings of the current research provided further evidence in the ongoing discussion of the role that network position

plays in innovation. First of all, most previous research studies had concluded that centrality has a significant positive relationship with innovation (Bell, 2005; Casanueva, Castro, & Galan, 2013; Chiu, 2008; Tsai & Ghoshal, 1998; Ibarra, 1993; Perry-Smith, 2006; Tseng et al., 2016). Based on the second multiple regression, In-Closeness Centrality predicts 7.24% of the variance in IIB and Out-Closeness Centrality predicts 7.34% of the variance in IIB. The evidence presented by the sample of the current research is in agreement with the majority of research studies, that the more central a location a network actor occupies, the more innovative behaviors the actor exhibits.

It is also worth noting which type of centrality was found to statistically significantly relate to innovative behaviors: closeness centrality, both in-bound and out-bound. Closeness centrality measures how close an actor is to all the other actors in the network. Different from degree centrality (which measures the extent of network ties of an actor among the rest of the network) and betweenness centrality (which measures the degree to which an actor “sits” in between two other random actors), closeness centrality measures how many “steps” it takes for any actor to reach all other actors in the same network. A prerequisite of innovation is effective information sharing, which, in its essence, is how well information is able to “travel” from one actor to the other. Closeness centrality measures how easy it is for a particular actor to receive and disseminate information to the rest of the network. Compared with the other two types of centrality, closeness centrality is the most useful in measuring the breadth and effectiveness of information sharing, which could explain why this particularly type of centrality was found to relate to innovation most significantly.

Different from centrality, the effect of tie strength (the accumulation of all relationship strengths measured by a 5-scale communication frequency ranking) on innovation is largely still in debate. On one hand, strong ties are more beneficial for the exchange of knowledge and information by creating trust and a willingness to share that consequently facilitates innovation (Albrecht & Ropp, 1984; Krackhardt, 1992; Bouty, 2000; Reagans & Zuckerman, 2001; Landry et al., 2002; Schulz, 2003; Uzzi & Lancaster, 2003; Levin & Cross, 2004; Moran, 2005; Smith et al., 2005; Fritsch & Kauffeld-Monz, 2010). On the other hand, the strength of weak ties is an often overlooked asset that brings actors access to disconnected partners and thus diverse information (Granovetter, 1973; Hansen, 1999; Perry-Smith, 2006; Yli-Renko, Autio, & Sapienza, 2001; Levin & Cross, 2004). To combine both perspectives, Leenders et al. (2003) discovered that it is an inverted U-shape relationship between tie strength inside a team and team creativity. They suggested that both too low and too high a level of interaction frequency impeded team creativity, while moderate frequency produce the highest level of creativity. McFadyen and Cannella (2004) had a similar finding, that after a certain point, more interactions seem to bring diminishing benefits.

The findings of the current research supported the argument for strong ties: the stronger the tie strengths are, the stronger the exhibition of innovative behaviors. As Hansen (1999) pointed out, innovation is a complex phenomenon, and requires intensive exchanges, deep trust with each other, and a motivation to act to be able to bring innovation to fruition, which weak ties cannot offer. In addition, strong ties could also create multiple layers of interpersonal exchange to facilitate the discussion of innovation in organizations (Albrecht & Ropp, 1984), which weak ties are also not able to provide.

For the current sample, based on the second multiple regression analysis, it could be concluded from our discussion that tie strength has a positive relationship to innovation.

Finally, one of the key purposes of the current study was to examine the relationship and dynamics between behavior and structure in a whole network. Unfortunately, the current study did not support the hypothesis. The interactional effects between network properties and NCB on IIB was not statistically significant. There seems to be no interactional relationship between network structural properties and NCB. Reasons for the lack of observation of this relationship is unclear. One key reason could be the less than satisfactory quality of the network data. Network data are sensitive to sample size and response rate. A good quality network research requires at least 90% of the participants in the network respond to the data collection survey. Such a requirement is not achieved in the current data collection effort. Only 31.32% of the invitees responded to the survey, which is consistent with the response rate found in other organizational research: a meta-analysis conducted by Cycyota and Harrison (2006) revealed that, on average, survey studies in organizations yield an average response rate of 32%. The full network population includes thousands of employees in the local government, and collecting professional relationships among such a large population was not realistic for a dissertation study. Because of the resource constraint, the proposed study sought to draw a convenient sample to provide a preliminary picture of the network structure of the population to lay a foundation for future research; however, it was proven to be not ideal. In a perfect world, if future research could collect better-quality network data on a smaller network, it would still be worth investigating the moderator effects of network structural properties on the relationship of IIB and NCB.

Structural versus behavioral approaches. In previous literature, the majority of social network researchers focused primarily on the structural approach, studying properties such as size, density, structural holes, and tie strengths and how they are related to innovation. Obsteld (2005) and Dolfsma and vander Eijk (2016) both called for calling attention to the “action problem” in social network research. The findings of the current study, however, suggest that the two approaches are not necessarily mutually exclusive, but rather supplementary. Both structural approach (network structural properties such as centrality and tie strength) as well as behavioral approach (NCB) impact IIB at the same time. The relationship between the two approaches are not necessarily either/or, but rather the combination of the two impact the outcome variable IIB simultaneously. The more specific relationship between the two approaches should be further examined in future research.

The results of the analysis in this research are presented in the conceptual framework in Figure 10, describing the relationships between NCB, network structural properties, and IIB.

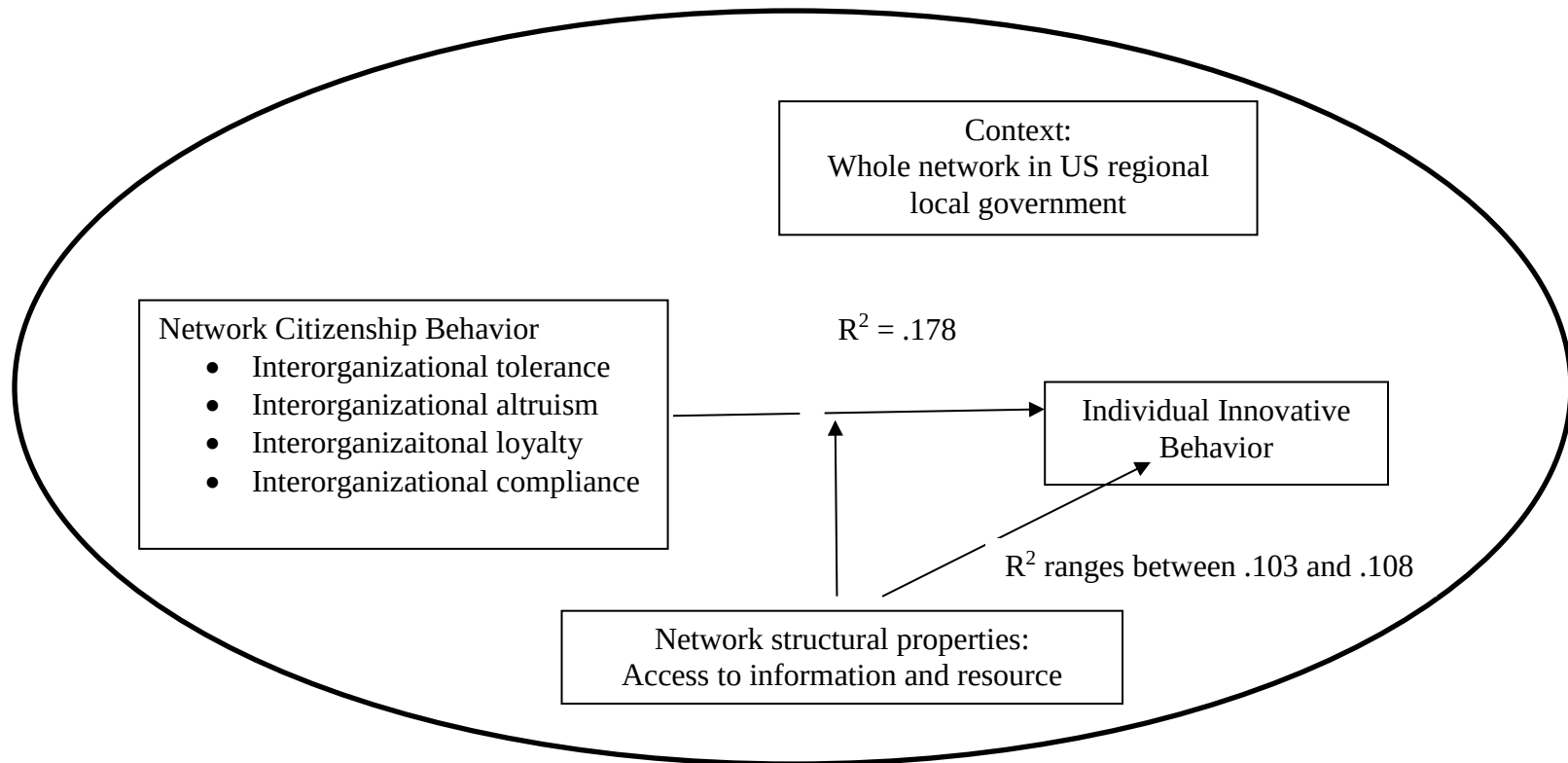


Figure 10: Conceptual Framework with Analysis Results

Conclusion 3: Instrument Validation

Although the current research did not set out to examine the reliability and validity of the instruments used, the data and analysis were able to provide additional evidence to support the use of the instrument. Based on the recommendations of CFA and Cronbach's alpha values, several modifications were made, and the reliability for both instruments was improved. This is an unexpected but supplementary value that the current research study was able to provide.

NCB 2.0. So far, extant research on NCB has been mostly either conceptual (Provan, Sydow, & Podsakoff, 2014) or qualitative (Gerke et al., 2017). The only measurement that has been developed, ICB by Skinner, Autry, and Lamb (2009), has received limited validation. Therefore, the first analysis that the current study conducted was a thorough validation of the NCB measurement instrument, and it was able to provide additional evidence for its reliability and validity.

The results of the current research analysis indicated that the Cronbach's alpha reliability score was able to be increased from .828 to .840. More importantly, the modified model significantly increased model fit indices. Most of the model fit indicators, including Goodness of Fit (GIF), Adjusted Goodness of Fit (AGIF), Comparative Fit Index (CFI), and Standardized Root Mean Square Residual (SRMR), all increased from not fit to fit after the modifications. Two other indicators, the Non-Normed Fit Index (NNFI), also known as Tucker-Lewis Index (TLI), and Root Mean Square Error of Approximation (RMSEA), still did not meet the criteria for being a fit model. However, they are both better and much closer to the cutoff value. The reason that RMSEA is still less than satisfactory could be because RMSEA tends to be sensitive

toward degrees of freedom, and a small sample size of less than 100 is likely to more negatively impact RMSEA than other indicators (Kline, 2016). The original measurement, for example, had a much larger sample ($n = 195$), and the reported RMSEA was much more significant ($RMSEA = .066$) (Skinner, Autry, & Lamb, 2009).

Based on the CFA analysis for NCB validation, three items were deleted for having low Cronbach's alpha values or factor loadings, and one item was reloaded to a different latent variable. The three deleted items were item 1, "we accept the inevitable mistakes that our partner sometimes makes without complaining"; item 5, "we sometimes volunteer our resources in order to lighten our partner's workload"; and item 7, "we sometimes sacrifice our own best interests in order to provide additional benefits to our partner." Item 1 has a negative connotation in its wording ("inevitable mistakes"). Item 5 and item 7 to some extent are asking for too much from partner organizations, such as "volunteering" resources or "sacrificing" self-interest, which are not likely to be acceptable in any organization. The remaining items that provided a higher Cronbach's alpha reliability score took a more positive tone, and discussed the services and assistance that are offered to partnership organizations instead of loss and damage.

Based on the recommendation of CFA of the IIB instrument, item 9 was reloaded to Altruism instead of Compliance. Item 9 stated "we attempt to meet or exceed any standards that our partner is also required or expected to meet." At first glance, it seems that this item discusses complying to the standards of partner organizations. However, different from the other two items that measure Compliance, which ask if employees "comply" or "abide by" rules and regulations of other partnership organizations, item 9 specifically asks if employees meet "or exceed" partnership organizations' standards. The

word “exceed” implies that it is no longer required, but is above and beyond what is required, suggesting such behaviors are no longer compliance. This item therefore provides better measurement for Altruism than does Compliance.

The improvement of the instrument and the development of NCB 2.0 is one of the most important contributions of the current research study. Before the current study, no empirical research had examined the quality of the instrument. The current research was able to evaluate its reliability, pinpoint its deficiencies, and provide recommendations for modification. A reliable instrument is critical for future research studies to continue examining the construct. The current study was not only able to provide an improved version of the instrument for future research studies to be built upon, but also to inspire further theory-building to better understand the details of the phenomenon. Admittedly one limitation remains, in that the Tolerance subscale was left with only one item, and future research is needed to develop more items for it.

IIB as a unidimensional construct. The validation for the IIB measurement instrument, developed originally by Kleysen and Street (2001), confirmed the findings from prior research that IIB is a unidimensional variable and cannot be further dissected into substeps or processes. At first glance, it could be counterintuitive and appear to be against existing theoretical arguments. Innovation has been described as a process that goes through several steps (De Jong & Den Hartog, 2010; Janssen, 2000; Kanter, 1988; Luke & Stephan, 2017; Messman & Mulder, 2012; Scott & Bruce, 1994). The fact that this instrument and several other innovative behavior instruments (Janssen, 2000; De Jong & Den Hartog, 2010) were not able to distinguish behaviors related to different subprocesses could be suggestive of a gap between the theoretical foundation and the

empirical examination. One possible explanation could be that the process view of innovation usually describes one particular innovative idea, product, or outcome, from beginning to end in a linear fashion. However, in reality, innovative practices in organizations are chaotic and complex and do not necessarily have clear-cut boundaries between each other. When it comes to innovative behaviors, it is not only an iterative process for any single innovation, but multiple innovations could also be entangled with one another and thus interact with each other in their developments. Furthermore, behaviors from different stages of innovation could be exhibited at the same time, thus making innovative behaviors difficult, if not impossible, to be separated into substeps.

Confirming that IIB is a unidimensional construct was critical to the current research so that IIB could be used as one single dependent variable without the distraction of subscales, and thus more emphasis could be placed upon the independent variables of NCB and the network structural properties. It does pose a constraint to future studies that focus more on innovative behavior as the construct of interest, which might not be able to provide as detailed insights as a multidimensional instrument could.

Theoretical Implications

The first significant contribution the current research is making toward theory is to bring attention to NCB, an understudied phenomenon in interorganizational collaborations. All too often, researchers focus on the network structure in organizations to explain innovation, financial performance, and other outputs, whereas behaviors above and beyond the employees' job requirements are largely ignored. Ever since its first development, there has been very, very limited research examining the construct. The

current research suggested that these behaviors not only exist, but they contribute to a significant portion of innovation in government organizations.

Secondly, the current research study provided additional data for the evaluation of the NCB measurement instrument. Previously, very little scholarly evidence was provided to support the reliability and validity of the NCB instrument. Only the original researchers conducted a systematic and thorough validation and reported on the reliability and validity of the instrument. Since then, the instrument has not undergone any empirical, quantitative testing. Even with the validation of the original article, the researchers were only able to report Cronbach's alpha values that ranged between .702 and .915 for its four subscales, and no overall Cronbach's alpha value for the entire measurement was mentioned. The current study provided additional, critical evidence of the validity and reliability of the instrument using firsthand, empirical data and extensive statistical analysis. With very few modifications, the current research identified items that do not measure NCB very well and items that should be loaded to different subscales. Using 8 out of the 11 original items, the current research increased the Cronbach's alpha value of the instrument from .828 to .840 using the updated model.

In addition, the current research also advanced our understanding of innovation in government organizations. Among many other factors that contribute to innovation, NCB has received very little attention, yet it has been proven to be a significant contributor. NCB should be incorporated into models or frameworks that seek to deeply understand innovation and its antecedents in organizations.

Practical Implications

All too often we hear managers restrict their employees from participating in interorganizational activities, citing reasons such as less time to be dedicated to their “real” job, fear of losing their staff to other organizations, or lack of value in engaging in these kinds of activities. The benefit of participating in interorganizational events and supporting other organizations with the same goal is rarely documented or appreciated. The current research presented at least one benefit that organizations and managers should care about: based on the first simple regression analysis, if you encourage your staff to work across organizational boundaries (exhibition of NCB), it is likely to make your staff more innovative (exhibition of IIB), which is highly valued and sought after in today’s environment (McNabb, 2006). This is especially true for non–boundary spanning employees: if it is not a responsibility for you to work across organizations, why “waste” time on it? However, the current research suggests that it is these non–boundary spanning employees who are more likely to translate interjurisdictional work into innovation, which further reassures managers to not only allow but also encourage their employees to go out and work across boundaries in order to achieve organizational goals.

In addition to establishing the importance of interorganizational engagement, the current research further outlines the types of behavior that are most likely to create more innovation. Among the four types of behaviors studied in this research – Tolerance, Altruism, Loyalty, and Compliance – Loyalty was found to be the most significant predictor of IIB. This implies that managers and organizations should allow employees to be proud of the work of partnership organizations, and recommend their products or services to the general public. In other words, instead of being competitive with one

another, becoming cooperative within the network helps to increase innovation within the network's membership organizations.

Limitation and Delimitation

Limitations. The current study does pose several limitations. First, even though the number of responses collected through the data collection met the required sample size calculated through power analysis for regression, it is still a small sample size ($N = 89$). Generally speaking, for a regression analysis, a minimum sample size of 100 or more participants is believed to yield more reliable results. In addition, the small sample size is particularly problematic for CFA. Kline (2016) recommended at least $n = 200$, or 10 for each item being measured. However, the current study did not set out to examine the validity of instruments. Though the conclusions drawn from CFA should be interpreted with caution as a result of the small size, it serves as a first step to inform future research studies to further examine the construct and further improve the instrument.

A second limitation of the current research is related to the nature of the survey research. Self-reported survey studies suffer from the lack of ability to control the accuracy of the participants' responses, and the results might be biased by individuals' feelings at the time of completing the survey. For example, some participants might go through the survey very fast and may not spend enough time on each item to provide a good-quality response. Some respondents selected the same option throughout the survey or for most of the items in the survey. This is especially the case toward the later part of the survey, as respondents became tired of answering the questions. For example, respondents 51 and 53 provided exactly the same answer (6 – Always) for all 11 items for IIB. These responses were included in the data analysis, because their responses to the

first half of the survey (network and NCB) were varied and valuable; however, it is difficult to estimate if their responses to the second half of the survey were a true reflection of reality. Also, the respondents might “predict” the answers the researcher is looking for based on preconceived perceptions of the purpose of the survey. However, the impact of this limitation should be minimal, because the researcher has no coercive power over the employment and performance of the respondents, thus the respondents should have no intention or motivation to answer the questions in order to please the researcher.

Another limitation of the survey method relates to the items and the respondents’ interpretation of them. It is entirely possible that issues such as item ambiguity, personal bias, and a lack of knowledge by respondents could lead to inaccurate responses. This was particularly noticeable in the collection of network data. The survey question stated, “Thinking about the top five people from other jurisdictions with whom you usually communicate regarding work issues, which jurisdictions are they from, in the order of frequency of communication?” However, there are several responses in which the respondents clearly did not understand the nature of the question. Responses such as “elected officials,” “my police chief,” “congress,” “state of Maryland,” or the name of a specific person are some examples of ineligible answers that were collected.

In addition, due to the issue of feasibility, the current research used a convenience sample instead of a more rigorous sampling strategy, such as random sampling or stratified sampling. One downside of a convenience sample in the current study is that the sample skews toward people whose jobs include some exposure to cross-jurisdictional responsibilities. For example, when answering the question “In your role, are you

required to work with people from other jurisdictions?” almost 80% of the participants responded yes. Anecdotal observation suggests, however, that in reality the percentage of employees who normally are required to work across organizational boundaries would be much lower. As a result, the exhibition of NCB in the sample might be higher than the exhibition of NCB in the general population. However, the impact of this biased sampling strategy is unknown. Based on the regression analysis, people without boundary-spanning responsibilities exhibit a higher level of IIB, so this convenience sample might possibly have lowered the effect of NCB over IIB. In any case, because the sample is not a perfect representation of the population, any conclusion and generalization drawn from the analyses of the current study should be interpreted with caution.

Moreover, even though the study was intended to be applied to other contexts in the future, currently, it only studied one whole network under one specific context. Although it could be informative to other whole networks of a similar size and under a similar industrial background, it can hardly be generalized beyond this context. However, the current study deliberately chose to focus on its specific context, and its generalizability can and should be tested in follow-up research.

Finally, there may be other contributing variables that were not the focus of this study and cannot be controlled for in this study due to a lack of observation. As indicated earlier, innovation was suggested to be predicted by a variety of factors in organizations. If and how NCB interacts with these other factors to predict IIB is unknown. However, this is a necessary first step in examining a novel construct, NCB, and any unexplainable variance can and should be explored by future research.

Delimitations. The current research intentionally chose the Metropolitan Washington Council of Governments (MWCOC) as its site, thus impacting the generalizability of its findings. First of all, MWCOC is a network of local government organizations, thus the findings might not be directly applicable to the private or nonprofit sector. Secondly, the research site is located on the east coast of the United States; as a result, no generalizations could be made about other nationalities or regions, even though cultural differences may have a strong impact on the exhibition of NCB or IIB. Even though it is the intention of the researcher to study NCB in a wide variety of populations, the current research specifically focused on regional local government in the United States, and the results could only be applied to other organizations of similar size, makeup, and industry.

Second, in addition to the factors that are being studied, there are many other organizational phenomena that could play a role. Many factors could lead to innovation, and NCB could lead to many other factors. However, it is the intent of the current research to focus only on these two variables, to serve as a foundation for future research to build more complicated models.

Future Research

Even though the current research was able to answer two research questions, there are more questions raised as a result. First of all, despite the improvements, the NCB instrument is still far from perfect. Among the several network citizenship behaviors that have been identified, only four types are included in the existing measurement. In addition, one particular type of NCB, Tolerance, was measured by only one item after deleting items with questionable loadings, calling into question the reliability of this

latent variable. Finally, the new instrument needs to be validated by a sample large enough for CFA.

A more robust, rigorous NCB measurement is desperately needed to continue this stream of research. In this process, it is critical to go back to the theories, and build up a stronger theoretical foundation for the construct. There are two ways to support the theory-building of NCB: first, to use an inductive approach and in-depth qualitative research to understand what other NCB behaviors are being practiced in organizations through interviews, observations, and focus groups. Second, to return to the theoretical roots, OCB and its components to build up a sound argument for the application of OCB in a whole network context.

Once a robust measurement instrument is developed and validated, NCB could be examined in more detail. So far in the current research, only one type of potential consequence of NCB, innovative behaviors, was suggested. Further research on antecedents, additional consequences, and other factors that interact with NCB in organizations should be looked at. Specifically, in their conceptual work, Provan, Sydow, and Podsakoff (2017) already proposed eight factors that could serve as antecedents for the existence of NCB. Figure 11 demonstrates these variables and their relationship to NCB. Their conceptual model is worth an empirical test.

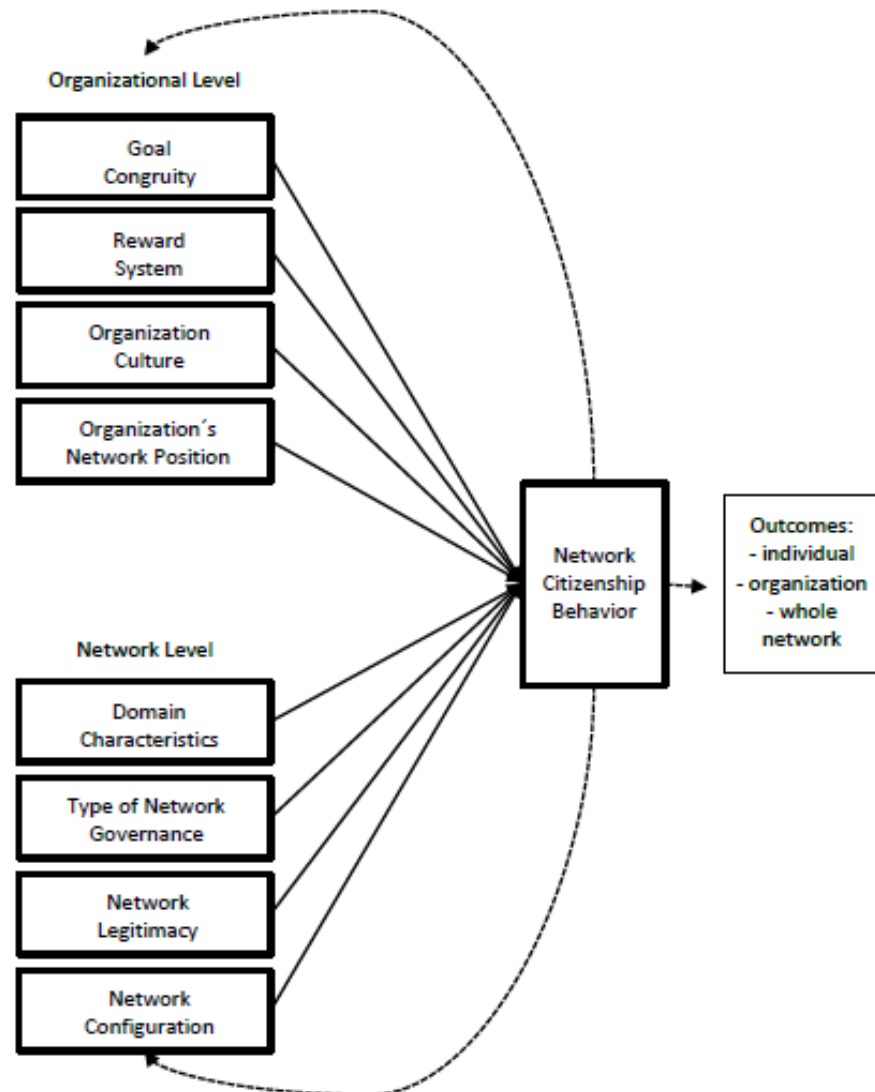


Figure 11: *Antecedents of NCB (Provan, Sydow, & Podsakoff, 2017)*

Moreover, the current research only hypothesized one type of relationship between NCB and IIB, and only one type of interactional relationship between structural properties and NCB. It is entirely possible that their relationship is far more complex than that. The relationship between NCB and IIB might not be linear; instead, it could be curvilinear or U-shaped. Network structural properties might not be a moderator between

NCB and IIB; instead, it is possible that network structure serves as a mediator or plays a completely different role in the relationship between NCB and IIB.

In addition, despite its intent, the current research did not find any evidence to support the interaction between behavior and network structural properties. However, that is not to say that such dynamics do not exist in organizations. Prior researchers had called for examining the “action problem” in network research (Dolfsma & van der Eijk, 2016; Obsteld, 2005), and the relationship between behavior and network positions definitely deserve more scrutiny.

What is more, another potential stream of study for future researchers is to apply the concept to other populations. Such populations could include private sector manufacturing or technology companies, networks in other regions and countries, or networks of different sizes.

Finally, the current study is limited by its resources, and was not able to conduct a true “whole-network” research. It merely utilized the concept of a whole network as the background of its research context, but was not able to produce dependent or independent variables on a network constellation level. In the future, if resources are available, it would be extremely valuable to design and conduct research studies on a whole-network level, comparing the exhibition of NCBs of different whole networks to each other, perhaps using other regional councils of governments across the U.S. to expand on the findings of this study.

Appendix: Survey sample

(page 1)

Informed Consent Form

Title of Study: For the Greater Good: Network Citizenship Behavior and Innovative Behaviors in American Local Government

IRB #: NCR191399

Principal Investigator: Dr. Diana Burley

INTRODUCTION:

You are invited to participate in a research study conducted by Jing Burgi-Tian, a doctoral candidate at The George Washington University, Graduate School of Education and Human Development, Human and Organizational Learning Program. The study is under the direction of Dr. Diana Burley, Professor of the Human and Organization Learning Department.

Your participation in this research is entirely voluntary. Your supervisor or employer will not be informed of your responses, and your employment and performance will not be affected in any way by participating in this study or by the way you answer the questions.

PURPOSE

The purpose of this research study is to examine the relationship between individual innovative behaviors and network citizenship behavior in the context of whole networks in local government in the United States.

PROCEDURE

If you choose to take part in this study, you will be asked to complete a survey that seeks to understand frequencies of certain behaviors. All survey items are on a frequency scale and the survey will not take you more than 15 minutes.

RISKS

Participating in this study poses minimal risk in your professional or personal life. If you experience any stress or discomfort answering any specific question, you may choose to skip that question or end your participation in the research at any time.

BENEFITS

Participating in this study does not benefit you directly. However, the findings of this research will inform both scholars and practitioners about behaviors needed in an interorganizational collaboration context pertinent to innovation. These findings can be further translated into management practices that enhance innovation and effectiveness in multiorganizational networks that will contribute to the well-being of our region.

CONFIDENTIALITY

All data will be retrieved only by the researcher and will be deleted from the online platform directly upon completion of the data collection process. We will not disclose

any personal information or survey responses to any employer, supervisor, peer, colleague, or teammate. Only aggregate data will be reported in the dissertation and any subsequent publications. All information will be handled in a confidential manner, stored at a password-protected hard drive, and every effort will be made to ensure your personal information will not be identified in the findings.

QUESTIONS:

If you have any questions, concerns, or complaints about the study, please contact the student researcher, Jing Burgi-Tian, at 215-200-9943 (cell), 202-994-5384 (office), or jingtian@gwu.edu

For questions regarding your rights as a participant in human research, please contact the Office of Human Research of George Washington University, at 202-994-2715.

DOCUMENTATION OF CONSENT

If you have read and understand the information above and you are willing to participate in this research, please click on the “next” button below.

Yours Sincerely,
Jing Burgi-Tian
Doctoral candidate, Human and Organizational Learning
Graduate School of Education and Human Development
George Washington University

Demographic Information:

In this section, we will ask you a list of demographic questions about yourself. None of the information will be capable of identifying you among the respondents, and you may choose not to answer any question if you do not wish to.

1. What is your age?
 - a. Below 30
 - b. 30-40
 - c. 40-50
 - d. 50-60
 - e. 60 and above
2. How long have you been working in local government?
 - a. Less than 5 years
 - b. 5-10 years
 - c. 10-20 years
 - d. 20-30 years
 - e. 30 years and above
3. Which jurisdiction are you from?
 - a. District of Columbia
 - b. Town of Bladensburg
 - c. City of Bowie
 - d. City of College Park
 - e. Charles County
 - f. City of Frederick
 - g. Frederick County
 - h. City of Gaithersburg
 - i. City of Greenbelt
 - j. City of Hyattsville
 - k. City of Laurel
 - l. Montgomery County
 - m. Prince George's County
 - n. City of Rockville
 - o. City of Takoma Park
 - p. City of Alexandria
 - q. Arlington County
 - r. City of Fairfax
 - s. Fairfax County
 - t. City of Falls Church
 - u. Loudoun County
 - v. City of Manassas
 - w. City of Manassas Park
 - x. Prince William County
4. What is your professional background?
 - a. Social Services
 - b. Public Safety
 - c. Administration
 - d. Human Resources
 - e. Public Work
 - f. Parks and Recreations
 - g. Budgeting and Finance
 - h. IT and Information Systems
 - i. Other
5. In your role, are you required to work with people from other jurisdictions?
 - a. Yes
 - b. No
6. Do you consider yourself to be
 - a. Entry-Level
 - b. Front-Line Supervisory
 - c. Middle Management
 - d. Strategic Leadership

7. Thinking about the top five people from other jurisdictions with whom you usually communicate regarding work issues, which jurisdictions are they from, in the order of frequency of communication?
 - a. The person I communicate most frequently with is from:
 - b. The person I communicate second most frequently with is from:
 - c. The person I communicate third most frequently with is from:
 - d. The person I communicate fourth most frequently with is from:
 - e. The person I communicate fifth most frequently with is from:

(Page 3)

In this section, we will ask you about your behaviors when working with interorganizational partners. When answering these questions, think of “partner” as being your colleagues in other jurisdictions in the National Capital Region.

On a scale of 1 to 7, to what degree do you agree with the following statement:

1	2	3	4	5	6	7
Strongly disagree	Disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Agree	Strongly agree

1. We accept the inevitable mistakes that our partner sometimes makes without complaining.
2. When our partner is unable to fulfill certain responsibilities, we try to be understanding.
3. We go out of our way to help our partner with business-related issues if we sense that they are in need.
4. We provide assistance to our partner if they ever have a problem in an area where we have expertise.
5. We sometimes volunteer our resources in order to lighten our partner’s workload.
6. We tell others that our partner is a good organization with whom to do business.
7. We sometimes sacrifice our own best interests in order to provide additional benefits to our partner.
8. We promote our partner’s services to others.
9. We attempt to meet or exceed any standards that our partner is also required or expected to meet.
10. When dealing with our partner, we comply with expected standards of social behavior.

11. We show respect for our partner's organizational policies and try to abide by them ourselves.

(Page 4)

In this section, we will ask you how often you exhibit certain behaviors in your workplace.

In your current job, on a scale of 1-6, how often do you:

1	2	3	4	5	6
Never	Rarely	Occasionally	Sometimes	Frequently	Always

1. Look for opportunities to improve an existing process, technology, product, service, or work relationship?
2. Recognize opportunities to make a positive difference in your work, department, organization, or with customers?
3. Pay attention to nonroutine issues in your work, department, organization, or the marketplace?
4. Generate ideas or solutions to address problems?
5. Define problems more broadly in order to gain greater insight into them?
6. Experiment with new ideas and solutions?
7. Test out ideas or solutions to address unmet needs?
8. Evaluate the strengths and weaknesses of new ideas?
9. Try to persuade others of the importance of a new idea or solution?
10. Push ideas forward so that they have a chance to become implemented?
11. Take the risk to support new ideas?
12. Implement changes that seem to be beneficial?
13. Work the bugs out of new approaches when applying them to an existing process, technology, product, or service?
14. Incorporate new ideas for improving an existing process, technology, product, or service into daily routines?

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